CA1 – Human Centered Deep Learning

Name: Saketh Reddy Chandupatla

Student Number: x00231080

Section 1: Introduction

This dataset uses the diabetes health indicators dataset which was derived from the 2015 Behavioral Risk Factor Surveillance System(BRFSS), which is one of the world's largest ongoing health surveys conducted by Centers for Disease and Prevention(CDC) containing health, lifestyle and demographic features collected within this survey.

The dataset used in this project is a structured tabular dataset intended wheter a person has diabetes(Diabetes_binary =1) or (Diabetes_binary =0). All features in this dataset are numerical making this very suitable for classification using neural networks and other supervising algorithms.For this project a subset of the actual complete dataset was assigned to each student to ensure there is fair division of work and computational power management.

Our goal is to explore and build a neural network based on model which classifies individuals as diabetic or not using these features. This involves understanding the dataset and its structure, exploring the relationship of different features and testing multiple model architecture and choosing the best one.

Each row in the dataset represents an individual participant where as each column is representing an individual charcteristic. This is my breakdown of the 22 columns/features, followed by the target variable.

Column Name	Description
`HighBP`	Binary: 1 if the individual has high blood pressure, else 0
`HighChol`	Binary: 1 if the individual has high cholesterol, else 0
`CholCheck`	Binary: 1 if cholesterol has been checked in the last 5 years, else 0
`BMI`	Continuous: Body Mass Index (a weight/height measure of body fat)
`Smoker`	Binary: 1 if the individual has smoked at least 100 cigarettes in lifetime
`Stroke`	Binary: 1 if the individual has ever had a stroke, else 0
`HeartDiseaseorAttack`	Binary: 1 if had coronary heart disease or myocardial infarction
`PhysActivity`	Binary: 1 if participated in any physical activity/exercise in past 30 days

Column Name	Description
`Fruits`	Binary: 1 if consumes fruits at least once per day
`Veggies`	Binary: 1 if consumes vegetables at least once per day
`HvyAlcoholConsump`	Binary: 1 if consumes >14 (men) or >7 (women) alcoholic drinks/ week
`AnyHealthcare`	Binary: 1 if has any form of health coverage
`NoDocbcCost`	Binary: 1 if could not see doctor due to cost in the past year
`GenHlth`	Ordinal (1–5): Self-reported general health (1 = Excellent, 5 = Poor)
`MentHlth`	Integer: Number of mentally unhealthy days in past 30 days
`PhysHlth`	Integer: Number of physically unhealthy days in past 30 days
`DiffWalk`	Binary: 1 if serious difficulty walking or climbing stairs
`Sex`	Binary: 0 = Female, 1 = Male
`Age`	Ordinal (1–13): Age category (e.g., 1 = 18–24, 13 = 80+ years)
`Education`	Ordinal (1–6): Level of education (1 = Never attended, 6 = College graduate)
`Income`	Ordinal (1–8): Income level (1 = Less than $10k, 8 = More than $75k)
`Diabetes_binary`	**Target variable**: 1 = Diabetic, 0 = Non-diabetic

The key observations from this dataset are that it contains a mix of

1. Binary values like HighBP, Smoker
2. Ordinal categories like GenHlth, Age, Income
3. Continuous or numeric data like BMI, MentHlth

Section 2 : Opening the dataset, Brief data exploration and Data pre-processing(if required)

This section explores the dataset structure, class distribution, summary statistics, and checks for missing values. Initial preprocessing decisions are also discussed.

In [1]:
```
# Step 1: Loading the dataset
import pandas as pd
# Load the CSV file
df = pd.read_csv('X00231080.csv')
# Displaying shape and the first few rows
print("Dataset shape:", df.shape)
df.head()
```

```
Dataset shape: (42415, 22)
```

Out[1]:

	HighBP	HighChol	CholCheck	BMI	Smoker	Stroke	HeartDiseaseorAttack	PhysActivity
0	0	0	0	21	1	0	0	
1	1	1	1	23	1	0	0	
2	1	1	1	36	0	0	0	
3	1	1	1	34	0	0	0	
4	1	1	1	45	1	0	0	

5 rows × 22 columns

This dataset contains 42,415 rows and 22 columns. each row representing an individual paricipant where each column represents a indicator like blood pressure, bmi and others discussed above.

In [2]:
```
import matplotlib.pyplot as plt
import seaborn as sns

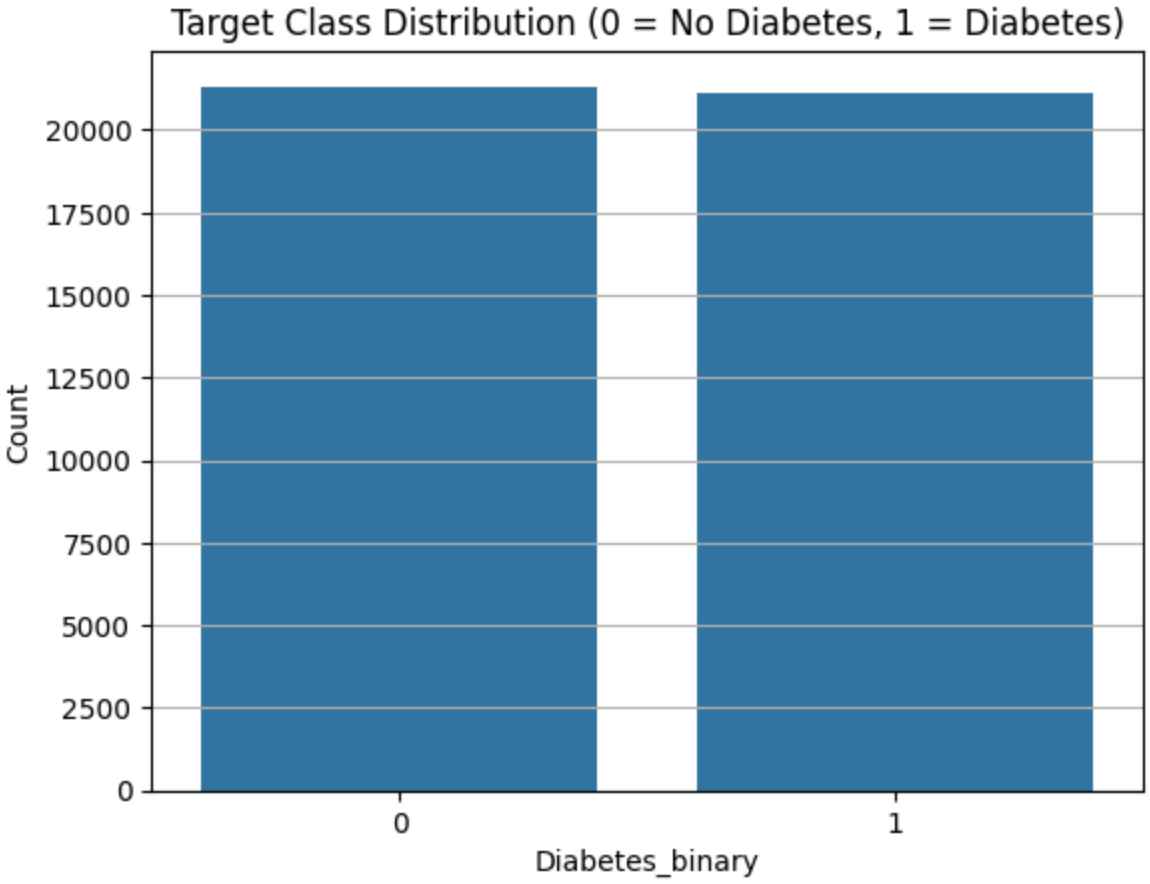
# Showing value counts and normalized percentages
target_counts = df['Diabetes_binary'].value_counts()
target_percentages = df['Diabetes_binary'].value_counts(normalize=True) * 100

print("Target class counts:")
print(target_counts)
print("\nTarget class distribution (%):")
print(target_percentages)

# Plot class distribution
sns.countplot(x='Diabetes_binary', data=df)
plt.title("Target Class Distribution (0 = No Diabetes, 1 = Diabetes)")
plt.xlabel("Diabetes_binary")
plt.ylabel("Count")
plt.grid(axis='y')
plt.show()
```

```
Target class counts:
Diabetes_binary
0    21326
1    21089
Name: count, dtype: int64

Target class distribution (%):
Diabetes_binary
0    50.279382
1    49.720618
Name: proportion, dtype: float64
```



The dataset is almost perfectly balanced with roughly 50% of instances in each class. This balance ensures that standard classification metrics like accuracy, precision and recall will be ,meaningful without the need for resampling or class-weight adjustments.

In [3]:
```
# Summary statistics for all columns
df.describe()
```

Out[3]:

	HighBP	HighChol	CholCheck	BMI	Smoker	Stroke
count	42415.000000	42415.000000	42415.000000	42415.000000	42415.000000	42415.000000
mean	0.563928	0.525781	0.975245	29.842768	0.475704	0.062973
std	0.495902	0.499341	0.155381	7.116605	0.499415	0.242917
min	0.000000	0.000000	0.000000	13.000000	0.000000	0.000000
25%	0.000000	0.000000	1.000000	25.000000	0.000000	0.000000
50%	1.000000	1.000000	1.000000	29.000000	0.000000	0.000000
75%	1.000000	1.000000	1.000000	33.000000	1.000000	0.000000
max	1.000000	1.000000	1.000000	98.000000	1.000000	1.000000

8 rows × 22 columns

From the summary statistics few things we can say are

1. BMI ranges from 13 to 98 with a mean of roughly 29.8, indicating a mix of healthy and overweight individuals
2. PhysActivity has a mean of 0.70 which makes the dataset being little biased towards a more health aware people
3. Age ranges only between 1 to 13 with a mean of 8.6

In [4]:
```
# Checking for missing values
missing_values = df.isnull().sum()
print("Missing values per column:")
print(missing_values[missing_values > 0])

# Confirming total
print("\nTotal missing values in dataset:", missing_values.sum())
```

```
Missing values per column:
Series([], dtype: int64)

Total missing values in dataset: 0
```

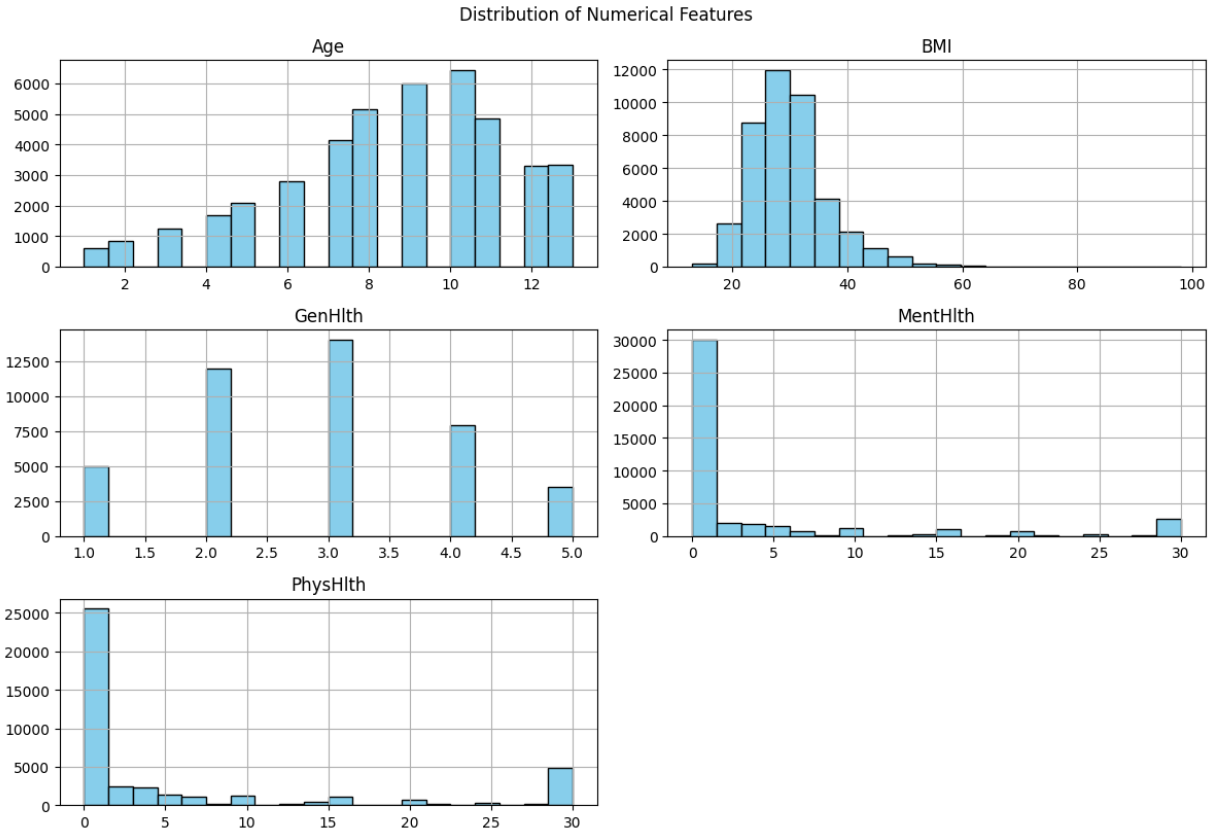
The dataset contains no missing values, which simplifies the pre processing step. All the 42,415 rows are complete and can be used directly for training and evaluation.

Numerical feature distribution

To better understand the dataset's numerical variables histograms were plotted for the following columns : Age, BMI, GenHlth, MentHlth, and PhysHlth.

In [5]:
```
numeric_cols = ['Age', 'BMI', 'GenHlth', 'MentHlth', 'PhysHlth']

# Histograms
df[numeric_cols].hist(bins=20, figsize=(12, 8), color='skyblue', edgecolor='black')
plt.tight_layout()
plt.suptitle("Distribution of Numerical Features", y=1.02)
plt.show()
```



1. Age: The age variable is encoded into groups (1 to 13) where the distribution was skewed towards the older population. The most frequent age groups lie between 9 and 11 indicating a large population is middle aged or older.
2. BMI: BMI shows a roughly normal shape mostly between 25 and 30 which aligns with overweight cateogary.
3. GenHlth: This measure is concentrated in middle values(2 to 4) suggesting most participants rate their health fairly good(on average).
4. MentHlth (Mental Health) and PhysHlth (Physical Health): In both these variables cases most people reported zero days and steeply decreasing for higer values which implies that most users consider themselves having good health both physically and mentally.

Binary Feature Distributions

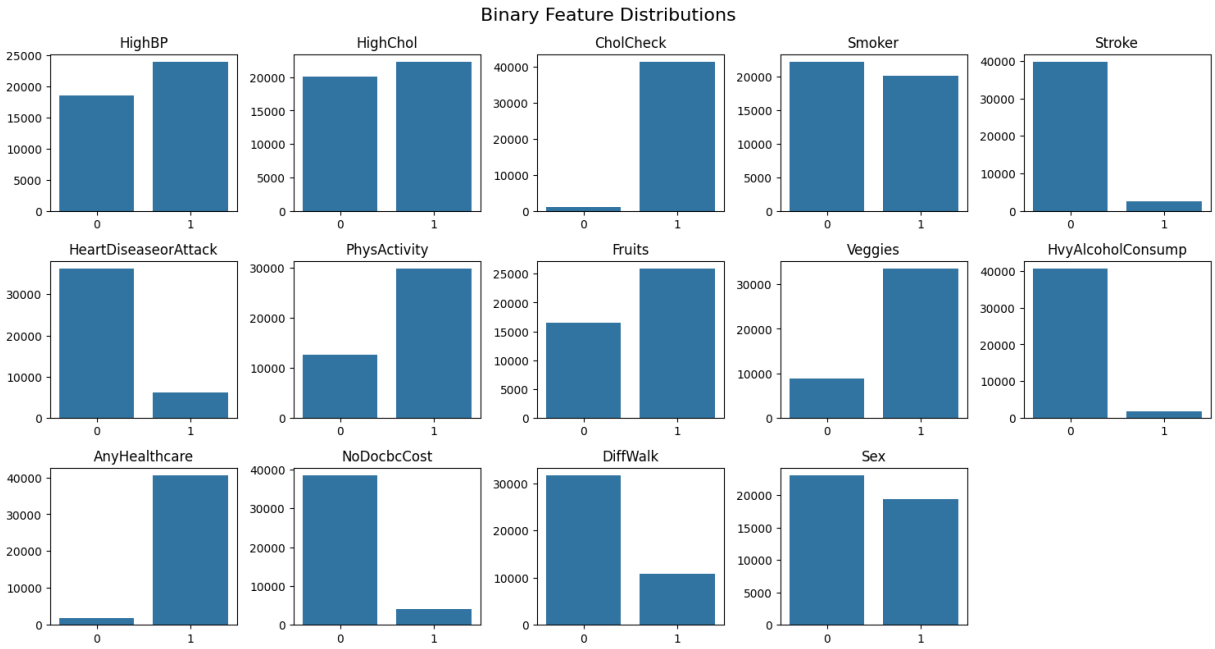
Same for binary features, to better visualize them count plots were plotted where key obervations are recorded.

In [6]:
```
# Showing counts for binary features
binary_cols = [col for col in df.columns if df[col].nunique() == 2 and col != 'Diab

plt.figure(figsize=(15, 10))
for i, col in enumerate(binary_cols, 1):
    plt.subplot(4, 5, i)
    sns.countplot(x=col, data=df)
    plt.title(col)
    plt.xlabel('')
```

```
    plt.ylabel('')
    plt.tight_layout()

plt.suptitle("Binary Feature Distributions", y=1.02, fontsize=16)
plt.show()
```



1. HighBP & HighChol: A higher proportion of individuals reported high blood pressure and cholesterol, both major risk factors for diabetes.
2. CholCheck: Nearly all participants reported having had their cholesterol checked, indicating good preventive healthcare access.
3. Smoker: Smoking behavior is relatively balanced, though non-smokers are slightly more prevalent.
4. Stroke and HeartDiseaseorAttack: Both are rare events in the dataset. The majority did not report experiencing these conditions.
5. PhysActivity: A majority of the population is engaging in some form of physical activity, which reflects positively on general lifestyle practices.
6. Fruits & Veggies: Most respondents report consuming fruits and vegetables regularly, with vegetable consumption being particularly high.
7. HvyAlcoholConsump: Very few individuals reported heavy alcohol consumption, which is a positive lifestyle indicator in this context.
8. AnyHealthcare: Nearly the entire population has access to healthcare, with very few lacking it.
9. NoDocbcCost: Most participants did not report skipping doctor visits due to cost which again reflets a good access to care.
10. DiffWalk: Most individuals do not report difficulty walking, suggesting relatively good mobility across the population.
11. Sex: The dataset has a very slightly uneven distribution between the two gender categories, with a slight male majority

Data Preprocessing

To prepare the data for modelling

- the target variable (Diabetes_binary) was separated from the features
- Numerical features like Age, BMI, GenHlth, MentHlth, PhysHlth, Education, and Income were identified for scaling which were then standardized using StandardScaler for ensuring consistent ranges across features.
- Binary and categorical features were left unchanged using 'passthrough' in the pipeline, to make them preserve their interpretability and logical structure
- the data was then split into training and testing sets using an 80:20 split, with startification on the target variable to preserve the class imbalance.

In [7]:
```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# Define features and target
X = df.drop('Diabetes_binary', axis=1)
y = df['Diabetes_binary']

# Reclassifying numeric and binary columns
numerical_cols = ['Age', 'BMI', 'GenHlth', 'MentHlth', 'PhysHlth', 'Education', 'In
categorical_cols = [col for col in X.columns if col not in numerical_cols]

# Defining the preprocessing pipeline
preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numerical_cols),
        ('cat', 'passthrough', categorical_cols)
    ]
)

# Spliting into train/test (80/20), stratified to preserve class distribution
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)

# Fitting transformer on training data, apply to train and test
X_train_processed = preprocessor.fit_transform(X_train)
X_test_processed = preprocessor.transform(X_test)
```

In [8]:
```
# Show shape and sample of processed data
print("Processed train shape:", X_train_processed.shape)
print("Processed test shape:", X_test_processed.shape)
print("First 3 samples (scaled + binary passthrough):")
print(X_train_processed[:3])
```

```
Processed train shape: (33932, 21)
Processed test shape: (8483, 21)
First 3 samples (scaled + binary passthrough):
[[ 0.48982465 -1.38947861 -1.65060949 -0.45865786 -0.58006365  1.04700799
   1.0599396   0.          1.          1.          0.          0.
   0.          1.          0.          1.          0.          1.
   0.          0.          0.        ]
 [ 0.48982465  0.87278271  1.0348758  -0.45865786 -0.48096429  1.04700799
  -1.69163798  1.          1.          1.          1.          0.
   0.          1.          0.          1.          0.          1.
   0.          0.          1.        ]
 [ 0.48982465 -0.25834795  0.13971404 -0.21256858 -0.08456682 -0.88691189
   0.14274707  1.          1.          1.          1.          0.
   0.          1.          1.          1.          0.          1.
   0.          0.          1.        ]]
```

After the transformation, both training and test datatsets had 21 feature each (7 scaled numerical values and 14 binary features) and were ready to be fed into the models

Correlation matrix

To further explore the relationship between the features and diabetes, a correlation matrix was computed using all numeric features with binary target variable Diabetes_binary.

In [9]:
```
# Computing correlation matrix (includes target)
correlation_matrix = df.corr(numeric_only=True)

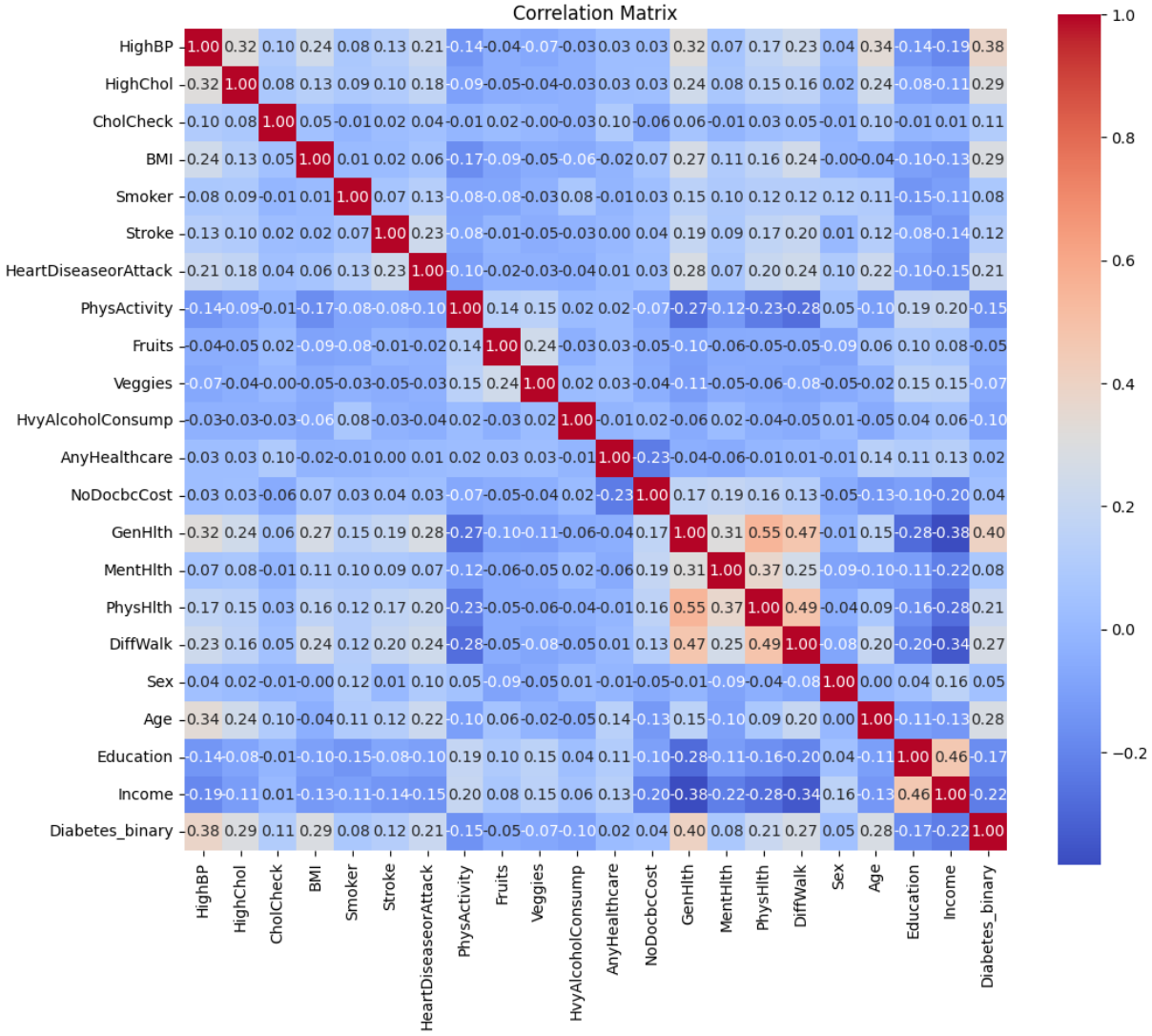
# Displaying top correlations with the target variable
target_corr = correlation_matrix['Diabetes_binary'].sort_values(ascending=False)
print(target_corr.head(6))

print("\nTop features negatively correlated with Diabetes_binary:")
print(target_corr.tail(5))
```

```
Diabetes_binary    1.000000
GenHlth            0.403320
HighBP             0.384567
HighChol           0.293386
BMI                0.288792
Age                0.276682
Name: Diabetes_binary, dtype: float64

Top features negatively correlated with Diabetes_binary:
Veggies             -0.074976
HvyAlcoholConsump   -0.096577
PhysActivity        -0.154211
Education           -0.169359
Income              -0.223148
Name: Diabetes_binary, dtype: float64
```

In [10]:
```
plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, fmt=".2f", cmap='coolwarm', square=True
plt.title("Correlation Matrix")
plt.show()
```



The results from the correlation matrix are

- Positive correlation

Feature	Correlation
GenHlth	0.403
HighBP	0.385
HighChol	0.293
BMI	0.289
Age	0.277

These findings say what we already know from known medical assosciations like individuals with higher BMI, old age, poor general health amd conditions like blood pressure and cholestrol are more likely to have diabetes.

- Negative correlation

Feature	Correlation
Income	-0.223
Education	-0.169
PhysActivity	-0.154
HvyAlcoholConsump	-0.097
Veggies	-0.075

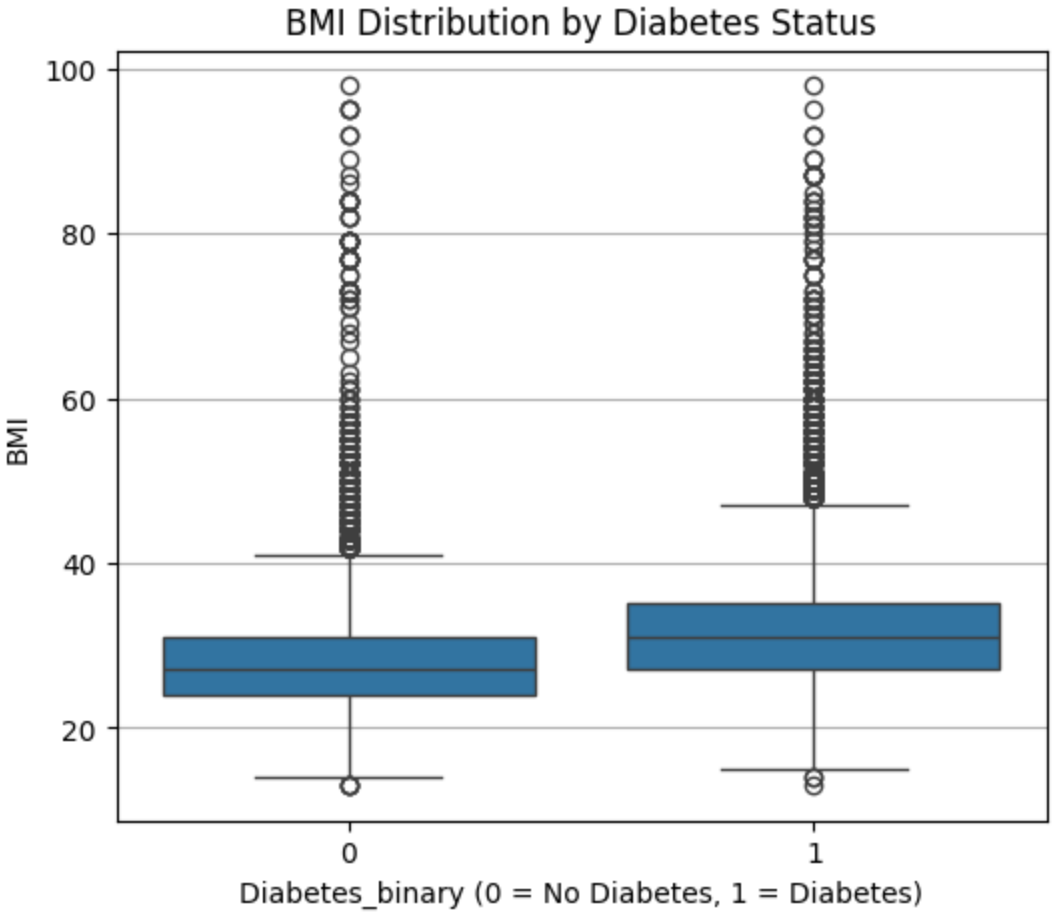
This suggests that higer income and education levels along with physical activities and nutritional conscious lifestyle are associated with lower diabetes prevalence.

The heatmap of the correlation matrix visually highlights these relationships which shows the strength and direction of feature interaction.

Relation between BMI and diabetes

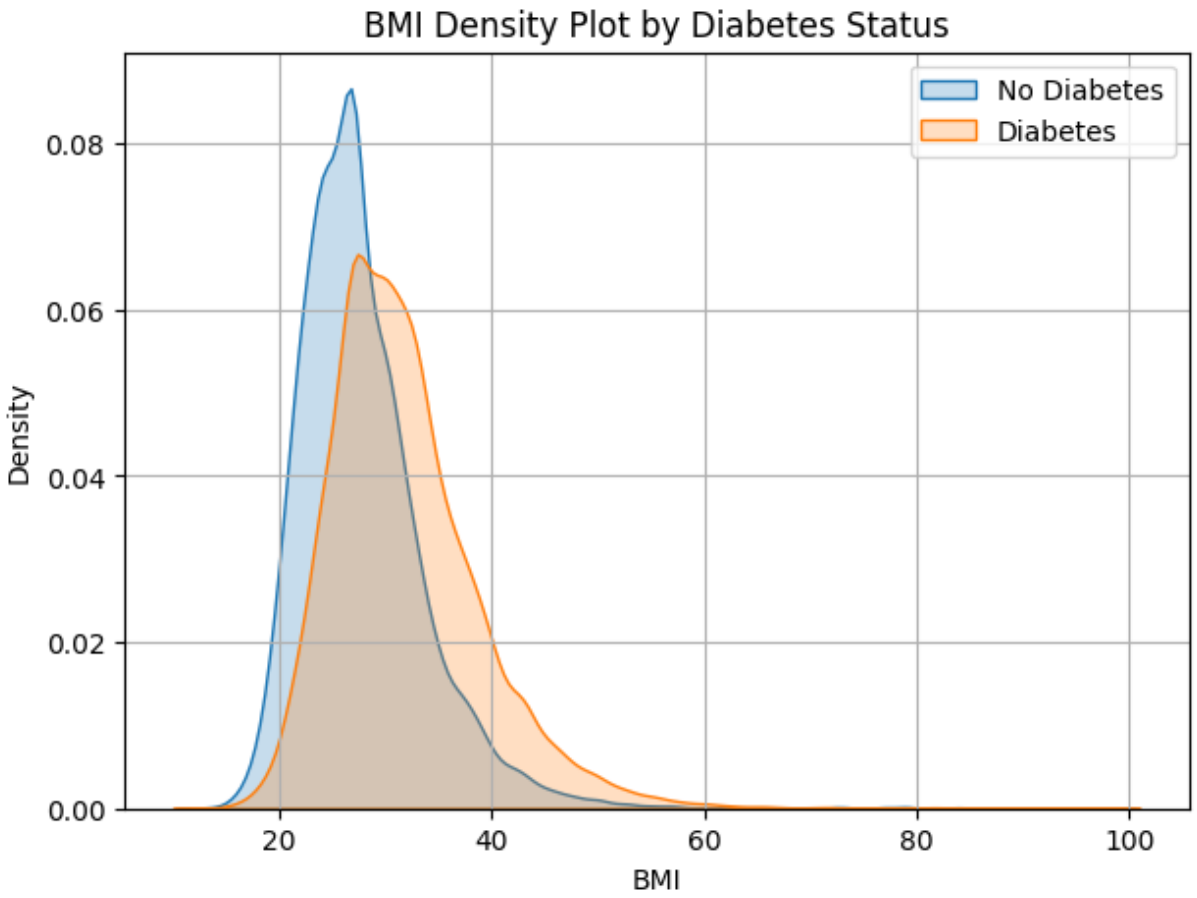
To understand the relation between target variable and one of the features we took BMI as the feature and looked into it

In [11]:
```
plt.figure(figsize=(6, 5))
sns.boxplot(x='Diabetes_binary', y='BMI', data=df)
plt.title("BMI Distribution by Diabetes Status")
plt.xlabel("Diabetes_binary (0 = No Diabetes, 1 = Diabetes)")
plt.ylabel("BMI")
plt.grid(True, axis='y')
plt.show()
```



This boxplot comparing BMI distributions across diabetic and non diabetic groups showed that people with diabetes have a notably higher median BMI than those without.

In [12]:
```
plt.figure(figsize=(7, 5))
sns.kdeplot(data=df[df['Diabetes_binary'] == 0], x='BMI', label='No Diabetes', fill
sns.kdeplot(data=df[df['Diabetes_binary'] == 1], x='BMI', label='Diabetes', fill=Tr
plt.title("BMI Density Plot by Diabetes Status")
plt.xlabel("BMI")
plt.ylabel("Density")
plt.legend()
plt.grid(True)
plt.show()
```



Also, the KDE(density) plot is showing a clear right shift in BMI distribution for diabetic individuals. While there is a slight overlap between the curves, the diabetic group is concentrated more in the obese range showing a meaningful relation between BMI and diabetes.

In [13]:
```
mean_bmi = df.groupby('Diabetes_binary')['BMI'].mean()
print("Mean BMI by Diabetes Status:")
print(mean_bmi)
```

```
Mean BMI by Diabetes Status:
Diabetes_binary
0    27.799025
1    31.909479
Name: BMI, dtype: float64
```

The mean BMI values also confirm this

- diabetic : 31.9
- non diabetic : 27.8

the roughly 4 average increase supports the strong relation between higher BMI in diabetic population. Also the broader BMI spread in the diabetic population suggests that while BMI is significant predictive factor it alone dosen't determine diabetes.

Section 3: Model exploration to determine network topology (8 required)

Before the model training, the dataset must be split into training and testing subsets as this ensures the model's performance can be evaluated on data it hasn't seen during training. Also, checking class imbalance is crucial for binary classification problems like diabetes prediction as an imbalanced dataset can bias the mojarity class.

In [14]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Input
from tensorflow.keras.callbacks import EarlyStopping
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
import random

seed = 1
np.random.seed(seed)
random.seed(seed)
tf.random.set_seed(seed)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.33, random_state=seed, stratify=y, shuffle=True
)

# Displaying class balance
print("Training dataset length:", len(X_train))
print("Testing dataset length:", len(X_test))
print("\n% of '1' in Training:", round(np.mean(y_train) * 100, 2), "%")
print("% of '1' in Test    :", round(np.mean(y_test) * 100, 2), "%")
```

```
Training dataset length: 28418
Testing dataset length: 13997

% of '1' in Training: 49.72 %
% of '1' in Test    : 49.72 %
```

The dataset is spllit into training(67 %) and testing (33 %) sets.

I used stratify = y to preserve the original ratio of diabetic and non-diabetic cases across both subsets. Without any startification, the model could recieeve a skewed distribution in the training or testing set.

The training set contains 28,418 samples and the test set contatins 13,997 samples and the class distribution in both thes sets is roughly 49.72 % for the positive class(1/diabetic), which is also almost close to a perfect 50-50 balance. This helps the model learn both classes fairly and makes metrics like accuracy more meaningful.++

3.1 Main Objective

The goal of this section is to experiment with a variety of neural network topologies and identify an architecture that can generalize well for the diabetes classification task. We test diffrerent combinations of hidden layers (both shallow and deep) to examine their effects on the model performance which ultimately selects the best performing modelbased on validation accuracy and other key metrics.

We explored eight different topologies (M1-M8) with incresing depth and width to observe underfitting, optimal capacirty and overfitting. These range from a simple 1 layer model(M1) to a deep 5 layer model(7)

3.2 Model Configuration

layer_configs = [

```
    [4],                    # M1 – Smallest network

    [8, 4],                 # M2 – Slightly deeper

    [16, 8, 4],             # M3

    [32, 16, 8],            # M4 – Best model

    [64, 32, 16, 8],        # M5

    [128, 64, 32, 16],      # M6

    [256, 128, 64, 32, 16], # M7 – Deepest network

    [32, 32, 32]            # M8 – Wide and balanced
```

]

These configuration allows us to :

1. Start from a minimal baseline(m1) to understand how a small model behaves.
2. Gradually increase depth to explore the effect of network complexity.
3. Introduce wider architecture(m8) to test horizontal capacity over depth.
4. Observe signs of underfitting, good generalization or overfitting

3.3 Training strategy

Each model was trained using:

```
    Loss function: binary_crossentropy

    Optimizer: adam
```

```
    Early stopping (patience=10)

    Max epochs: 80

    Train/validation split: 80/20
```

The goal is to balance performance training stability while avoiding unnecessary overfitting. Also the training history being tracked for all models.

In [15]:
```
# Defining 8 model configurations to explore
layer_configs = [
    [4],                      # M1 — Minimal network
    [8, 4],                   # M2
    [16, 8, 4],               # M3
    [32, 16, 8],              # M4 — Moderate depth
    [64, 32, 16, 8],          # M5
    [128, 64, 32, 16],        # M6
    [256, 128, 64, 32, 16],   # M7 — Deep
    [32, 32, 32]              # M8 — Wide and balanced
]

results = []
best_val_accuracy = 0
best_config = None

for i, config in enumerate(layer_configs, start=1):
    print(f"\n Training Model M{i} with layers: {config}")

    # Building the model
    model = Sequential()
    model.add(Input(shape=(X_train.shape[1],)))
    for units in config:
        model.add(Dense(units, activation='relu'))
    model.add(Dense(1, activation='sigmoid'))  # Binary classification

    model.compile(
        optimizer='adam',
        loss='binary_crossentropy',
        metrics=['accuracy']
    )

    # Using Early stopping for avoiding overfitting
    early_stop = EarlyStopping(monitor='val_accuracy', patience=10, restore_best_we

    history = model.fit(
        X_train, y_train,
        validation_data=(X_test, y_test),
        epochs=80,
        batch_size=32,
        callbacks=[early_stop],
        verbose=1
    )
```

```
    # Evaluating best validation accuracy
    max_val_acc = max(history.history['val_accuracy'])
    print(f" Max Validation Accuracy for Model M{i}: {max_val_acc:.4f}")

    # Predict on test data
    y_pred_prob = model.predict(X_test)
    y_pred = (y_pred_prob > 0.5).astype(int)

    # Metrics
    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred)
    rec = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    results.append({
        'Model': f'M{i}',
        'Layers': str(config),
        'Val_Accuracy': max_val_acc,
        'Test_Accuracy': acc,
        'Precision': prec,
        'Recall': rec,
        'F1_Score': f1
    })

    # Plot training curves
    plt.figure(figsize=(12, 4))

    # Loss
    plt.subplot(1, 2, 1)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')
    plt.title(f'Model M{i} Loss')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
    plt.legend()

    # Accuracy
    plt.subplot(1, 2, 2)
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.title(f'Model M{i} Accuracy')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()

    plt.suptitle(f"Training Curves for Model M{i} ({config})")
    plt.tight_layout()
    plt.show()

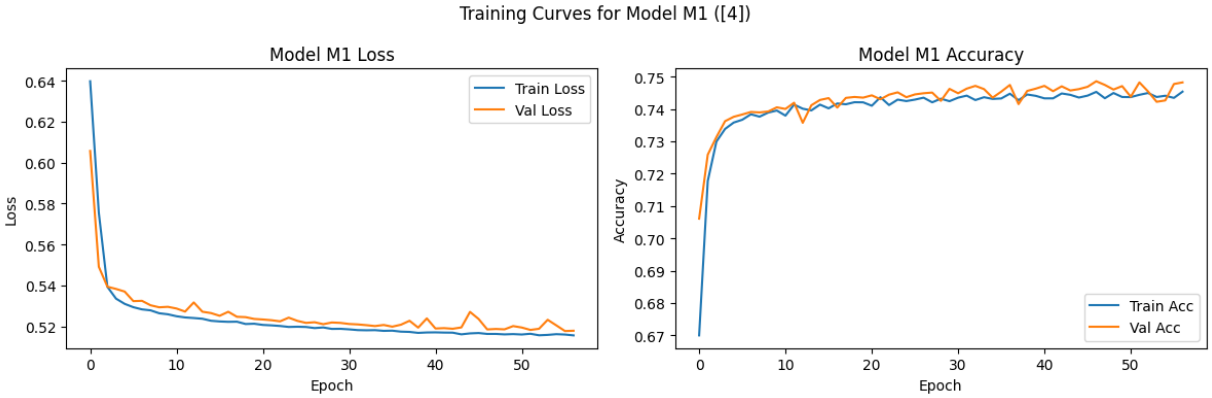
    # Track best model
    if max_val_acc > best_val_accuracy:
        best_val_accuracy = max_val_acc
        best_config = config
```

```
 Training Model M1 with layers: [4]
Epoch 1/80
889/889 [==============================] - 1s 894us/step - loss: 0.6397 - accuracy:
0.6700 - val_loss: 0.6057 - val_accuracy: 0.7061
Epoch 2/80
889/889 [==============================] - 1s 765us/step - loss: 0.5755 - accuracy:
0.7177 - val_loss: 0.5492 - val_accuracy: 0.7259
Epoch 3/80
889/889 [==============================] - 1s 755us/step - loss: 0.5393 - accuracy:
0.7299 - val_loss: 0.5394 - val_accuracy: 0.7313
Epoch 4/80
889/889 [==============================] - 1s 775us/step - loss: 0.5336 - accuracy:
0.7338 - val_loss: 0.5383 - val_accuracy: 0.7362
Epoch 5/80
889/889 [==============================] - 1s 770us/step - loss: 0.5311 - accuracy:
0.7358 - val_loss: 0.5371 - val_accuracy: 0.7376
Epoch 6/80
889/889 [==============================] - 1s 748us/step - loss: 0.5294 - accuracy:
0.7367 - val_loss: 0.5324 - val_accuracy: 0.7383
Epoch 7/80
889/889 [==============================] - 1s 766us/step - loss: 0.5284 - accuracy:
0.7384 - val_loss: 0.5325 - val_accuracy: 0.7391
Epoch 8/80
889/889 [==============================] - 1s 766us/step - loss: 0.5279 - accuracy:
0.7376 - val_loss: 0.5304 - val_accuracy: 0.7389
Epoch 9/80
889/889 [==============================] - 1s 849us/step - loss: 0.5265 - accuracy:
0.7389 - val_loss: 0.5294 - val_accuracy: 0.7392
Epoch 10/80
889/889 [==============================] - 1s 809us/step - loss: 0.5260 - accuracy:
0.7395 - val_loss: 0.5296 - val_accuracy: 0.7405
Epoch 11/80
889/889 [==============================] - 1s 775us/step - loss: 0.5250 - accuracy:
0.7379 - val_loss: 0.5288 - val_accuracy: 0.7400
Epoch 12/80
889/889 [==============================] - 1s 776us/step - loss: 0.5244 - accuracy:
0.7414 - val_loss: 0.5273 - val_accuracy: 0.7419
Epoch 13/80
889/889 [==============================] - 1s 765us/step - loss: 0.5241 - accuracy:
0.7401 - val_loss: 0.5318 - val_accuracy: 0.7357
Epoch 14/80
889/889 [==============================] - 1s 788us/step - loss: 0.5238 - accuracy:
0.7396 - val_loss: 0.5272 - val_accuracy: 0.7412
Epoch 15/80
889/889 [==============================] - 1s 776us/step - loss: 0.5228 - accuracy:
0.7414 - val_loss: 0.5266 - val_accuracy: 0.7428
Epoch 16/80
889/889 [==============================] - 1s 795us/step - loss: 0.5225 - accuracy:
0.7402 - val_loss: 0.5252 - val_accuracy: 0.7434
Epoch 17/80
889/889 [==============================] - 1s 796us/step - loss: 0.5223 - accuracy:
0.7417 - val_loss: 0.5272 - val_accuracy: 0.7404
Epoch 18/80
889/889 [==============================] - 1s 787us/step - loss: 0.5224 - accuracy:
0.7415 - val_loss: 0.5248 - val_accuracy: 0.7434
Epoch 19/80
```

```
889/889 [==============================] - 1s 778us/step - loss: 0.5212 - accuracy:
0.7421 - val_loss: 0.5246 - val_accuracy: 0.7437
Epoch 20/80
889/889 [==============================] - 1s 798us/step - loss: 0.5214 - accuracy:
0.7421 - val_loss: 0.5237 - val_accuracy: 0.7435
Epoch 21/80
889/889 [==============================] - 1s 784us/step - loss: 0.5208 - accuracy:
0.7410 - val_loss: 0.5234 - val_accuracy: 0.7442
Epoch 22/80
889/889 [==============================] - 1s 786us/step - loss: 0.5206 - accuracy:
0.7436 - val_loss: 0.5231 - val_accuracy: 0.7430
Epoch 23/80
889/889 [==============================] - 1s 764us/step - loss: 0.5203 - accuracy:
0.7412 - val_loss: 0.5225 - val_accuracy: 0.7444
Epoch 24/80
889/889 [==============================] - 1s 789us/step - loss: 0.5198 - accuracy:
0.7429 - val_loss: 0.5243 - val_accuracy: 0.7452
Epoch 25/80
889/889 [==============================] - 1s 789us/step - loss: 0.5199 - accuracy:
0.7425 - val_loss: 0.5228 - val_accuracy: 0.7437
Epoch 26/80
889/889 [==============================] - 1s 772us/step - loss: 0.5198 - accuracy:
0.7429 - val_loss: 0.5218 - val_accuracy: 0.7445
Epoch 27/80
889/889 [==============================] - 1s 778us/step - loss: 0.5192 - accuracy:
0.7435 - val_loss: 0.5221 - val_accuracy: 0.7449
Epoch 28/80
889/889 [==============================] - 1s 779us/step - loss: 0.5195 - accuracy:
0.7421 - val_loss: 0.5211 - val_accuracy: 0.7451
Epoch 29/80
889/889 [==============================] - 1s 788us/step - loss: 0.5188 - accuracy:
0.7432 - val_loss: 0.5220 - val_accuracy: 0.7425
Epoch 30/80
889/889 [==============================] - 1s 767us/step - loss: 0.5189 - accuracy:
0.7424 - val_loss: 0.5218 - val_accuracy: 0.7462
Epoch 31/80
889/889 [==============================] - 1s 763us/step - loss: 0.5186 - accuracy:
0.7435 - val_loss: 0.5212 - val_accuracy: 0.7449
Epoch 32/80
889/889 [==============================] - 1s 758us/step - loss: 0.5183 - accuracy:
0.7441 - val_loss: 0.5210 - val_accuracy: 0.7463
Epoch 33/80
889/889 [==============================] - 1s 801us/step - loss: 0.5182 - accuracy:
0.7428 - val_loss: 0.5207 - val_accuracy: 0.7472
Epoch 34/80
889/889 [==============================] - 1s 763us/step - loss: 0.5182 - accuracy:
0.7436 - val_loss: 0.5202 - val_accuracy: 0.7462
Epoch 35/80
889/889 [==============================] - 1s 754us/step - loss: 0.5179 - accuracy:
0.7431 - val_loss: 0.5208 - val_accuracy: 0.7435
Epoch 36/80
889/889 [==============================] - 1s 770us/step - loss: 0.5180 - accuracy:
0.7433 - val_loss: 0.5199 - val_accuracy: 0.7454
Epoch 37/80
889/889 [==============================] - 1s 765us/step - loss: 0.5175 - accuracy:
0.7447 - val_loss: 0.5209 - val_accuracy: 0.7474
```

```
Epoch 38/80
889/889 [==============================] - 1s 766us/step - loss: 0.5174 - accuracy:
0.7427 - val_loss: 0.5228 - val_accuracy: 0.7415
Epoch 39/80
889/889 [==============================] - 1s 751us/step - loss: 0.5169 - accuracy:
0.7445 - val_loss: 0.5195 - val_accuracy: 0.7456
Epoch 40/80
889/889 [==============================] - 1s 762us/step - loss: 0.5171 - accuracy:
0.7441 - val_loss: 0.5240 - val_accuracy: 0.7463
Epoch 41/80
889/889 [==============================] - 1s 765us/step - loss: 0.5171 - accuracy:
0.7433 - val_loss: 0.5190 - val_accuracy: 0.7472
Epoch 42/80
889/889 [==============================] - 1s 767us/step - loss: 0.5170 - accuracy:
0.7433 - val_loss: 0.5192 - val_accuracy: 0.7455
Epoch 43/80
889/889 [==============================] - 1s 788us/step - loss: 0.5170 - accuracy:
0.7448 - val_loss: 0.5189 - val_accuracy: 0.7470
Epoch 44/80
889/889 [==============================] - 1s 796us/step - loss: 0.5162 - accuracy:
0.7444 - val_loss: 0.5196 - val_accuracy: 0.7457
Epoch 45/80
889/889 [==============================] - 1s 783us/step - loss: 0.5166 - accuracy:
0.7435 - val_loss: 0.5271 - val_accuracy: 0.7462
Epoch 46/80
889/889 [==============================] - 1s 768us/step - loss: 0.5168 - accuracy:
0.7441 - val_loss: 0.5237 - val_accuracy: 0.7469
Epoch 47/80
889/889 [==============================] - 1s 784us/step - loss: 0.5164 - accuracy:
0.7453 - val_loss: 0.5186 - val_accuracy: 0.7486
Epoch 48/80
889/889 [==============================] - 1s 763us/step - loss: 0.5164 - accuracy:
0.7433 - val_loss: 0.5188 - val_accuracy: 0.7474
Epoch 49/80
889/889 [==============================] - 1s 766us/step - loss: 0.5162 - accuracy:
0.7450 - val_loss: 0.5186 - val_accuracy: 0.7460
Epoch 50/80
889/889 [==============================] - 1s 782us/step - loss: 0.5163 - accuracy:
0.7437 - val_loss: 0.5202 - val_accuracy: 0.7471
Epoch 51/80
889/889 [==============================] - 1s 761us/step - loss: 0.5161 - accuracy:
0.7437 - val_loss: 0.5195 - val_accuracy: 0.7438
Epoch 52/80
889/889 [==============================] - 1s 769us/step - loss: 0.5165 - accuracy:
0.7444 - val_loss: 0.5183 - val_accuracy: 0.7482
Epoch 53/80
889/889 [==============================] - 1s 770us/step - loss: 0.5157 - accuracy:
0.7450 - val_loss: 0.5189 - val_accuracy: 0.7454
Epoch 54/80
889/889 [==============================] - 1s 785us/step - loss: 0.5159 - accuracy:
0.7437 - val_loss: 0.5233 - val_accuracy: 0.7422
Epoch 55/80
889/889 [==============================] - 1s 755us/step - loss: 0.5163 - accuracy:
0.7441 - val_loss: 0.5206 - val_accuracy: 0.7427
Epoch 56/80
889/889 [==============================] - 1s 785us/step - loss: 0.5161 - accuracy:
```

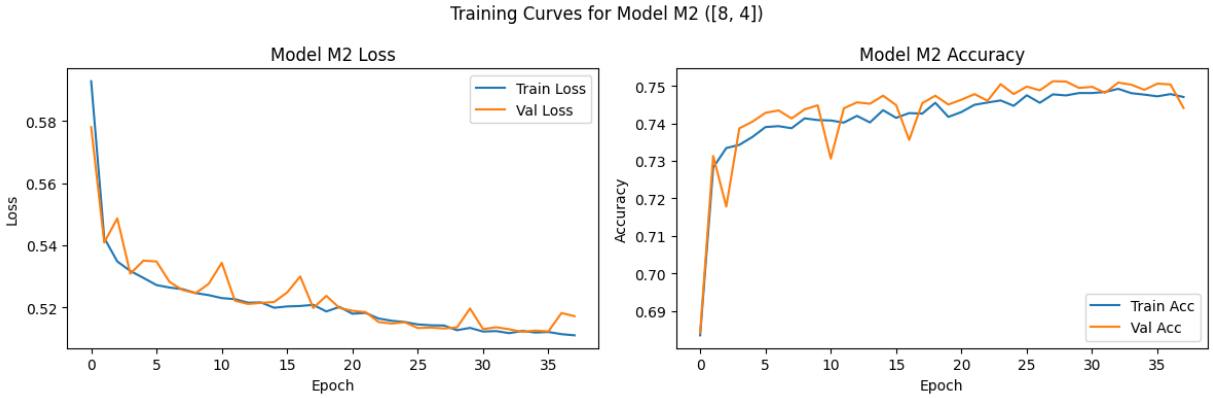
```
0.7434 - val_loss: 0.5178 - val_accuracy: 0.7477
Epoch 57/80
889/889 [==============================] - 1s 765us/step - loss: 0.5157 - accuracy:
0.7454 - val_loss: 0.5179 - val_accuracy: 0.7482
 Max Validation Accuracy for Model M1: 0.7486
438/438 [==============================] - 0s 398us/step
```



```
 Training Model M2 with layers: [8, 4]
Epoch 1/80
889/889 [==============================] - 1s 894us/step - loss: 0.5929 - accuracy:
0.6834 - val_loss: 0.5781 - val_accuracy: 0.6841
Epoch 2/80
889/889 [==============================] - 1s 812us/step - loss: 0.5423 - accuracy:
0.7282 - val_loss: 0.5409 - val_accuracy: 0.7313
Epoch 3/80
889/889 [==============================] - 1s 796us/step - loss: 0.5348 - accuracy:
0.7334 - val_loss: 0.5487 - val_accuracy: 0.7178
Epoch 4/80
889/889 [==============================] - 1s 802us/step - loss: 0.5317 - accuracy:
0.7343 - val_loss: 0.5308 - val_accuracy: 0.7387
Epoch 5/80
889/889 [==============================] - 1s 797us/step - loss: 0.5294 - accuracy:
0.7364 - val_loss: 0.5350 - val_accuracy: 0.7404
Epoch 6/80
889/889 [==============================] - 1s 784us/step - loss: 0.5271 - accuracy:
0.7390 - val_loss: 0.5347 - val_accuracy: 0.7428
Epoch 7/80
889/889 [==============================] - 1s 811us/step - loss: 0.5263 - accuracy:
0.7392 - val_loss: 0.5281 - val_accuracy: 0.7434
Epoch 8/80
889/889 [==============================] - 1s 799us/step - loss: 0.5258 - accuracy:
0.7387 - val_loss: 0.5256 - val_accuracy: 0.7413
Epoch 9/80
889/889 [==============================] - 1s 796us/step - loss: 0.5246 - accuracy:
0.7413 - val_loss: 0.5245 - val_accuracy: 0.7437
Epoch 10/80
889/889 [==============================] - 1s 787us/step - loss: 0.5239 - accuracy:
0.7409 - val_loss: 0.5276 - val_accuracy: 0.7448
Epoch 11/80
889/889 [==============================] - 1s 795us/step - loss: 0.5229 - accuracy:
0.7408 - val_loss: 0.5343 - val_accuracy: 0.7306
Epoch 12/80
889/889 [==============================] - 1s 791us/step - loss: 0.5226 - accuracy:
0.7402 - val_loss: 0.5221 - val_accuracy: 0.7440
Epoch 13/80
889/889 [==============================] - 1s 801us/step - loss: 0.5215 - accuracy:
0.7420 - val_loss: 0.5210 - val_accuracy: 0.7456
Epoch 14/80
889/889 [==============================] - 1s 801us/step - loss: 0.5215 - accuracy:
0.7402 - val_loss: 0.5214 - val_accuracy: 0.7452
Epoch 15/80
889/889 [==============================] - 1s 792us/step - loss: 0.5198 - accuracy:
0.7435 - val_loss: 0.5217 - val_accuracy: 0.7474
Epoch 16/80
889/889 [==============================] - 1s 790us/step - loss: 0.5202 - accuracy:
0.7415 - val_loss: 0.5247 - val_accuracy: 0.7449
Epoch 17/80
889/889 [==============================] - 1s 796us/step - loss: 0.5204 - accuracy:
0.7427 - val_loss: 0.5299 - val_accuracy: 0.7356
Epoch 18/80
889/889 [==============================] - 1s 797us/step - loss: 0.5207 - accuracy:
0.7426 - val_loss: 0.5198 - val_accuracy: 0.7454
Epoch 19/80
```

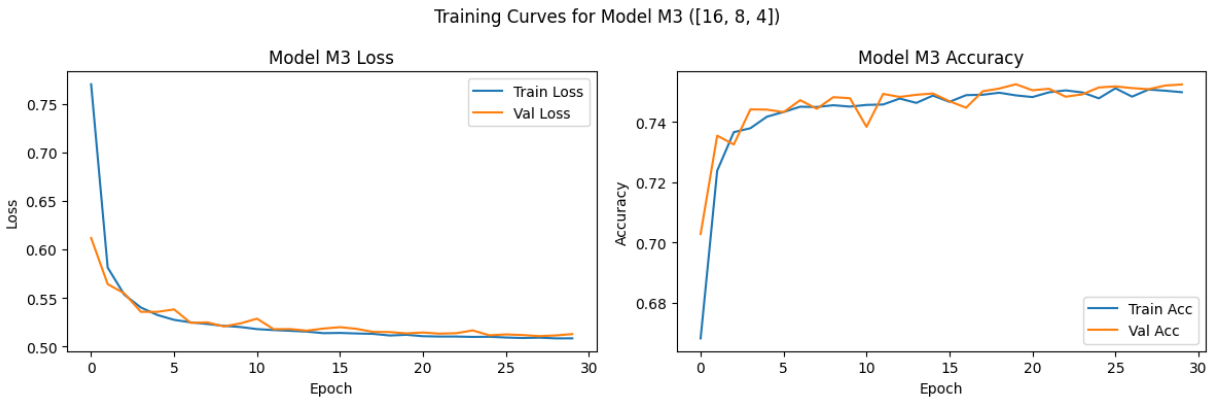
```
889/889 [==============================] - 1s 801us/step - loss: 0.5186 - accuracy:
0.7455 - val_loss: 0.5237 - val_accuracy: 0.7474
Epoch 20/80
889/889 [==============================] - 1s 821us/step - loss: 0.5201 - accuracy:
0.7417 - val_loss: 0.5198 - val_accuracy: 0.7450
Epoch 21/80
889/889 [==============================] - 1s 797us/step - loss: 0.5179 - accuracy:
0.7430 - val_loss: 0.5188 - val_accuracy: 0.7463
Epoch 22/80
889/889 [==============================] - 1s 814us/step - loss: 0.5182 - accuracy:
0.7450 - val_loss: 0.5184 - val_accuracy: 0.7478
Epoch 23/80
889/889 [==============================] - 1s 784us/step - loss: 0.5163 - accuracy:
0.7455 - val_loss: 0.5152 - val_accuracy: 0.7460
Epoch 24/80
889/889 [==============================] - 1s 812us/step - loss: 0.5156 - accuracy:
0.7461 - val_loss: 0.5147 - val_accuracy: 0.7504
Epoch 25/80
889/889 [==============================] - 1s 791us/step - loss: 0.5152 - accuracy:
0.7447 - val_loss: 0.5151 - val_accuracy: 0.7478
Epoch 26/80
889/889 [==============================] - 1s 803us/step - loss: 0.5144 - accuracy:
0.7475 - val_loss: 0.5133 - val_accuracy: 0.7498
Epoch 27/80
889/889 [==============================] - 1s 797us/step - loss: 0.5142 - accuracy:
0.7455 - val_loss: 0.5134 - val_accuracy: 0.7488
Epoch 28/80
889/889 [==============================] - 1s 800us/step - loss: 0.5141 - accuracy:
0.7477 - val_loss: 0.5131 - val_accuracy: 0.7512
Epoch 29/80
889/889 [==============================] - 1s 800us/step - loss: 0.5126 - accuracy:
0.7474 - val_loss: 0.5135 - val_accuracy: 0.7512
Epoch 30/80
889/889 [==============================] - 1s 803us/step - loss: 0.5133 - accuracy:
0.7481 - val_loss: 0.5196 - val_accuracy: 0.7494
Epoch 31/80
889/889 [==============================] - 1s 799us/step - loss: 0.5121 - accuracy:
0.7481 - val_loss: 0.5129 - val_accuracy: 0.7497
Epoch 32/80
889/889 [==============================] - 1s 799us/step - loss: 0.5123 - accuracy:
0.7484 - val_loss: 0.5135 - val_accuracy: 0.7482
Epoch 33/80
889/889 [==============================] - 1s 797us/step - loss: 0.5116 - accuracy:
0.7492 - val_loss: 0.5129 - val_accuracy: 0.7509
Epoch 34/80
889/889 [==============================] - 1s 799us/step - loss: 0.5123 - accuracy:
0.7480 - val_loss: 0.5120 - val_accuracy: 0.7503
Epoch 35/80
889/889 [==============================] - 1s 805us/step - loss: 0.5118 - accuracy:
0.7477 - val_loss: 0.5125 - val_accuracy: 0.7489
Epoch 36/80
889/889 [==============================] - 1s 808us/step - loss: 0.5120 - accuracy:
0.7472 - val_loss: 0.5122 - val_accuracy: 0.7506
Epoch 37/80
889/889 [==============================] - 1s 803us/step - loss: 0.5113 - accuracy:
0.7478 - val_loss: 0.5181 - val_accuracy: 0.7504
```

```
Epoch 38/80
889/889 [==============================] - 1s 817us/step - loss: 0.5109 - accuracy:
0.7470 - val_loss: 0.5171 - val_accuracy: 0.7441
 Max Validation Accuracy for Model M2: 0.7512
438/438 [==============================] - 0s 442us/step
```



```
 Training Model M3 with layers: [16, 8, 4]
Epoch 1/80
889/889 [==============================] - 1s 925us/step - loss: 0.7701 - accuracy:
0.6682 - val_loss: 0.6117 - val_accuracy: 0.7029
Epoch 2/80
889/889 [==============================] - 1s 844us/step - loss: 0.5812 - accuracy:
0.7239 - val_loss: 0.5642 - val_accuracy: 0.7355
Epoch 3/80
889/889 [==============================] - 1s 828us/step - loss: 0.5533 - accuracy:
0.7367 - val_loss: 0.5546 - val_accuracy: 0.7326
Epoch 4/80
889/889 [==============================] - 1s 824us/step - loss: 0.5400 - accuracy:
0.7380 - val_loss: 0.5356 - val_accuracy: 0.7442
Epoch 5/80
889/889 [==============================] - 1s 842us/step - loss: 0.5323 - accuracy:
0.7418 - val_loss: 0.5356 - val_accuracy: 0.7442
Epoch 6/80
889/889 [==============================] - 1s 839us/step - loss: 0.5273 - accuracy:
0.7434 - val_loss: 0.5381 - val_accuracy: 0.7433
Epoch 7/80
889/889 [==============================] - 1s 837us/step - loss: 0.5248 - accuracy:
0.7451 - val_loss: 0.5243 - val_accuracy: 0.7473
Epoch 8/80
889/889 [==============================] - 1s 839us/step - loss: 0.5231 - accuracy:
0.7450 - val_loss: 0.5248 - val_accuracy: 0.7444
Epoch 9/80
889/889 [==============================] - 1s 832us/step - loss: 0.5211 - accuracy:
0.7456 - val_loss: 0.5206 - val_accuracy: 0.7483
Epoch 10/80
889/889 [==============================] - 1s 826us/step - loss: 0.5200 - accuracy:
0.7452 - val_loss: 0.5237 - val_accuracy: 0.7479
Epoch 11/80
889/889 [==============================] - 1s 830us/step - loss: 0.5178 - accuracy:
0.7457 - val_loss: 0.5286 - val_accuracy: 0.7384
Epoch 12/80
889/889 [==============================] - 1s 817us/step - loss: 0.5168 - accuracy:
0.7459 - val_loss: 0.5177 - val_accuracy: 0.7494
Epoch 13/80
889/889 [==============================] - 1s 842us/step - loss: 0.5160 - accuracy:
0.7478 - val_loss: 0.5178 - val_accuracy: 0.7484
Epoch 14/80
889/889 [==============================] - 1s 834us/step - loss: 0.5152 - accuracy:
0.7464 - val_loss: 0.5162 - val_accuracy: 0.7491
Epoch 15/80
889/889 [==============================] - 1s 836us/step - loss: 0.5136 - accuracy:
0.7488 - val_loss: 0.5184 - val_accuracy: 0.7495
Epoch 16/80
889/889 [==============================] - 1s 838us/step - loss: 0.5139 - accuracy:
0.7467 - val_loss: 0.5198 - val_accuracy: 0.7469
Epoch 17/80
889/889 [==============================] - 1s 824us/step - loss: 0.5133 - accuracy:
0.7490 - val_loss: 0.5181 - val_accuracy: 0.7448
Epoch 18/80
889/889 [==============================] - 1s 819us/step - loss: 0.5129 - accuracy:
0.7491 - val_loss: 0.5149 - val_accuracy: 0.7502
Epoch 19/80
```

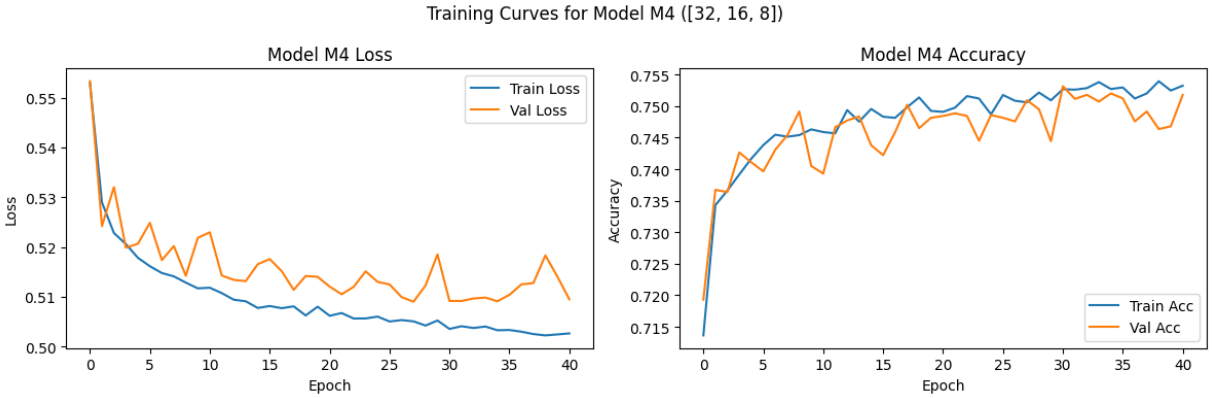
```
889/889 [==============================] - 1s 854us/step - loss: 0.5112 - accuracy:
0.7498 - val_loss: 0.5148 - val_accuracy: 0.7512
Epoch 20/80
889/889 [==============================] - 1s 839us/step - loss: 0.5119 - accuracy:
0.7489 - val_loss: 0.5134 - val_accuracy: 0.7526
Epoch 21/80
889/889 [==============================] - 1s 835us/step - loss: 0.5105 - accuracy:
0.7483 - val_loss: 0.5143 - val_accuracy: 0.7506
Epoch 22/80
889/889 [==============================] - 1s 836us/step - loss: 0.5102 - accuracy:
0.7499 - val_loss: 0.5131 - val_accuracy: 0.7511
Epoch 23/80
889/889 [==============================] - 1s 833us/step - loss: 0.5101 - accuracy:
0.7505 - val_loss: 0.5136 - val_accuracy: 0.7484
Epoch 24/80
889/889 [==============================] - 1s 820us/step - loss: 0.5098 - accuracy:
0.7499 - val_loss: 0.5164 - val_accuracy: 0.7492
Epoch 25/80
889/889 [==============================] - 1s 828us/step - loss: 0.5099 - accuracy:
0.7479 - val_loss: 0.5114 - val_accuracy: 0.7515
Epoch 26/80
889/889 [==============================] - 1s 817us/step - loss: 0.5092 - accuracy:
0.7512 - val_loss: 0.5124 - val_accuracy: 0.7519
Epoch 27/80
889/889 [==============================] - 1s 820us/step - loss: 0.5086 - accuracy:
0.7485 - val_loss: 0.5117 - val_accuracy: 0.7513
Epoch 28/80
889/889 [==============================] - 1s 814us/step - loss: 0.5090 - accuracy:
0.7508 - val_loss: 0.5106 - val_accuracy: 0.7509
Epoch 29/80
889/889 [==============================] - 1s 824us/step - loss: 0.5083 - accuracy:
0.7504 - val_loss: 0.5113 - val_accuracy: 0.7522
Epoch 30/80
889/889 [==============================] - 1s 818us/step - loss: 0.5083 - accuracy:
0.7499 - val_loss: 0.5127 - val_accuracy: 0.7525
 Max Validation Accuracy for Model M3: 0.7526
438/438 [==============================] - 0s 452us/step
```



```
 Training Model M4 with layers: [32, 16, 8]
Epoch 1/80
889/889 [==============================] - 1s 909us/step - loss: 0.5529 - accuracy:
0.7136 - val_loss: 0.5533 - val_accuracy: 0.7193
Epoch 2/80
889/889 [==============================] - 1s 829us/step - loss: 0.5289 - accuracy:
0.7343 - val_loss: 0.5242 - val_accuracy: 0.7367
Epoch 3/80
889/889 [==============================] - 1s 818us/step - loss: 0.5228 - accuracy:
0.7366 - val_loss: 0.5320 - val_accuracy: 0.7364
Epoch 4/80
889/889 [==============================] - 1s 822us/step - loss: 0.5206 - accuracy:
0.7392 - val_loss: 0.5199 - val_accuracy: 0.7427
Epoch 5/80
889/889 [==============================] - 1s 824us/step - loss: 0.5178 - accuracy:
0.7417 - val_loss: 0.5206 - val_accuracy: 0.7411
Epoch 6/80
889/889 [==============================] - 1s 824us/step - loss: 0.5161 - accuracy:
0.7438 - val_loss: 0.5249 - val_accuracy: 0.7397
Epoch 7/80
889/889 [==============================] - 1s 820us/step - loss: 0.5148 - accuracy:
0.7455 - val_loss: 0.5174 - val_accuracy: 0.7431
Epoch 8/80
889/889 [==============================] - 1s 798us/step - loss: 0.5141 - accuracy:
0.7452 - val_loss: 0.5202 - val_accuracy: 0.7454
Epoch 9/80
889/889 [==============================] - 1s 821us/step - loss: 0.5128 - accuracy:
0.7454 - val_loss: 0.5142 - val_accuracy: 0.7492
Epoch 10/80
889/889 [==============================] - 1s 818us/step - loss: 0.5117 - accuracy:
0.7463 - val_loss: 0.5218 - val_accuracy: 0.7405
Epoch 11/80
889/889 [==============================] - 1s 806us/step - loss: 0.5118 - accuracy:
0.7459 - val_loss: 0.5229 - val_accuracy: 0.7393
Epoch 12/80
889/889 [==============================] - 1s 821us/step - loss: 0.5107 - accuracy:
0.7457 - val_loss: 0.5143 - val_accuracy: 0.7467
Epoch 13/80
889/889 [==============================] - 1s 815us/step - loss: 0.5094 - accuracy:
0.7494 - val_loss: 0.5134 - val_accuracy: 0.7477
Epoch 14/80
889/889 [==============================] - 1s 811us/step - loss: 0.5091 - accuracy:
0.7476 - val_loss: 0.5131 - val_accuracy: 0.7484
Epoch 15/80
889/889 [==============================] - 1s 818us/step - loss: 0.5077 - accuracy:
0.7496 - val_loss: 0.5165 - val_accuracy: 0.7438
Epoch 16/80
889/889 [==============================] - 1s 811us/step - loss: 0.5081 - accuracy:
0.7483 - val_loss: 0.5176 - val_accuracy: 0.7422
Epoch 17/80
889/889 [==============================] - 1s 828us/step - loss: 0.5077 - accuracy:
0.7482 - val_loss: 0.5151 - val_accuracy: 0.7459
Epoch 18/80
889/889 [==============================] - 1s 816us/step - loss: 0.5081 - accuracy:
0.7498 - val_loss: 0.5114 - val_accuracy: 0.7502
Epoch 19/80
```

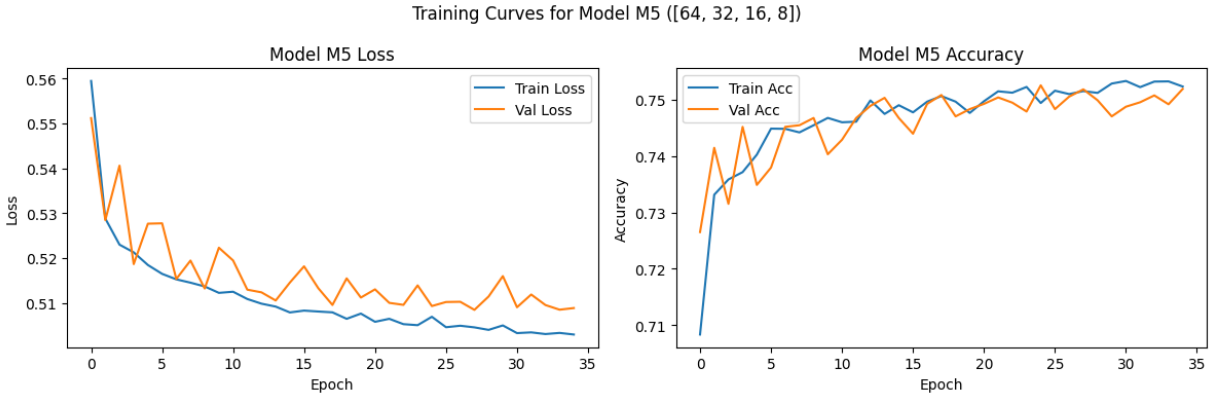
```
889/889 [==============================] - 1s 810us/step - loss: 0.5062 - accuracy:
0.7514 - val_loss: 0.5142 - val_accuracy: 0.7465
Epoch 20/80
889/889 [==============================] - 1s 816us/step - loss: 0.5080 - accuracy:
0.7492 - val_loss: 0.5140 - val_accuracy: 0.7482
Epoch 21/80
889/889 [==============================] - 1s 816us/step - loss: 0.5062 - accuracy:
0.7491 - val_loss: 0.5120 - val_accuracy: 0.7484
Epoch 22/80
889/889 [==============================] - 1s 819us/step - loss: 0.5067 - accuracy:
0.7498 - val_loss: 0.5105 - val_accuracy: 0.7489
Epoch 23/80
889/889 [==============================] - 1s 820us/step - loss: 0.5056 - accuracy:
0.7516 - val_loss: 0.5120 - val_accuracy: 0.7484
Epoch 24/80
889/889 [==============================] - 1s 827us/step - loss: 0.5056 - accuracy:
0.7512 - val_loss: 0.5151 - val_accuracy: 0.7445
Epoch 25/80
889/889 [==============================] - 1s 823us/step - loss: 0.5060 - accuracy:
0.7487 - val_loss: 0.5130 - val_accuracy: 0.7486
Epoch 26/80
889/889 [==============================] - 1s 808us/step - loss: 0.5050 - accuracy:
0.7518 - val_loss: 0.5125 - val_accuracy: 0.7482
Epoch 27/80
889/889 [==============================] - 1s 821us/step - loss: 0.5053 - accuracy:
0.7509 - val_loss: 0.5099 - val_accuracy: 0.7476
Epoch 28/80
889/889 [==============================] - 1s 818us/step - loss: 0.5051 - accuracy:
0.7506 - val_loss: 0.5090 - val_accuracy: 0.7509
Epoch 29/80
889/889 [==============================] - 1s 817us/step - loss: 0.5042 - accuracy:
0.7522 - val_loss: 0.5123 - val_accuracy: 0.7495
Epoch 30/80
889/889 [==============================] - 1s 823us/step - loss: 0.5052 - accuracy:
0.7509 - val_loss: 0.5185 - val_accuracy: 0.7444
Epoch 31/80
889/889 [==============================] - 1s 816us/step - loss: 0.5035 - accuracy:
0.7527 - val_loss: 0.5092 - val_accuracy: 0.7532
Epoch 32/80
889/889 [==============================] - 1s 821us/step - loss: 0.5041 - accuracy:
0.7526 - val_loss: 0.5091 - val_accuracy: 0.7512
Epoch 33/80
889/889 [==============================] - 1s 820us/step - loss: 0.5037 - accuracy:
0.7529 - val_loss: 0.5096 - val_accuracy: 0.7518
Epoch 34/80
889/889 [==============================] - 1s 821us/step - loss: 0.5040 - accuracy:
0.7538 - val_loss: 0.5098 - val_accuracy: 0.7507
Epoch 35/80
889/889 [==============================] - 1s 816us/step - loss: 0.5033 - accuracy:
0.7527 - val_loss: 0.5091 - val_accuracy: 0.7520
Epoch 36/80
889/889 [==============================] - 1s 821us/step - loss: 0.5033 - accuracy:
0.7530 - val_loss: 0.5104 - val_accuracy: 0.7512
Epoch 37/80
889/889 [==============================] - 1s 825us/step - loss: 0.5030 - accuracy:
0.7512 - val_loss: 0.5125 - val_accuracy: 0.7476
```

```
Epoch 38/80
889/889 [==============================] - 1s 820us/step - loss: 0.5025 - accuracy:
0.7520 - val_loss: 0.5127 - val_accuracy: 0.7492
Epoch 39/80
889/889 [==============================] - 1s 811us/step - loss: 0.5022 - accuracy:
0.7540 - val_loss: 0.5183 - val_accuracy: 0.7464
Epoch 40/80
889/889 [==============================] - 1s 812us/step - loss: 0.5024 - accuracy:
0.7525 - val_loss: 0.5141 - val_accuracy: 0.7468
Epoch 41/80
889/889 [==============================] - 1s 843us/step - loss: 0.5026 - accuracy:
0.7533 - val_loss: 0.5095 - val_accuracy: 0.7518
 Max Validation Accuracy for Model M4: 0.7532
438/438 [==============================] - 0s 434us/step
```



```
 Training Model M5 with layers: [64, 32, 16, 8]
Epoch 1/80
889/889 [==============================] - 1s 938us/step - loss: 0.5595 - accuracy:
0.7084 - val_loss: 0.5512 - val_accuracy: 0.7265
Epoch 2/80
889/889 [==============================] - 1s 859us/step - loss: 0.5288 - accuracy:
0.7331 - val_loss: 0.5284 - val_accuracy: 0.7414
Epoch 3/80
889/889 [==============================] - 1s 848us/step - loss: 0.5230 - accuracy:
0.7358 - val_loss: 0.5406 - val_accuracy: 0.7315
Epoch 4/80
889/889 [==============================] - 1s 855us/step - loss: 0.5213 - accuracy:
0.7371 - val_loss: 0.5186 - val_accuracy: 0.7452
Epoch 5/80
889/889 [==============================] - 1s 846us/step - loss: 0.5184 - accuracy:
0.7402 - val_loss: 0.5277 - val_accuracy: 0.7349
Epoch 6/80
889/889 [==============================] - 1s 848us/step - loss: 0.5165 - accuracy:
0.7448 - val_loss: 0.5277 - val_accuracy: 0.7379
Epoch 7/80
889/889 [==============================] - 1s 848us/step - loss: 0.5152 - accuracy:
0.7448 - val_loss: 0.5153 - val_accuracy: 0.7452
Epoch 8/80
889/889 [==============================] - 1s 857us/step - loss: 0.5145 - accuracy:
0.7442 - val_loss: 0.5194 - val_accuracy: 0.7454
Epoch 9/80
889/889 [==============================] - 1s 842us/step - loss: 0.5137 - accuracy:
0.7454 - val_loss: 0.5132 - val_accuracy: 0.7467
Epoch 10/80
889/889 [==============================] - 1s 847us/step - loss: 0.5122 - accuracy:
0.7467 - val_loss: 0.5223 - val_accuracy: 0.7403
Epoch 11/80
889/889 [==============================] - 1s 846us/step - loss: 0.5125 - accuracy:
0.7460 - val_loss: 0.5195 - val_accuracy: 0.7429
Epoch 12/80
889/889 [==============================] - 1s 843us/step - loss: 0.5109 - accuracy:
0.7461 - val_loss: 0.5129 - val_accuracy: 0.7467
Epoch 13/80
889/889 [==============================] - 1s 858us/step - loss: 0.5098 - accuracy:
0.7498 - val_loss: 0.5124 - val_accuracy: 0.7489
Epoch 14/80
889/889 [==============================] - 1s 848us/step - loss: 0.5092 - accuracy:
0.7474 - val_loss: 0.5105 - val_accuracy: 0.7503
Epoch 15/80
889/889 [==============================] - 1s 846us/step - loss: 0.5079 - accuracy:
0.7490 - val_loss: 0.5145 - val_accuracy: 0.7467
Epoch 16/80
889/889 [==============================] - 1s 854us/step - loss: 0.5083 - accuracy:
0.7477 - val_loss: 0.5182 - val_accuracy: 0.7439
Epoch 17/80
889/889 [==============================] - 1s 856us/step - loss: 0.5081 - accuracy:
0.7496 - val_loss: 0.5133 - val_accuracy: 0.7492
Epoch 18/80
889/889 [==============================] - 1s 854us/step - loss: 0.5079 - accuracy:
0.7506 - val_loss: 0.5095 - val_accuracy: 0.7508
Epoch 19/80
```

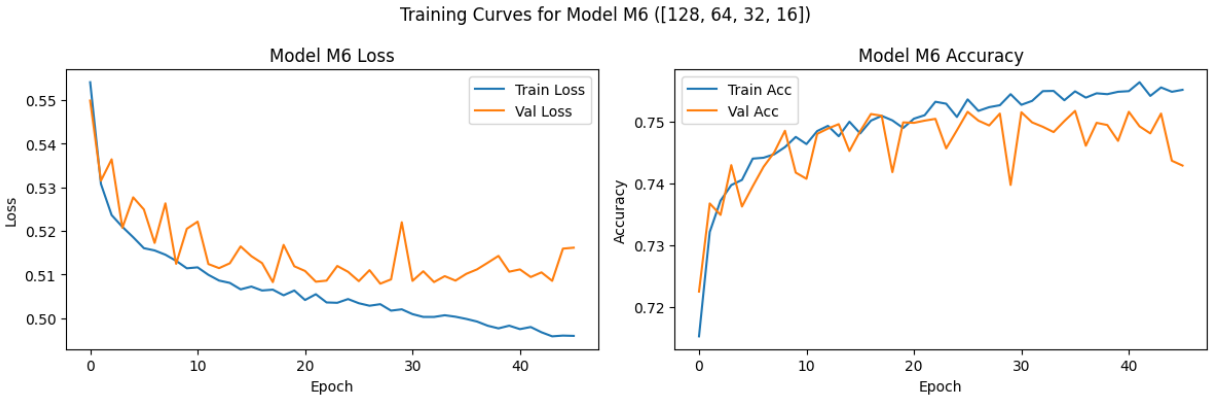
```
889/889 [==============================] - 1s 851us/step - loss: 0.5064 - accuracy:
0.7496 - val_loss: 0.5155 - val_accuracy: 0.7470
Epoch 20/80
889/889 [==============================] - 1s 853us/step - loss: 0.5076 - accuracy:
0.7476 - val_loss: 0.5112 - val_accuracy: 0.7483
Epoch 21/80
889/889 [==============================] - 1s 853us/step - loss: 0.5058 - accuracy:
0.7497 - val_loss: 0.5130 - val_accuracy: 0.7492
Epoch 22/80
889/889 [==============================] - 1s 853us/step - loss: 0.5065 - accuracy:
0.7515 - val_loss: 0.5100 - val_accuracy: 0.7504
Epoch 23/80
889/889 [==============================] - 1s 852us/step - loss: 0.5053 - accuracy:
0.7512 - val_loss: 0.5096 - val_accuracy: 0.7494
Epoch 24/80
889/889 [==============================] - 1s 846us/step - loss: 0.5050 - accuracy:
0.7522 - val_loss: 0.5139 - val_accuracy: 0.7479
Epoch 25/80
889/889 [==============================] - 1s 863us/step - loss: 0.5069 - accuracy:
0.7494 - val_loss: 0.5093 - val_accuracy: 0.7525
Epoch 26/80
889/889 [==============================] - 1s 876us/step - loss: 0.5046 - accuracy:
0.7516 - val_loss: 0.5102 - val_accuracy: 0.7483
Epoch 27/80
889/889 [==============================] - 1s 895us/step - loss: 0.5049 - accuracy:
0.7510 - val_loss: 0.5102 - val_accuracy: 0.7505
Epoch 28/80
889/889 [==============================] - 1s 852us/step - loss: 0.5045 - accuracy:
0.7515 - val_loss: 0.5084 - val_accuracy: 0.7518
Epoch 29/80
889/889 [==============================] - 1s 862us/step - loss: 0.5040 - accuracy:
0.7512 - val_loss: 0.5114 - val_accuracy: 0.7499
Epoch 30/80
889/889 [==============================] - 1s 872us/step - loss: 0.5050 - accuracy:
0.7528 - val_loss: 0.5160 - val_accuracy: 0.7470
Epoch 31/80
889/889 [==============================] - 1s 855us/step - loss: 0.5033 - accuracy:
0.7533 - val_loss: 0.5090 - val_accuracy: 0.7487
Epoch 32/80
889/889 [==============================] - 1s 859us/step - loss: 0.5034 - accuracy:
0.7522 - val_loss: 0.5119 - val_accuracy: 0.7495
Epoch 33/80
889/889 [==============================] - 1s 859us/step - loss: 0.5031 - accuracy:
0.7532 - val_loss: 0.5095 - val_accuracy: 0.7507
Epoch 34/80
889/889 [==============================] - 1s 848us/step - loss: 0.5033 - accuracy:
0.7532 - val_loss: 0.5085 - val_accuracy: 0.7492
Epoch 35/80
889/889 [==============================] - 1s 866us/step - loss: 0.5029 - accuracy:
0.7523 - val_loss: 0.5088 - val_accuracy: 0.7519
 Max Validation Accuracy for Model M5: 0.7525
438/438 [==============================] - 0s 432us/step
```


Training Curves for Model M5 ([64, 32, 16, 8])
Model M5 Loss
Train Loss
Val Loss
Loss
Epoch
Model M5 Accuracy
Train Acc
Val Acc
Accuracy
Epoch

```
 Training Model M6 with layers: [128, 64, 32, 16]
Epoch 1/80
889/889 [==============================] - 1s 976us/step - loss: 0.5540 - accuracy:
0.7152 - val_loss: 0.5498 - val_accuracy: 0.7224
Epoch 2/80
889/889 [==============================] - 1s 873us/step - loss: 0.5307 - accuracy:
0.7321 - val_loss: 0.5315 - val_accuracy: 0.7367
Epoch 3/80
889/889 [==============================] - 1s 863us/step - loss: 0.5236 - accuracy:
0.7372 - val_loss: 0.5364 - val_accuracy: 0.7349
Epoch 4/80
889/889 [==============================] - 1s 897us/step - loss: 0.5209 - accuracy:
0.7397 - val_loss: 0.5207 - val_accuracy: 0.7429
Epoch 5/80
889/889 [==============================] - 1s 867us/step - loss: 0.5186 - accuracy:
0.7406 - val_loss: 0.5277 - val_accuracy: 0.7362
Epoch 6/80
889/889 [==============================] - 1s 868us/step - loss: 0.5160 - accuracy:
0.7440 - val_loss: 0.5249 - val_accuracy: 0.7395
Epoch 7/80
889/889 [==============================] - 1s 870us/step - loss: 0.5155 - accuracy:
0.7441 - val_loss: 0.5172 - val_accuracy: 0.7427
Epoch 8/80
889/889 [==============================] - 1s 889us/step - loss: 0.5145 - accuracy:
0.7447 - val_loss: 0.5263 - val_accuracy: 0.7450
Epoch 9/80
889/889 [==============================] - 1s 870us/step - loss: 0.5131 - accuracy:
0.7459 - val_loss: 0.5125 - val_accuracy: 0.7485
Epoch 10/80
889/889 [==============================] - 1s 877us/step - loss: 0.5114 - accuracy:
0.7475 - val_loss: 0.5205 - val_accuracy: 0.7417
Epoch 11/80
889/889 [==============================] - 1s 872us/step - loss: 0.5117 - accuracy:
0.7464 - val_loss: 0.5221 - val_accuracy: 0.7407
Epoch 12/80
889/889 [==============================] - 1s 879us/step - loss: 0.5099 - accuracy:
0.7485 - val_loss: 0.5124 - val_accuracy: 0.7480
Epoch 13/80
889/889 [==============================] - 1s 862us/step - loss: 0.5086 - accuracy:
0.7493 - val_loss: 0.5115 - val_accuracy: 0.7489
Epoch 14/80
889/889 [==============================] - 1s 870us/step - loss: 0.5081 - accuracy:
0.7476 - val_loss: 0.5126 - val_accuracy: 0.7496
Epoch 15/80
889/889 [==============================] - 1s 871us/step - loss: 0.5066 - accuracy:
0.7500 - val_loss: 0.5164 - val_accuracy: 0.7452
Epoch 16/80
889/889 [==============================] - 1s 871us/step - loss: 0.5073 - accuracy:
0.7481 - val_loss: 0.5142 - val_accuracy: 0.7484
Epoch 17/80
889/889 [==============================] - 1s 872us/step - loss: 0.5064 - accuracy:
0.7502 - val_loss: 0.5126 - val_accuracy: 0.7512
Epoch 18/80
889/889 [==============================] - 1s 888us/step - loss: 0.5065 - accuracy:
0.7509 - val_loss: 0.5083 - val_accuracy: 0.7509
Epoch 19/80
```

```
889/889 [==============================] - 1s 870us/step - loss: 0.5053 - accuracy:
0.7502 - val_loss: 0.5168 - val_accuracy: 0.7418
Epoch 20/80
889/889 [==============================] - 1s 871us/step - loss: 0.5063 - accuracy:
0.7490 - val_loss: 0.5119 - val_accuracy: 0.7499
Epoch 21/80
889/889 [==============================] - 1s 872us/step - loss: 0.5042 - accuracy:
0.7505 - val_loss: 0.5108 - val_accuracy: 0.7498
Epoch 22/80
889/889 [==============================] - 1s 890us/step - loss: 0.5055 - accuracy:
0.7510 - val_loss: 0.5084 - val_accuracy: 0.7502
Epoch 23/80
889/889 [==============================] - 1s 890us/step - loss: 0.5036 - accuracy:
0.7532 - val_loss: 0.5086 - val_accuracy: 0.7504
Epoch 24/80
889/889 [==============================] - 1s 889us/step - loss: 0.5035 - accuracy:
0.7529 - val_loss: 0.5120 - val_accuracy: 0.7457
Epoch 25/80
889/889 [==============================] - 1s 881us/step - loss: 0.5044 - accuracy:
0.7508 - val_loss: 0.5106 - val_accuracy: 0.7486
Epoch 26/80
889/889 [==============================] - 1s 862us/step - loss: 0.5034 - accuracy:
0.7536 - val_loss: 0.5085 - val_accuracy: 0.7516
Epoch 27/80
889/889 [==============================] - 1s 890us/step - loss: 0.5029 - accuracy:
0.7517 - val_loss: 0.5110 - val_accuracy: 0.7502
Epoch 28/80
889/889 [==============================] - 1s 868us/step - loss: 0.5032 - accuracy:
0.7523 - val_loss: 0.5079 - val_accuracy: 0.7494
Epoch 29/80
889/889 [==============================] - 1s 871us/step - loss: 0.5017 - accuracy:
0.7527 - val_loss: 0.5089 - val_accuracy: 0.7513
Epoch 30/80
889/889 [==============================] - 1s 888us/step - loss: 0.5020 - accuracy:
0.7545 - val_loss: 0.5220 - val_accuracy: 0.7397
Epoch 31/80
889/889 [==============================] - 1s 890us/step - loss: 0.5009 - accuracy:
0.7527 - val_loss: 0.5086 - val_accuracy: 0.7515
Epoch 32/80
889/889 [==============================] - 1s 871us/step - loss: 0.5003 - accuracy:
0.7534 - val_loss: 0.5108 - val_accuracy: 0.7499
Epoch 33/80
889/889 [==============================] - 1s 909us/step - loss: 0.5003 - accuracy:
0.7549 - val_loss: 0.5083 - val_accuracy: 0.7492
Epoch 34/80
889/889 [==============================] - 1s 888us/step - loss: 0.5007 - accuracy:
0.7550 - val_loss: 0.5096 - val_accuracy: 0.7483
Epoch 35/80
889/889 [==============================] - 1s 876us/step - loss: 0.5003 - accuracy:
0.7535 - val_loss: 0.5086 - val_accuracy: 0.7501
Epoch 36/80
889/889 [==============================] - 1s 868us/step - loss: 0.4998 - accuracy:
0.7549 - val_loss: 0.5102 - val_accuracy: 0.7517
Epoch 37/80
889/889 [==============================] - 1s 874us/step - loss: 0.4992 - accuracy:
0.7539 - val_loss: 0.5112 - val_accuracy: 0.7461
```

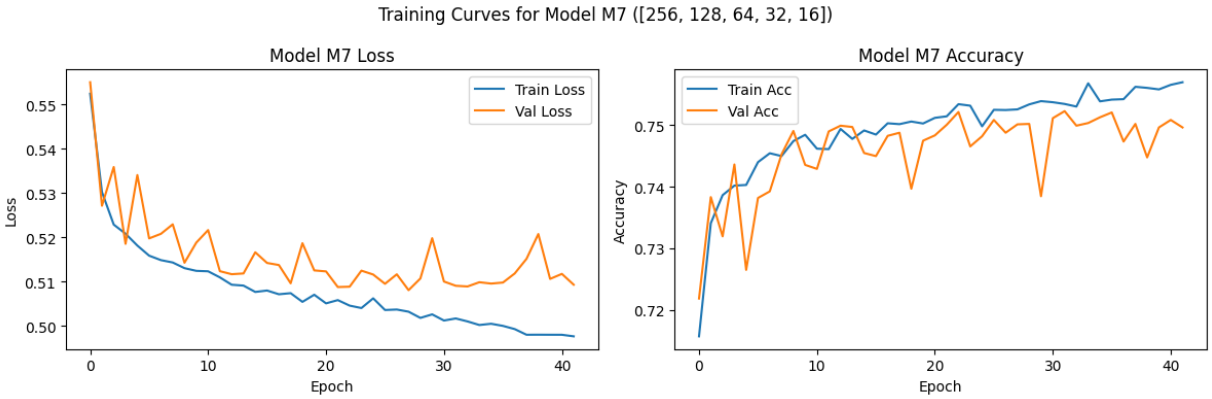
```
Epoch 38/80
889/889 [==============================] - 1s 869us/step - loss: 0.4983 - accuracy:
0.7546 - val_loss: 0.5127 - val_accuracy: 0.7498
Epoch 39/80
889/889 [==============================] - 1s 864us/step - loss: 0.4977 - accuracy:
0.7545 - val_loss: 0.5143 - val_accuracy: 0.7494
Epoch 40/80
889/889 [==============================] - 1s 855us/step - loss: 0.4983 - accuracy:
0.7548 - val_loss: 0.5107 - val_accuracy: 0.7469
Epoch 41/80
889/889 [==============================] - 1s 892us/step - loss: 0.4975 - accuracy:
0.7549 - val_loss: 0.5112 - val_accuracy: 0.7516
Epoch 42/80
889/889 [==============================] - 1s 873us/step - loss: 0.4980 - accuracy:
0.7564 - val_loss: 0.5094 - val_accuracy: 0.7492
Epoch 43/80
889/889 [==============================] - 1s 915us/step - loss: 0.4968 - accuracy:
0.7542 - val_loss: 0.5105 - val_accuracy: 0.7481
Epoch 44/80
889/889 [==============================] - 1s 900us/step - loss: 0.4959 - accuracy:
0.7555 - val_loss: 0.5086 - val_accuracy: 0.7513
Epoch 45/80
889/889 [==============================] - 1s 869us/step - loss: 0.4960 - accuracy:
0.7548 - val_loss: 0.5159 - val_accuracy: 0.7437
Epoch 46/80
889/889 [==============================] - 1s 887us/step - loss: 0.4960 - accuracy:
0.7552 - val_loss: 0.5161 - val_accuracy: 0.7429
 Max Validation Accuracy for Model M6: 0.7517
438/438 [==============================] - 0s 442us/step
```



```
 Training Model M7 with layers: [256, 128, 64, 32, 16]
Epoch 1/80
889/889 [==============================] - 2s 1ms/step - loss: 0.5525 - accuracy: 0.
7157 - val_loss: 0.5551 - val_accuracy: 0.7219
Epoch 2/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5302 - accuracy: 0.
7341 - val_loss: 0.5271 - val_accuracy: 0.7383
Epoch 3/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5229 - accuracy: 0.
7386 - val_loss: 0.5359 - val_accuracy: 0.7319
Epoch 4/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5208 - accuracy: 0.
7402 - val_loss: 0.5185 - val_accuracy: 0.7436
Epoch 5/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5181 - accuracy: 0.
7403 - val_loss: 0.5341 - val_accuracy: 0.7265
Epoch 6/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5159 - accuracy: 0.
7440 - val_loss: 0.5198 - val_accuracy: 0.7382
Epoch 7/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5148 - accuracy: 0.
7454 - val_loss: 0.5208 - val_accuracy: 0.7392
Epoch 8/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5143 - accuracy: 0.
7450 - val_loss: 0.5229 - val_accuracy: 0.7452
Epoch 9/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5130 - accuracy: 0.
7474 - val_loss: 0.5143 - val_accuracy: 0.7490
Epoch 10/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5124 - accuracy: 0.
7484 - val_loss: 0.5188 - val_accuracy: 0.7435
Epoch 11/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5123 - accuracy: 0.
7461 - val_loss: 0.5216 - val_accuracy: 0.7429
Epoch 12/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5109 - accuracy: 0.
7461 - val_loss: 0.5123 - val_accuracy: 0.7489
Epoch 13/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5093 - accuracy: 0.
7493 - val_loss: 0.5117 - val_accuracy: 0.7499
Epoch 14/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5091 - accuracy: 0.
7477 - val_loss: 0.5118 - val_accuracy: 0.7497
Epoch 15/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5077 - accuracy: 0.
7491 - val_loss: 0.5166 - val_accuracy: 0.7454
Epoch 16/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5080 - accuracy: 0.
7484 - val_loss: 0.5142 - val_accuracy: 0.7449
Epoch 17/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5071 - accuracy: 0.
7503 - val_loss: 0.5137 - val_accuracy: 0.7482
Epoch 18/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5074 - accuracy: 0.
7501 - val_loss: 0.5096 - val_accuracy: 0.7487
Epoch 19/80
```

```
889/889 [==============================] - 1s 1ms/step - loss: 0.5054 - accuracy: 0.
7505 - val_loss: 0.5187 - val_accuracy: 0.7397
Epoch 20/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5070 - accuracy: 0.
7502 - val_loss: 0.5125 - val_accuracy: 0.7474
Epoch 21/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5051 - accuracy: 0.
7511 - val_loss: 0.5123 - val_accuracy: 0.7483
Epoch 22/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5058 - accuracy: 0.
7514 - val_loss: 0.5088 - val_accuracy: 0.7500
Epoch 23/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5046 - accuracy: 0.
7534 - val_loss: 0.5088 - val_accuracy: 0.7521
Epoch 24/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5040 - accuracy: 0.
7531 - val_loss: 0.5125 - val_accuracy: 0.7465
Epoch 25/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5062 - accuracy: 0.
7498 - val_loss: 0.5116 - val_accuracy: 0.7482
Epoch 26/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5036 - accuracy: 0.
7524 - val_loss: 0.5095 - val_accuracy: 0.7508
Epoch 27/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5037 - accuracy: 0.
7524 - val_loss: 0.5116 - val_accuracy: 0.7487
Epoch 28/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5032 - accuracy: 0.
7525 - val_loss: 0.5081 - val_accuracy: 0.7501
Epoch 29/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5018 - accuracy: 0.
7533 - val_loss: 0.5107 - val_accuracy: 0.7502
Epoch 30/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5026 - accuracy: 0.
7539 - val_loss: 0.5198 - val_accuracy: 0.7384
Epoch 31/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5012 - accuracy: 0.
7537 - val_loss: 0.5100 - val_accuracy: 0.7511
Epoch 32/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5017 - accuracy: 0.
7534 - val_loss: 0.5091 - val_accuracy: 0.7522
Epoch 33/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5010 - accuracy: 0.
7530 - val_loss: 0.5089 - val_accuracy: 0.7499
Epoch 34/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5002 - accuracy: 0.
7567 - val_loss: 0.5099 - val_accuracy: 0.7503
Epoch 35/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5005 - accuracy: 0.
7538 - val_loss: 0.5095 - val_accuracy: 0.7512
Epoch 36/80
889/889 [==============================] - 1s 1ms/step - loss: 0.5000 - accuracy: 0.
7541 - val_loss: 0.5098 - val_accuracy: 0.7520
Epoch 37/80
889/889 [==============================] - 1s 1ms/step - loss: 0.4993 - accuracy: 0.
7542 - val_loss: 0.5118 - val_accuracy: 0.7473
```

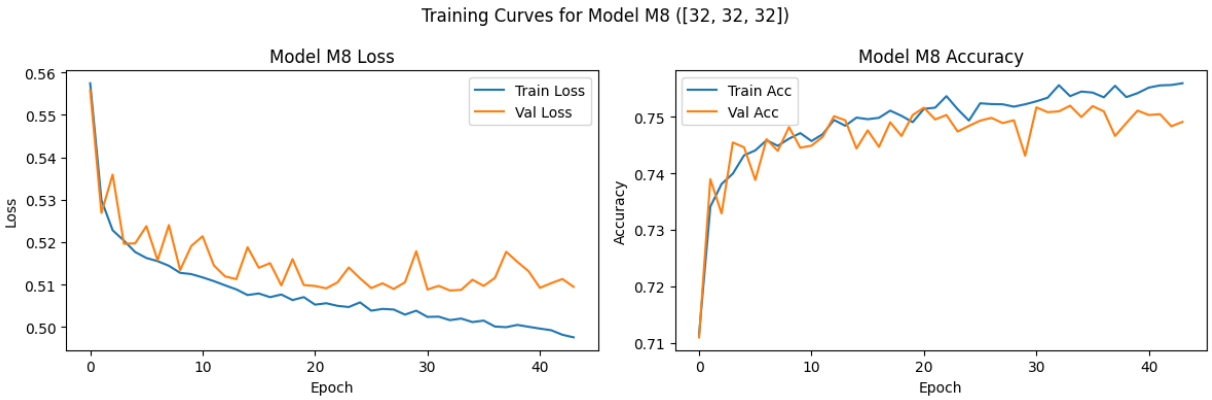
```
Epoch 38/80
889/889 [==============================] - 1s 1ms/step - loss: 0.4980 - accuracy: 0.
7562 - val_loss: 0.5151 - val_accuracy: 0.7502
Epoch 39/80
889/889 [==============================] - 1s 1ms/step - loss: 0.4980 - accuracy: 0.
7560 - val_loss: 0.5207 - val_accuracy: 0.7447
Epoch 40/80
889/889 [==============================] - 1s 1ms/step - loss: 0.4980 - accuracy: 0.
7558 - val_loss: 0.5106 - val_accuracy: 0.7496
Epoch 41/80
889/889 [==============================] - 1s 1ms/step - loss: 0.4980 - accuracy: 0.
7565 - val_loss: 0.5117 - val_accuracy: 0.7508
Epoch 42/80
889/889 [==============================] - 1s 1ms/step - loss: 0.4976 - accuracy: 0.
7569 - val_loss: 0.5093 - val_accuracy: 0.7496
 Max Validation Accuracy for Model M7: 0.7522
438/438 [==============================] - 0s 541us/step
```



```
 Training Model M8 with layers: [32, 32, 32]
Epoch 1/80
889/889 [==============================] - 1s 910us/step - loss: 0.5574 - accuracy:
0.7115 - val_loss: 0.5558 - val_accuracy: 0.7109
Epoch 2/80
889/889 [==============================] - 1s 936us/step - loss: 0.5299 - accuracy:
0.7341 - val_loss: 0.5269 - val_accuracy: 0.7389
Epoch 3/80
889/889 [==============================] - 1s 996us/step - loss: 0.5228 - accuracy:
0.7381 - val_loss: 0.5359 - val_accuracy: 0.7329
Epoch 4/80
889/889 [==============================] - 1s 862us/step - loss: 0.5204 - accuracy:
0.7399 - val_loss: 0.5196 - val_accuracy: 0.7454
Epoch 5/80
889/889 [==============================] - 1s 832us/step - loss: 0.5177 - accuracy:
0.7432 - val_loss: 0.5197 - val_accuracy: 0.7446
Epoch 6/80
889/889 [==============================] - 1s 835us/step - loss: 0.5163 - accuracy:
0.7440 - val_loss: 0.5238 - val_accuracy: 0.7388
Epoch 7/80
889/889 [==============================] - 1s 821us/step - loss: 0.5155 - accuracy:
0.7458 - val_loss: 0.5157 - val_accuracy: 0.7460
Epoch 8/80
889/889 [==============================] - 1s 873us/step - loss: 0.5144 - accuracy:
0.7448 - val_loss: 0.5240 - val_accuracy: 0.7439
Epoch 9/80
889/889 [==============================] - 1s 822us/step - loss: 0.5128 - accuracy:
0.7461 - val_loss: 0.5134 - val_accuracy: 0.7482
Epoch 10/80
889/889 [==============================] - 1s 823us/step - loss: 0.5125 - accuracy:
0.7471 - val_loss: 0.5191 - val_accuracy: 0.7445
Epoch 11/80
889/889 [==============================] - 1s 820us/step - loss: 0.5117 - accuracy:
0.7457 - val_loss: 0.5214 - val_accuracy: 0.7449
Epoch 12/80
889/889 [==============================] - 1s 839us/step - loss: 0.5108 - accuracy:
0.7469 - val_loss: 0.5145 - val_accuracy: 0.7464
Epoch 13/80
889/889 [==============================] - 1s 855us/step - loss: 0.5098 - accuracy:
0.7494 - val_loss: 0.5119 - val_accuracy: 0.7501
Epoch 14/80
889/889 [==============================] - 1s 840us/step - loss: 0.5089 - accuracy:
0.7484 - val_loss: 0.5113 - val_accuracy: 0.7494
Epoch 15/80
889/889 [==============================] - 1s 821us/step - loss: 0.5075 - accuracy:
0.7498 - val_loss: 0.5188 - val_accuracy: 0.7444
Epoch 16/80
889/889 [==============================] - 1s 822us/step - loss: 0.5079 - accuracy:
0.7496 - val_loss: 0.5140 - val_accuracy: 0.7476
Epoch 17/80
889/889 [==============================] - 1s 822us/step - loss: 0.5070 - accuracy:
0.7498 - val_loss: 0.5150 - val_accuracy: 0.7447
Epoch 18/80
889/889 [==============================] - 1s 891us/step - loss: 0.5077 - accuracy:
0.7511 - val_loss: 0.5098 - val_accuracy: 0.7490
Epoch 19/80
```

```
889/889 [==============================] - 1s 821us/step - loss: 0.5064 - accuracy:
0.7502 - val_loss: 0.5160 - val_accuracy: 0.7466
Epoch 20/80
889/889 [==============================] - 1s 847us/step - loss: 0.5071 - accuracy:
0.7490 - val_loss: 0.5099 - val_accuracy: 0.7503
Epoch 21/80
889/889 [==============================] - 1s 830us/step - loss: 0.5053 - accuracy:
0.7514 - val_loss: 0.5097 - val_accuracy: 0.7516
Epoch 22/80
889/889 [==============================] - 1s 837us/step - loss: 0.5056 - accuracy:
0.7516 - val_loss: 0.5091 - val_accuracy: 0.7495
Epoch 23/80
889/889 [==============================] - 1s 872us/step - loss: 0.5050 - accuracy:
0.7536 - val_loss: 0.5105 - val_accuracy: 0.7503
Epoch 24/80
889/889 [==============================] - 1s 824us/step - loss: 0.5047 - accuracy:
0.7513 - val_loss: 0.5140 - val_accuracy: 0.7474
Epoch 25/80
889/889 [==============================] - 1s 823us/step - loss: 0.5058 - accuracy:
0.7493 - val_loss: 0.5115 - val_accuracy: 0.7484
Epoch 26/80
889/889 [==============================] - 1s 823us/step - loss: 0.5038 - accuracy:
0.7524 - val_loss: 0.5092 - val_accuracy: 0.7493
Epoch 27/80
889/889 [==============================] - 1s 875us/step - loss: 0.5043 - accuracy:
0.7522 - val_loss: 0.5103 - val_accuracy: 0.7498
Epoch 28/80
889/889 [==============================] - 1s 839us/step - loss: 0.5041 - accuracy:
0.7522 - val_loss: 0.5090 - val_accuracy: 0.7489
Epoch 29/80
889/889 [==============================] - 1s 838us/step - loss: 0.5029 - accuracy:
0.7518 - val_loss: 0.5105 - val_accuracy: 0.7494
Epoch 30/80
889/889 [==============================] - 1s 845us/step - loss: 0.5039 - accuracy:
0.7522 - val_loss: 0.5178 - val_accuracy: 0.7431
Epoch 31/80
889/889 [==============================] - 1s 822us/step - loss: 0.5024 - accuracy:
0.7527 - val_loss: 0.5088 - val_accuracy: 0.7517
Epoch 32/80
889/889 [==============================] - 1s 871us/step - loss: 0.5024 - accuracy:
0.7534 - val_loss: 0.5097 - val_accuracy: 0.7508
Epoch 33/80
889/889 [==============================] - 1s 838us/step - loss: 0.5016 - accuracy:
0.7556 - val_loss: 0.5086 - val_accuracy: 0.7509
Epoch 34/80
889/889 [==============================] - 1s 838us/step - loss: 0.5020 - accuracy:
0.7536 - val_loss: 0.5088 - val_accuracy: 0.7519
Epoch 35/80
889/889 [==============================] - 1s 838us/step - loss: 0.5012 - accuracy:
0.7545 - val_loss: 0.5111 - val_accuracy: 0.7499
Epoch 36/80
889/889 [==============================] - 1s 822us/step - loss: 0.5015 - accuracy:
0.7543 - val_loss: 0.5097 - val_accuracy: 0.7519
Epoch 37/80
889/889 [==============================] - 1s 838us/step - loss: 0.5001 - accuracy:
0.7534 - val_loss: 0.5116 - val_accuracy: 0.7509
```

```
Epoch 38/80
889/889 [==============================] - 1s 820us/step - loss: 0.5000 - accuracy:
0.7555 - val_loss: 0.5177 - val_accuracy: 0.7466
Epoch 39/80
889/889 [==============================] - 1s 851us/step - loss: 0.5005 - accuracy:
0.7535 - val_loss: 0.5154 - val_accuracy: 0.7489
Epoch 40/80
889/889 [==============================] - 1s 826us/step - loss: 0.5001 - accuracy:
0.7542 - val_loss: 0.5131 - val_accuracy: 0.7511
Epoch 41/80
889/889 [==============================] - 1s 813us/step - loss: 0.4996 - accuracy:
0.7552 - val_loss: 0.5092 - val_accuracy: 0.7503
Epoch 42/80
889/889 [==============================] - 1s 841us/step - loss: 0.4993 - accuracy:
0.7555 - val_loss: 0.5103 - val_accuracy: 0.7504
Epoch 43/80
889/889 [==============================] - 1s 853us/step - loss: 0.4982 - accuracy:
0.7556 - val_loss: 0.5113 - val_accuracy: 0.7483
Epoch 44/80
889/889 [==============================] - 1s 839us/step - loss: 0.4976 - accuracy:
0.7559 - val_loss: 0.5095 - val_accuracy: 0.7491
 Max Validation Accuracy for Model M8: 0.7519
438/438 [==============================] - 0s 436us/step
```



3.4 Performance Summary

After training, we collected the best validation and test metrics for each model:

In [16]:
```
# Showing the best model info
print(f"\n Best model: Layers {best_config} with validation accuracy: {best_val_acc

# Displaying summary table
results_df = pd.DataFrame(results)
results_df = results_df.sort_values(by='Val_Accuracy', ascending=False)
print("\n Model Performance Summary:")
print(results_df)
```

```
Best model: Layers [32, 16, 8] with validation accuracy: 0.7532

Model Performance Summary:
  Model                  Layers  Val_Accuracy  Test_Accuracy  Precision  \
3    M4             [32, 16, 8]      0.753161       0.753161   0.727355
2    M3              [16, 8, 4]      0.752590       0.752590   0.732942
4    M5         [64, 32, 16, 8]      0.752518       0.752518   0.719618
6    M7  [256, 128, 64, 32, 16]      0.752233       0.752233   0.727013
7    M8            [32, 32, 32]      0.751947       0.751947   0.724129
5    M6       [128, 64, 32, 16]      0.751733       0.751733   0.727772
1    M2                  [8, 4]      0.751232       0.751232   0.728720
0    M1                     [4]      0.748589       0.748589   0.734205

     Recall  F1_Score
3  0.805432  0.764405
2  0.790343  0.760561
4  0.822819  0.767766
6  0.803276  0.763244
7  0.809455  0.764419
5  0.799828  0.762100
1  0.795948  0.760852
0  0.774824  0.753968
```

3.5 Analysis And interpretation

1. M1:[4]
    - Acts as the baseline model.
    - achieves roughly 74.86% validation accuracy.
    - highesh precision which suggests good performance on positive class.
    - but the recall and F1 are lower which indicates its missing more positives.
2. M2:[8,4]
    - small increase in performance.
    - smoother training curves and less noise
    - modest improvement in recall and F1.
3. M3: [16,8,4]
    - first significantly better model.
    - Accuracy and F1 both improve, reflecting increased capacuity.
4. M4: [32,16,8]
    - Achieved the best validation accuracy: 0.7532.
    - strong performance across all metrics.
    - Smooth and stable learning.
    - no signs of overfitting, unlike deeper models.
5. M5: [64,32,16,8]
    - similar accuracy to m4, but slightly better F1 score.
    - Incresed depth gives slight gains but is adding training complexity.

6. M6 and M8: Deep Networks.
    - Performance is starting to go plateau.
    - validation accuracy doesn'[t improve.
    - Training is becoming less stable, with some overfitting signs in loss curve.
7. M8: [32,32,32]
    - Wide network
    - Good f1 and recall but no clear or visible advantage over M4/M5

Section 4: Hyperparameter Investigation.

4.1 Batch Size Investigation

Batch size determines how many samples are processed before a model is updated. To explore its impact,we are testing a range of batch sizes: 16,32,64,128,256,512,1024 and 2048. These batches choosen to be spanned across small(16-64), medium(128-512) and large 91024_2048) batches where i can capture variations in training stability, convergence speed, and generaliztion. Small batches often improve generalizatio but can be noisy whereas large ones being faster but risk overfitting. each setup used same architecture,early stopping and upto 80 epochs. Reslts were evaluated absed on validation accuracy, learning trnds and final performance to identify the best batch size.

In [17]:
```
import matplotlib.pyplot as plt
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, Input
from tensorflow.keras.callbacks import EarlyStopping
from tensorflow.keras.optimizers import Adam, SGD, RMSprop
from tensorflow.keras.regularizers import l2
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# --- 4.1 Batch Size Investigation ---
batch_sizes = [16, 32, 64, 128, 256, 512, 1024, 2048]

best_val_accuracy = 0
best_batch = None

print("### Batch Size Exploration ###")
for bs in batch_sizes:
    print(f"\nTraining with batch size: {bs}")

    model = Sequential([
    Input(shape=(X_train.shape[1],)),
    Dense(32, activation='relu'),
    Dense(16, activation='relu'),
    Dense(8, activation='relu'),
    Dense(1, activation='sigmoid')
    ])


    model.compile(
        optimizer='adam',
        loss='binary_crossentropy',
        metrics=['accuracy']
    )

    early_stop = EarlyStopping(monitor='val_accuracy', patience=10, restore_best_we

    history = model.fit(
        X_train, y_train,
        validation_data=(X_test, y_test),
        epochs=80,
        batch_size=bs,
        callbacks=[early_stop],
        verbose=1
    )

    max_val_acc = max(history.history['val_accuracy'])
    print(f"Max validation accuracy for batch size {bs}: {max_val_acc:.4f}")

    if max_val_acc > best_val_accuracy:
        best_val_accuracy = max_val_acc
        best_batch = bs

    # Plotting
    plt.figure(figsize=(12, 4))
    plt.subplot(1, 2, 1)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')
```

```
    plt.title(f'Loss — Batch Size {bs}')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
    plt.legend()

    plt.subplot(1, 2, 2)
    plt.plot(history.history['accuracy'], label='Train Accuracy')
    plt.plot(history.history['val_accuracy'], label='Val Accuracy')
    plt.title(f'Accuracy — Batch Size {bs}')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()

    plt.tight_layout()
    plt.show()

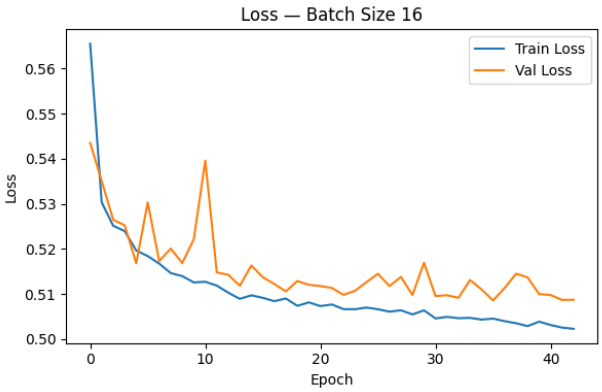
print(f"\nBest batch size: {best_batch} with Validation Accuracy: {best_val_accurac
```

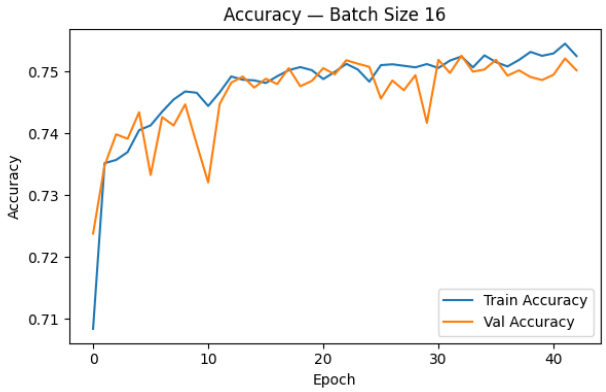
```
### Batch Size Exploration ###

Training with batch size: 16
Epoch 1/80
1777/1777 [==============================] - 2s 850us/step - loss: 0.5655 - accuracy
: 0.7084 - val_loss: 0.5435 - val_accuracy: 0.7238
Epoch 2/80
1777/1777 [==============================] - 1s 822us/step - loss: 0.5303 - accuracy
: 0.7352 - val_loss: 0.5350 - val_accuracy: 0.7348
Epoch 3/80
1777/1777 [==============================] - 1s 811us/step - loss: 0.5251 - accuracy
: 0.7357 - val_loss: 0.5265 - val_accuracy: 0.7399
Epoch 4/80
1777/1777 [==============================] - 1s 804us/step - loss: 0.5240 - accuracy
: 0.7370 - val_loss: 0.5252 - val_accuracy: 0.7392
Epoch 5/80
1777/1777 [==============================] - 1s 820us/step - loss: 0.5197 - accuracy
: 0.7406 - val_loss: 0.5168 - val_accuracy: 0.7434
Epoch 6/80
1777/1777 [==============================] - 1s 819us/step - loss: 0.5184 - accuracy
: 0.7413 - val_loss: 0.5303 - val_accuracy: 0.7333
Epoch 7/80
1777/1777 [==============================] - 1s 820us/step - loss: 0.5167 - accuracy
: 0.7435 - val_loss: 0.5173 - val_accuracy: 0.7427
Epoch 8/80
1777/1777 [==============================] - 1s 825us/step - loss: 0.5146 - accuracy
: 0.7455 - val_loss: 0.5201 - val_accuracy: 0.7413
Epoch 9/80
1777/1777 [==============================] - 1s 821us/step - loss: 0.5140 - accuracy
: 0.7468 - val_loss: 0.5169 - val_accuracy: 0.7447
Epoch 10/80
1777/1777 [==============================] - 1s 812us/step - loss: 0.5126 - accuracy
: 0.7466 - val_loss: 0.5221 - val_accuracy: 0.7383
Epoch 11/80
1777/1777 [==============================] - 1s 813us/step - loss: 0.5127 - accuracy
: 0.7445 - val_loss: 0.5396 - val_accuracy: 0.7321
Epoch 12/80
1777/1777 [==============================] - 1s 813us/step - loss: 0.5119 - accuracy
: 0.7467 - val_loss: 0.5148 - val_accuracy: 0.7448
Epoch 13/80
1777/1777 [==============================] - 1s 821us/step - loss: 0.5103 - accuracy
: 0.7492 - val_loss: 0.5143 - val_accuracy: 0.7482
Epoch 14/80
1777/1777 [==============================] - 1s 810us/step - loss: 0.5089 - accuracy
: 0.7487 - val_loss: 0.5118 - val_accuracy: 0.7492
Epoch 15/80
1777/1777 [==============================] - 1s 820us/step - loss: 0.5097 - accuracy
: 0.7486 - val_loss: 0.5163 - val_accuracy: 0.7474
Epoch 16/80
1777/1777 [==============================] - 1s 810us/step - loss: 0.5092 - accuracy
: 0.7482 - val_loss: 0.5137 - val_accuracy: 0.7489
Epoch 17/80
1777/1777 [==============================] - 1s 819us/step - loss: 0.5084 - accuracy
: 0.7493 - val_loss: 0.5122 - val_accuracy: 0.7480
Epoch 18/80
1777/1777 [==============================] - 1s 812us/step - loss: 0.5090 - accuracy
```

```
: 0.7503 - val_loss: 0.5106 - val_accuracy: 0.7506
Epoch 19/80
1777/1777 [==============================] - 1s 813us/step - loss: 0.5074 - accuracy
: 0.7508 - val_loss: 0.5129 - val_accuracy: 0.7477
Epoch 20/80
1777/1777 [==============================] - 1s 820us/step - loss: 0.5082 - accuracy
: 0.7503 - val_loss: 0.5120 - val_accuracy: 0.7485
Epoch 21/80
1777/1777 [==============================] - 1s 812us/step - loss: 0.5073 - accuracy
: 0.7488 - val_loss: 0.5118 - val_accuracy: 0.7506
Epoch 22/80
1777/1777 [==============================] - 1s 812us/step - loss: 0.5077 - accuracy
: 0.7500 - val_loss: 0.5113 - val_accuracy: 0.7496
Epoch 23/80
1777/1777 [==============================] - 1s 807us/step - loss: 0.5067 - accuracy
: 0.7513 - val_loss: 0.5098 - val_accuracy: 0.7519
Epoch 24/80
1777/1777 [==============================] - 1s 811us/step - loss: 0.5066 - accuracy
: 0.7504 - val_loss: 0.5107 - val_accuracy: 0.7513
Epoch 25/80
1777/1777 [==============================] - 1s 811us/step - loss: 0.5070 - accuracy
: 0.7484 - val_loss: 0.5126 - val_accuracy: 0.7508
Epoch 26/80
1777/1777 [==============================] - 1s 821us/step - loss: 0.5066 - accuracy
: 0.7511 - val_loss: 0.5145 - val_accuracy: 0.7457
Epoch 27/80
1777/1777 [==============================] - 1s 821us/step - loss: 0.5061 - accuracy
: 0.7512 - val_loss: 0.5118 - val_accuracy: 0.7486
Epoch 28/80
1777/1777 [==============================] - 1s 819us/step - loss: 0.5064 - accuracy
: 0.7510 - val_loss: 0.5138 - val_accuracy: 0.7470
Epoch 29/80
1777/1777 [==============================] - 1s 814us/step - loss: 0.5055 - accuracy
: 0.7507 - val_loss: 0.5098 - val_accuracy: 0.7494
Epoch 30/80
1777/1777 [==============================] - 1s 807us/step - loss: 0.5064 - accuracy
: 0.7512 - val_loss: 0.5169 - val_accuracy: 0.7417
Epoch 31/80
1777/1777 [==============================] - 1s 820us/step - loss: 0.5046 - accuracy
: 0.7506 - val_loss: 0.5096 - val_accuracy: 0.7519
Epoch 32/80
1777/1777 [==============================] - 1s 812us/step - loss: 0.5049 - accuracy
: 0.7518 - val_loss: 0.5097 - val_accuracy: 0.7498
Epoch 33/80
1777/1777 [==============================] - 1s 813us/step - loss: 0.5047 - accuracy
: 0.7525 - val_loss: 0.5092 - val_accuracy: 0.7526
Epoch 34/80
1777/1777 [==============================] - 1s 819us/step - loss: 0.5047 - accuracy
: 0.7507 - val_loss: 0.5131 - val_accuracy: 0.7500
Epoch 35/80
1777/1777 [==============================] - 1s 811us/step - loss: 0.5043 - accuracy
: 0.7527 - val_loss: 0.5110 - val_accuracy: 0.7504
Epoch 36/80
1777/1777 [==============================] - 1s 812us/step - loss: 0.5046 - accuracy
: 0.7516 - val_loss: 0.5086 - val_accuracy: 0.7519
Epoch 37/80
```

```
1777/1777 [==============================] - 1s 811us/step - loss: 0.5040 - accuracy
: 0.7509 - val_loss: 0.5113 - val_accuracy: 0.7494
Epoch 38/80
1777/1777 [==============================] - 1s 812us/step - loss: 0.5035 - accuracy
: 0.7519 - val_loss: 0.5145 - val_accuracy: 0.7502
Epoch 39/80
1777/1777 [==============================] - 1s 820us/step - loss: 0.5029 - accuracy
: 0.7532 - val_loss: 0.5137 - val_accuracy: 0.7492
Epoch 40/80
1777/1777 [==============================] - 1s 810us/step - loss: 0.5039 - accuracy
: 0.7526 - val_loss: 0.5100 - val_accuracy: 0.7487
Epoch 41/80
1777/1777 [==============================] - 1s 819us/step - loss: 0.5031 - accuracy
: 0.7530 - val_loss: 0.5098 - val_accuracy: 0.7495
Epoch 42/80
1777/1777 [==============================] - 1s 811us/step - loss: 0.5026 - accuracy
: 0.7546 - val_loss: 0.5087 - val_accuracy: 0.7522
Epoch 43/80
1777/1777 [==============================] - 1s 811us/step - loss: 0.5023 - accuracy
: 0.7526 - val_loss: 0.5087 - val_accuracy: 0.7502
Max validation accuracy for batch size 16: 0.7526
```



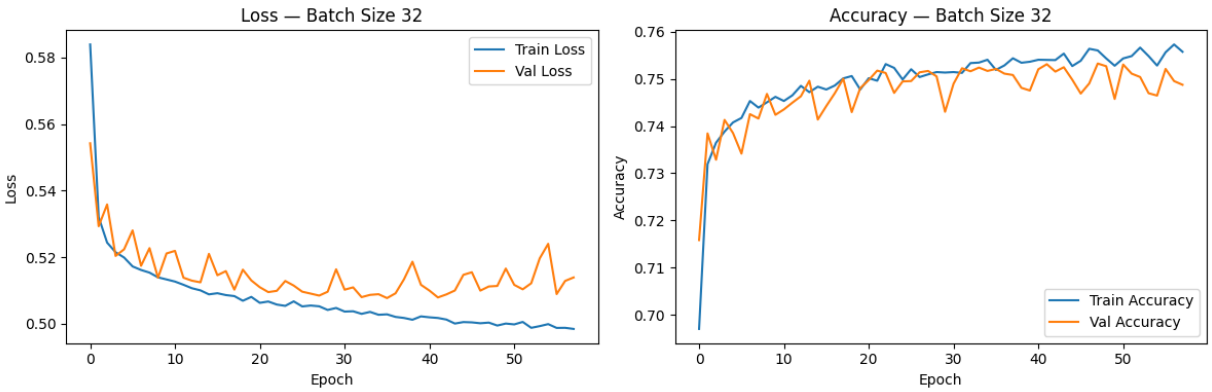


```
Training with batch size: 32
Epoch 1/80
889/889 [==============================] - 1s 927us/step - loss: 0.5839 - accuracy:
0.6971 - val_loss: 0.5542 - val_accuracy: 0.7159
Epoch 2/80
889/889 [==============================] - 1s 838us/step - loss: 0.5321 - accuracy:
0.7319 - val_loss: 0.5293 - val_accuracy: 0.7384
Epoch 3/80
889/889 [==============================] - 1s 821us/step - loss: 0.5244 - accuracy:
0.7365 - val_loss: 0.5359 - val_accuracy: 0.7329
Epoch 4/80
889/889 [==============================] - 1s 821us/step - loss: 0.5215 - accuracy:
0.7388 - val_loss: 0.5204 - val_accuracy: 0.7413
Epoch 5/80
889/889 [==============================] - 1s 840us/step - loss: 0.5199 - accuracy:
0.7408 - val_loss: 0.5224 - val_accuracy: 0.7385
Epoch 6/80
889/889 [==============================] - 1s 838us/step - loss: 0.5172 - accuracy:
0.7417 - val_loss: 0.5281 - val_accuracy: 0.7342
Epoch 7/80
889/889 [==============================] - 1s 835us/step - loss: 0.5162 - accuracy:
0.7453 - val_loss: 0.5174 - val_accuracy: 0.7425
Epoch 8/80
889/889 [==============================] - 1s 820us/step - loss: 0.5154 - accuracy:
0.7439 - val_loss: 0.5227 - val_accuracy: 0.7416
Epoch 9/80
889/889 [==============================] - 1s 812us/step - loss: 0.5140 - accuracy:
0.7450 - val_loss: 0.5138 - val_accuracy: 0.7468
Epoch 10/80
889/889 [==============================] - 1s 818us/step - loss: 0.5133 - accuracy:
0.7462 - val_loss: 0.5211 - val_accuracy: 0.7424
Epoch 11/80
889/889 [==============================] - 1s 837us/step - loss: 0.5127 - accuracy:
0.7453 - val_loss: 0.5219 - val_accuracy: 0.7435
Epoch 12/80
889/889 [==============================] - 1s 836us/step - loss: 0.5117 - accuracy:
0.7465 - val_loss: 0.5138 - val_accuracy: 0.7449
Epoch 13/80
889/889 [==============================] - 1s 819us/step - loss: 0.5107 - accuracy:
0.7485 - val_loss: 0.5129 - val_accuracy: 0.7463
Epoch 14/80
889/889 [==============================] - 1s 838us/step - loss: 0.5100 - accuracy:
0.7472 - val_loss: 0.5125 - val_accuracy: 0.7496
Epoch 15/80
889/889 [==============================] - 1s 836us/step - loss: 0.5088 - accuracy:
0.7483 - val_loss: 0.5210 - val_accuracy: 0.7414
Epoch 16/80
889/889 [==============================] - 1s 818us/step - loss: 0.5092 - accuracy:
0.7477 - val_loss: 0.5146 - val_accuracy: 0.7442
Epoch 17/80
889/889 [==============================] - 1s 838us/step - loss: 0.5086 - accuracy:
0.7486 - val_loss: 0.5158 - val_accuracy: 0.7469
Epoch 18/80
889/889 [==============================] - 1s 819us/step - loss: 0.5083 - accuracy:
0.7501 - val_loss: 0.5103 - val_accuracy: 0.7500
Epoch 19/80
```

```
889/889 [==============================] - 1s 839us/step - loss: 0.5069 - accuracy:
0.7506 - val_loss: 0.5163 - val_accuracy: 0.7429
Epoch 20/80
889/889 [==============================] - 1s 820us/step - loss: 0.5081 - accuracy:
0.7477 - val_loss: 0.5130 - val_accuracy: 0.7480
Epoch 21/80
889/889 [==============================] - 1s 821us/step - loss: 0.5063 - accuracy:
0.7501 - val_loss: 0.5110 - val_accuracy: 0.7497
Epoch 22/80
889/889 [==============================] - 1s 836us/step - loss: 0.5067 - accuracy:
0.7496 - val_loss: 0.5095 - val_accuracy: 0.7517
Epoch 23/80
889/889 [==============================] - 1s 821us/step - loss: 0.5058 - accuracy:
0.7531 - val_loss: 0.5099 - val_accuracy: 0.7512
Epoch 24/80
889/889 [==============================] - 1s 821us/step - loss: 0.5054 - accuracy:
0.7523 - val_loss: 0.5129 - val_accuracy: 0.7470
Epoch 25/80
889/889 [==============================] - 1s 838us/step - loss: 0.5067 - accuracy:
0.7498 - val_loss: 0.5115 - val_accuracy: 0.7494
Epoch 26/80
889/889 [==============================] - 1s 836us/step - loss: 0.5052 - accuracy:
0.7520 - val_loss: 0.5096 - val_accuracy: 0.7495
Epoch 27/80
889/889 [==============================] - 1s 819us/step - loss: 0.5055 - accuracy:
0.7504 - val_loss: 0.5091 - val_accuracy: 0.7514
Epoch 28/80
889/889 [==============================] - 1s 835us/step - loss: 0.5053 - accuracy:
0.7509 - val_loss: 0.5085 - val_accuracy: 0.7517
Epoch 29/80
889/889 [==============================] - 1s 821us/step - loss: 0.5042 - accuracy:
0.7514 - val_loss: 0.5096 - val_accuracy: 0.7505
Epoch 30/80
889/889 [==============================] - 1s 835us/step - loss: 0.5048 - accuracy:
0.7513 - val_loss: 0.5164 - val_accuracy: 0.7430
Epoch 31/80
889/889 [==============================] - 1s 821us/step - loss: 0.5037 - accuracy:
0.7514 - val_loss: 0.5102 - val_accuracy: 0.7489
Epoch 32/80
889/889 [==============================] - 1s 838us/step - loss: 0.5038 - accuracy:
0.7512 - val_loss: 0.5109 - val_accuracy: 0.7522
Epoch 33/80
889/889 [==============================] - 1s 818us/step - loss: 0.5030 - accuracy:
0.7533 - val_loss: 0.5080 - val_accuracy: 0.7516
Epoch 34/80
889/889 [==============================] - 1s 837us/step - loss: 0.5036 - accuracy:
0.7534 - val_loss: 0.5087 - val_accuracy: 0.7524
Epoch 35/80
889/889 [==============================] - 1s 820us/step - loss: 0.5027 - accuracy:
0.7540 - val_loss: 0.5089 - val_accuracy: 0.7517
Epoch 36/80
889/889 [==============================] - 1s 837us/step - loss: 0.5028 - accuracy:
0.7519 - val_loss: 0.5077 - val_accuracy: 0.7521
Epoch 37/80
889/889 [==============================] - 1s 820us/step - loss: 0.5021 - accuracy:
0.7528 - val_loss: 0.5092 - val_accuracy: 0.7511
```

```
Epoch 38/80
889/889 [==============================] - 1s 840us/step - loss: 0.5017 - accuracy:
0.7543 - val_loss: 0.5134 - val_accuracy: 0.7508
Epoch 39/80
889/889 [==============================] - 1s 827us/step - loss: 0.5012 - accuracy:
0.7534 - val_loss: 0.5186 - val_accuracy: 0.7481
Epoch 40/80
889/889 [==============================] - 1s 832us/step - loss: 0.5022 - accuracy:
0.7536 - val_loss: 0.5117 - val_accuracy: 0.7475
Epoch 41/80
889/889 [==============================] - 1s 829us/step - loss: 0.5019 - accuracy:
0.7540 - val_loss: 0.5100 - val_accuracy: 0.7520
Epoch 42/80
889/889 [==============================] - 1s 835us/step - loss: 0.5017 - accuracy:
0.7540 - val_loss: 0.5079 - val_accuracy: 0.7531
Epoch 43/80
889/889 [==============================] - 1s 825us/step - loss: 0.5013 - accuracy:
0.7540 - val_loss: 0.5088 - val_accuracy: 0.7515
Epoch 44/80
889/889 [==============================] - 1s 825us/step - loss: 0.5001 - accuracy:
0.7553 - val_loss: 0.5100 - val_accuracy: 0.7524
Epoch 45/80
889/889 [==============================] - 1s 827us/step - loss: 0.5005 - accuracy:
0.7527 - val_loss: 0.5147 - val_accuracy: 0.7498
Epoch 46/80
889/889 [==============================] - 1s 829us/step - loss: 0.5004 - accuracy:
0.7538 - val_loss: 0.5155 - val_accuracy: 0.7469
Epoch 47/80
889/889 [==============================] - 1s 825us/step - loss: 0.5002 - accuracy:
0.7564 - val_loss: 0.5100 - val_accuracy: 0.7490
Epoch 48/80
889/889 [==============================] - 1s 823us/step - loss: 0.5004 - accuracy:
0.7560 - val_loss: 0.5112 - val_accuracy: 0.7532
Epoch 49/80
889/889 [==============================] - 1s 842us/step - loss: 0.4995 - accuracy:
0.7543 - val_loss: 0.5114 - val_accuracy: 0.7527
Epoch 50/80
889/889 [==============================] - 1s 827us/step - loss: 0.5001 - accuracy:
0.7528 - val_loss: 0.5166 - val_accuracy: 0.7457
Epoch 51/80
889/889 [==============================] - 1s 823us/step - loss: 0.4998 - accuracy:
0.7543 - val_loss: 0.5117 - val_accuracy: 0.7530
Epoch 52/80
889/889 [==============================] - 1s 831us/step - loss: 0.5006 - accuracy:
0.7548 - val_loss: 0.5104 - val_accuracy: 0.7511
Epoch 53/80
889/889 [==============================] - 1s 833us/step - loss: 0.4988 - accuracy:
0.7566 - val_loss: 0.5122 - val_accuracy: 0.7504
Epoch 54/80
889/889 [==============================] - 1s 831us/step - loss: 0.4993 - accuracy:
0.7548 - val_loss: 0.5197 - val_accuracy: 0.7469
Epoch 55/80
889/889 [==============================] - 1s 826us/step - loss: 0.4999 - accuracy:
0.7528 - val_loss: 0.5240 - val_accuracy: 0.7464
Epoch 56/80
889/889 [==============================] - 1s 840us/step - loss: 0.4988 - accuracy:
```

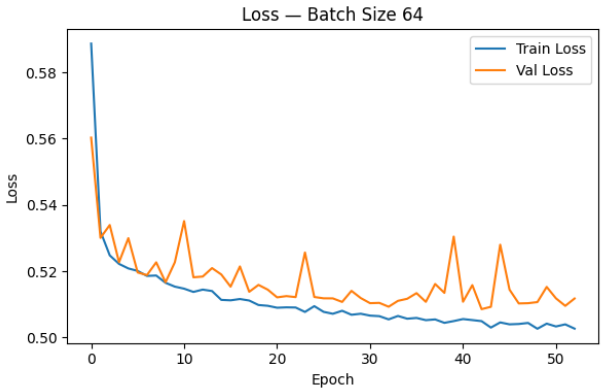
```
0.7556 - val_loss: 0.5089 - val_accuracy: 0.7521
Epoch 57/80
889/889 [==============================] - 1s 834us/step - loss: 0.4988 - accuracy:
0.7573 - val_loss: 0.5129 - val_accuracy: 0.7495
Epoch 58/80
889/889 [==============================] - 1s 831us/step - loss: 0.4985 - accuracy:
0.7557 - val_loss: 0.5139 - val_accuracy: 0.7487
Max validation accuracy for batch size 32: 0.7532
```

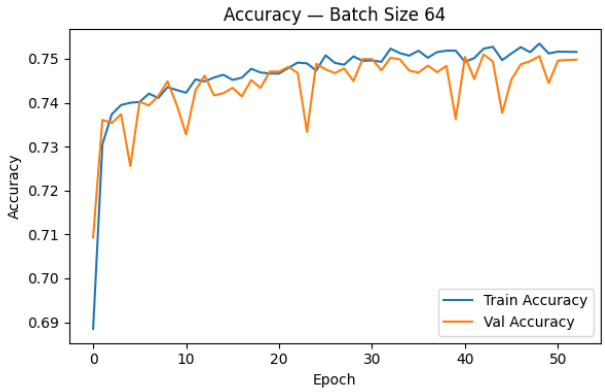


```
Training with batch size: 64
Epoch 1/80
445/445 [==============================] - 1s 1ms/step - loss: 0.5887 - accuracy: 0.
6885 - val_loss: 0.5603 - val_accuracy: 0.7094
Epoch 2/80
445/445 [==============================] - 0s 850us/step - loss: 0.5320 - accuracy:
0.7305 - val_loss: 0.5301 - val_accuracy: 0.7361
Epoch 3/80
445/445 [==============================] - 0s 867us/step - loss: 0.5248 - accuracy:
0.7373 - val_loss: 0.5339 - val_accuracy: 0.7354
Epoch 4/80
445/445 [==============================] - 0s 861us/step - loss: 0.5222 - accuracy:
0.7395 - val_loss: 0.5227 - val_accuracy: 0.7374
Epoch 5/80
445/445 [==============================] - 0s 865us/step - loss: 0.5208 - accuracy:
0.7400 - val_loss: 0.5300 - val_accuracy: 0.7256
Epoch 6/80
445/445 [==============================] - 0s 866us/step - loss: 0.5201 - accuracy:
0.7402 - val_loss: 0.5196 - val_accuracy: 0.7402
Epoch 7/80
445/445 [==============================] - 0s 862us/step - loss: 0.5186 - accuracy:
0.7421 - val_loss: 0.5188 - val_accuracy: 0.7394
Epoch 8/80
445/445 [==============================] - 0s 866us/step - loss: 0.5187 - accuracy:
0.7411 - val_loss: 0.5227 - val_accuracy: 0.7415
Epoch 9/80
445/445 [==============================] - 0s 849us/step - loss: 0.5165 - accuracy:
0.7435 - val_loss: 0.5167 - val_accuracy: 0.7449
Epoch 10/80
445/445 [==============================] - 0s 872us/step - loss: 0.5154 - accuracy:
0.7429 - val_loss: 0.5227 - val_accuracy: 0.7394
Epoch 11/80
445/445 [==============================] - 0s 849us/step - loss: 0.5147 - accuracy:
0.7423 - val_loss: 0.5351 - val_accuracy: 0.7327
Epoch 12/80
445/445 [==============================] - 0s 868us/step - loss: 0.5138 - accuracy:
0.7453 - val_loss: 0.5182 - val_accuracy: 0.7428
Epoch 13/80
445/445 [==============================] - 0s 862us/step - loss: 0.5145 - accuracy:
0.7448 - val_loss: 0.5184 - val_accuracy: 0.7462
Epoch 14/80
445/445 [==============================] - 0s 855us/step - loss: 0.5140 - accuracy:
0.7457 - val_loss: 0.5209 - val_accuracy: 0.7417
Epoch 15/80
445/445 [==============================] - 0s 853us/step - loss: 0.5114 - accuracy:
0.7464 - val_loss: 0.5190 - val_accuracy: 0.7422
Epoch 16/80
445/445 [==============================] - 0s 847us/step - loss: 0.5112 - accuracy:
0.7452 - val_loss: 0.5153 - val_accuracy: 0.7434
Epoch 17/80
445/445 [==============================] - 0s 872us/step - loss: 0.5116 - accuracy:
0.7457 - val_loss: 0.5214 - val_accuracy: 0.7414
Epoch 18/80
445/445 [==============================] - 0s 857us/step - loss: 0.5112 - accuracy:
0.7477 - val_loss: 0.5138 - val_accuracy: 0.7452
Epoch 19/80
```

```
445/445 [==============================] - 0s 868us/step - loss: 0.5098 - accuracy:
0.7469 - val_loss: 0.5159 - val_accuracy: 0.7434
Epoch 20/80
445/445 [==============================] - 0s 853us/step - loss: 0.5096 - accuracy:
0.7466 - val_loss: 0.5144 - val_accuracy: 0.7471
Epoch 21/80
445/445 [==============================] - 0s 862us/step - loss: 0.5090 - accuracy:
0.7466 - val_loss: 0.5121 - val_accuracy: 0.7471
Epoch 22/80
445/445 [==============================] - 0s 860us/step - loss: 0.5091 - accuracy:
0.7479 - val_loss: 0.5125 - val_accuracy: 0.7482
Epoch 23/80
445/445 [==============================] - 0s 846us/step - loss: 0.5091 - accuracy:
0.7491 - val_loss: 0.5122 - val_accuracy: 0.7468
Epoch 24/80
445/445 [==============================] - 0s 850us/step - loss: 0.5077 - accuracy:
0.7490 - val_loss: 0.5256 - val_accuracy: 0.7334
Epoch 25/80
445/445 [==============================] - 0s 849us/step - loss: 0.5095 - accuracy:
0.7473 - val_loss: 0.5122 - val_accuracy: 0.7489
Epoch 26/80
445/445 [==============================] - 0s 866us/step - loss: 0.5078 - accuracy:
0.7508 - val_loss: 0.5118 - val_accuracy: 0.7476
Epoch 27/80
445/445 [==============================] - 0s 861us/step - loss: 0.5072 - accuracy:
0.7491 - val_loss: 0.5118 - val_accuracy: 0.7467
Epoch 28/80
445/445 [==============================] - 0s 861us/step - loss: 0.5081 - accuracy:
0.7486 - val_loss: 0.5107 - val_accuracy: 0.7478
Epoch 29/80
445/445 [==============================] - 0s 850us/step - loss: 0.5069 - accuracy:
0.7505 - val_loss: 0.5141 - val_accuracy: 0.7449
Epoch 30/80
445/445 [==============================] - 0s 846us/step - loss: 0.5072 - accuracy:
0.7495 - val_loss: 0.5119 - val_accuracy: 0.7499
Epoch 31/80
445/445 [==============================] - 0s 842us/step - loss: 0.5066 - accuracy:
0.7496 - val_loss: 0.5104 - val_accuracy: 0.7499
Epoch 32/80
445/445 [==============================] - 0s 851us/step - loss: 0.5064 - accuracy:
0.7493 - val_loss: 0.5105 - val_accuracy: 0.7474
Epoch 33/80
445/445 [==============================] - 0s 845us/step - loss: 0.5055 - accuracy:
0.7523 - val_loss: 0.5093 - val_accuracy: 0.7502
Epoch 34/80
445/445 [==============================] - 0s 851us/step - loss: 0.5065 - accuracy:
0.7513 - val_loss: 0.5111 - val_accuracy: 0.7499
Epoch 35/80
445/445 [==============================] - 0s 861us/step - loss: 0.5057 - accuracy:
0.7507 - val_loss: 0.5117 - val_accuracy: 0.7473
Epoch 36/80
445/445 [==============================] - 0s 858us/step - loss: 0.5059 - accuracy:
0.7518 - val_loss: 0.5134 - val_accuracy: 0.7469
Epoch 37/80
445/445 [==============================] - 0s 875us/step - loss: 0.5052 - accuracy:
0.7502 - val_loss: 0.5108 - val_accuracy: 0.7484
```

```
Epoch 38/80
445/445 [==============================] - 0s 869us/step - loss: 0.5055 - accuracy:
0.7515 - val_loss: 0.5162 - val_accuracy: 0.7469
Epoch 39/80
445/445 [==============================] - 0s 849us/step - loss: 0.5044 - accuracy:
0.7518 - val_loss: 0.5135 - val_accuracy: 0.7484
Epoch 40/80
445/445 [==============================] - 0s 856us/step - loss: 0.5050 - accuracy:
0.7518 - val_loss: 0.5304 - val_accuracy: 0.7363
Epoch 41/80
445/445 [==============================] - 0s 849us/step - loss: 0.5056 - accuracy:
0.7493 - val_loss: 0.5108 - val_accuracy: 0.7504
Epoch 42/80
445/445 [==============================] - 0s 842us/step - loss: 0.5053 - accuracy:
0.7502 - val_loss: 0.5158 - val_accuracy: 0.7454
Epoch 43/80
445/445 [==============================] - 0s 852us/step - loss: 0.5049 - accuracy:
0.7523 - val_loss: 0.5086 - val_accuracy: 0.7510
Epoch 44/80
445/445 [==============================] - 0s 856us/step - loss: 0.5030 - accuracy:
0.7527 - val_loss: 0.5093 - val_accuracy: 0.7494
Epoch 45/80
445/445 [==============================] - 0s 820us/step - loss: 0.5046 - accuracy:
0.7497 - val_loss: 0.5280 - val_accuracy: 0.7377
Epoch 46/80
445/445 [==============================] - 0s 857us/step - loss: 0.5040 - accuracy:
0.7512 - val_loss: 0.5145 - val_accuracy: 0.7453
Epoch 47/80
445/445 [==============================] - 0s 857us/step - loss: 0.5041 - accuracy:
0.7527 - val_loss: 0.5103 - val_accuracy: 0.7487
Epoch 48/80
445/445 [==============================] - 0s 854us/step - loss: 0.5044 - accuracy:
0.7515 - val_loss: 0.5104 - val_accuracy: 0.7494
Epoch 49/80
445/445 [==============================] - 0s 857us/step - loss: 0.5027 - accuracy:
0.7535 - val_loss: 0.5107 - val_accuracy: 0.7506
Epoch 50/80
445/445 [==============================] - 0s 852us/step - loss: 0.5042 - accuracy:
0.7512 - val_loss: 0.5153 - val_accuracy: 0.7444
Epoch 51/80
445/445 [==============================] - 0s 855us/step - loss: 0.5033 - accuracy:
0.7516 - val_loss: 0.5118 - val_accuracy: 0.7496
Epoch 52/80
445/445 [==============================] - 0s 820us/step - loss: 0.5040 - accuracy:
0.7516 - val_loss: 0.5096 - val_accuracy: 0.7497
Epoch 53/80
445/445 [==============================] - 0s 887us/step - loss: 0.5027 - accuracy:
0.7516 - val_loss: 0.5118 - val_accuracy: 0.7498
Max validation accuracy for batch size 64: 0.7510
```


Loss — Batch Size 64
Train Loss
Val Loss
Loss
Epoch
0.58
0.56
0.54
0.52
0.50
0
10
20
30
40
50

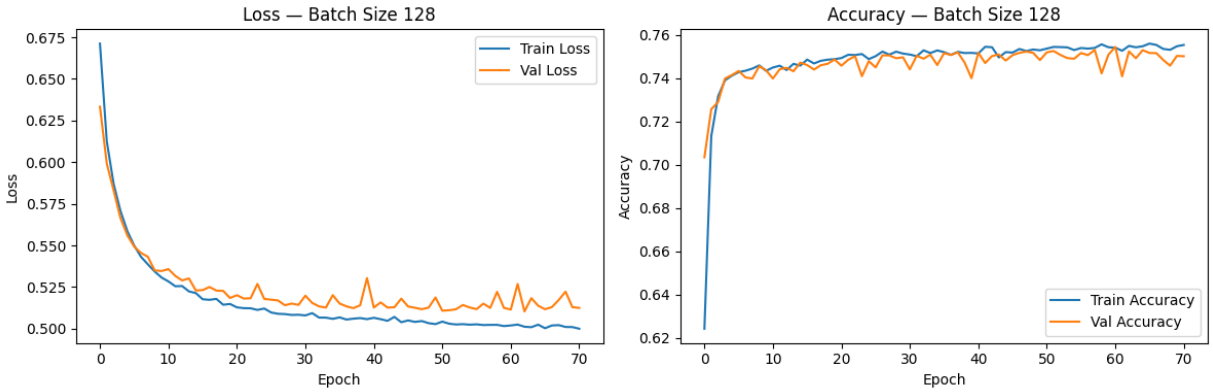

Accuracy — Batch Size 64
Train Accuracy
Val Accuracy
Accuracy
Epoch
0.75
0.74
0.73
0.72
0.71
0.70
0.69
0
10
20
30
40
50

```
Training with batch size: 128
Epoch 1/80
223/223 [==============================] - 1s 1ms/step - loss: 0.6712 - accuracy: 0.
6243 - val_loss: 0.6333 - val_accuracy: 0.7034
Epoch 2/80
223/223 [==============================] - 0s 979us/step - loss: 0.6125 - accuracy:
0.7132 - val_loss: 0.5992 - val_accuracy: 0.7257
Epoch 3/80
223/223 [==============================] - 0s 926us/step - loss: 0.5870 - accuracy:
0.7316 - val_loss: 0.5830 - val_accuracy: 0.7290
Epoch 4/80
223/223 [==============================] - 0s 858us/step - loss: 0.5707 - accuracy:
0.7390 - val_loss: 0.5664 - val_accuracy: 0.7398
Epoch 5/80
223/223 [==============================] - 0s 855us/step - loss: 0.5586 - accuracy:
0.7411 - val_loss: 0.5560 - val_accuracy: 0.7414
Epoch 6/80
223/223 [==============================] - 0s 924us/step - loss: 0.5499 - accuracy:
0.7427 - val_loss: 0.5493 - val_accuracy: 0.7433
Epoch 7/80
223/223 [==============================] - 0s 857us/step - loss: 0.5431 - accuracy:
0.7434 - val_loss: 0.5455 - val_accuracy: 0.7403
Epoch 8/80
223/223 [==============================] - 0s 925us/step - loss: 0.5386 - accuracy:
0.7444 - val_loss: 0.5432 - val_accuracy: 0.7398
Epoch 9/80
223/223 [==============================] - 0s 855us/step - loss: 0.5343 - accuracy:
0.7459 - val_loss: 0.5350 - val_accuracy: 0.7455
Epoch 10/80
223/223 [==============================] - 0s 855us/step - loss: 0.5307 - accuracy:
0.7434 - val_loss: 0.5346 - val_accuracy: 0.7436
Epoch 11/80
223/223 [==============================] - 0s 928us/step - loss: 0.5284 - accuracy:
0.7449 - val_loss: 0.5358 - val_accuracy: 0.7398
Epoch 12/80
223/223 [==============================] - 0s 855us/step - loss: 0.5255 - accuracy:
0.7457 - val_loss: 0.5317 - val_accuracy: 0.7441
Epoch 13/80
223/223 [==============================] - 0s 927us/step - loss: 0.5256 - accuracy:
0.7437 - val_loss: 0.5290 - val_accuracy: 0.7447
Epoch 14/80
223/223 [==============================] - 0s 924us/step - loss: 0.5224 - accuracy:
0.7466 - val_loss: 0.5302 - val_accuracy: 0.7432
Epoch 15/80
223/223 [==============================] - 0s 854us/step - loss: 0.5214 - accuracy:
0.7458 - val_loss: 0.5229 - val_accuracy: 0.7472
Epoch 16/80
223/223 [==============================] - 0s 928us/step - loss: 0.5177 - accuracy:
0.7485 - val_loss: 0.5232 - val_accuracy: 0.7460
Epoch 17/80
223/223 [==============================] - 0s 929us/step - loss: 0.5173 - accuracy:
0.7467 - val_loss: 0.5250 - val_accuracy: 0.7439
Epoch 18/80
223/223 [==============================] - 0s 855us/step - loss: 0.5179 - accuracy:
0.7480 - val_loss: 0.5229 - val_accuracy: 0.7459
Epoch 19/80
```

```
223/223 [==============================] - 0s 925us/step - loss: 0.5144 - accuracy:
0.7485 - val_loss: 0.5227 - val_accuracy: 0.7466
Epoch 20/80
223/223 [==============================] - 0s 926us/step - loss: 0.5148 - accuracy:
0.7487 - val_loss: 0.5184 - val_accuracy: 0.7486
Epoch 21/80
223/223 [==============================] - 0s 856us/step - loss: 0.5128 - accuracy:
0.7492 - val_loss: 0.5201 - val_accuracy: 0.7457
Epoch 22/80
223/223 [==============================] - 0s 858us/step - loss: 0.5123 - accuracy:
0.7508 - val_loss: 0.5181 - val_accuracy: 0.7484
Epoch 23/80
223/223 [==============================] - 0s 924us/step - loss: 0.5122 - accuracy:
0.7507 - val_loss: 0.5183 - val_accuracy: 0.7501
Epoch 24/80
223/223 [==============================] - 0s 932us/step - loss: 0.5113 - accuracy:
0.7511 - val_loss: 0.5269 - val_accuracy: 0.7409
Epoch 25/80
223/223 [==============================] - 0s 890us/step - loss: 0.5121 - accuracy:
0.7488 - val_loss: 0.5179 - val_accuracy: 0.7478
Epoch 26/80
223/223 [==============================] - 0s 928us/step - loss: 0.5097 - accuracy:
0.7501 - val_loss: 0.5174 - val_accuracy: 0.7449
Epoch 27/80
223/223 [==============================] - 0s 926us/step - loss: 0.5089 - accuracy:
0.7522 - val_loss: 0.5170 - val_accuracy: 0.7505
Epoch 28/80
223/223 [==============================] - 0s 860us/step - loss: 0.5088 - accuracy:
0.7508 - val_loss: 0.5141 - val_accuracy: 0.7504
Epoch 29/80
223/223 [==============================] - 0s 931us/step - loss: 0.5082 - accuracy:
0.7522 - val_loss: 0.5151 - val_accuracy: 0.7492
Epoch 30/80
223/223 [==============================] - 0s 930us/step - loss: 0.5084 - accuracy:
0.7513 - val_loss: 0.5143 - val_accuracy: 0.7496
Epoch 31/80
223/223 [==============================] - 0s 858us/step - loss: 0.5079 - accuracy:
0.7509 - val_loss: 0.5198 - val_accuracy: 0.7440
Epoch 32/80
223/223 [==============================] - 0s 926us/step - loss: 0.5093 - accuracy:
0.7501 - val_loss: 0.5155 - val_accuracy: 0.7503
Epoch 33/80
223/223 [==============================] - 0s 926us/step - loss: 0.5067 - accuracy:
0.7528 - val_loss: 0.5134 - val_accuracy: 0.7489
Epoch 34/80
223/223 [==============================] - 0s 926us/step - loss: 0.5066 - accuracy:
0.7516 - val_loss: 0.5127 - val_accuracy: 0.7508
Epoch 35/80
223/223 [==============================] - 0s 925us/step - loss: 0.5059 - accuracy:
0.7528 - val_loss: 0.5201 - val_accuracy: 0.7460
Epoch 36/80
223/223 [==============================] - 0s 901us/step - loss: 0.5068 - accuracy:
0.7520 - val_loss: 0.5151 - val_accuracy: 0.7515
Epoch 37/80
223/223 [==============================] - 0s 901us/step - loss: 0.5055 - accuracy:
0.7507 - val_loss: 0.5134 - val_accuracy: 0.7507
```

```
Epoch 38/80
223/223 [==============================] - 0s 892us/step - loss: 0.5060 - accuracy:
0.7521 - val_loss: 0.5123 - val_accuracy: 0.7519
Epoch 39/80
223/223 [==============================] - 0s 902us/step - loss: 0.5064 - accuracy:
0.7516 - val_loss: 0.5140 - val_accuracy: 0.7472
Epoch 40/80
223/223 [==============================] - 0s 901us/step - loss: 0.5057 - accuracy:
0.7516 - val_loss: 0.5304 - val_accuracy: 0.7399
Epoch 41/80
223/223 [==============================] - 0s 884us/step - loss: 0.5065 - accuracy:
0.7514 - val_loss: 0.5127 - val_accuracy: 0.7517
Epoch 42/80
223/223 [==============================] - 0s 903us/step - loss: 0.5057 - accuracy:
0.7545 - val_loss: 0.5158 - val_accuracy: 0.7469
Epoch 43/80
223/223 [==============================] - 0s 922us/step - loss: 0.5047 - accuracy:
0.7542 - val_loss: 0.5126 - val_accuracy: 0.7502
Epoch 44/80
223/223 [==============================] - 0s 880us/step - loss: 0.5071 - accuracy:
0.7495 - val_loss: 0.5129 - val_accuracy: 0.7509
Epoch 45/80
223/223 [==============================] - 0s 907us/step - loss: 0.5039 - accuracy:
0.7520 - val_loss: 0.5180 - val_accuracy: 0.7481
Epoch 46/80
223/223 [==============================] - 0s 897us/step - loss: 0.5050 - accuracy:
0.7518 - val_loss: 0.5133 - val_accuracy: 0.7506
Epoch 47/80
223/223 [==============================] - 0s 907us/step - loss: 0.5041 - accuracy:
0.7534 - val_loss: 0.5125 - val_accuracy: 0.7517
Epoch 48/80
223/223 [==============================] - 0s 881us/step - loss: 0.5046 - accuracy:
0.7525 - val_loss: 0.5117 - val_accuracy: 0.7522
Epoch 49/80
223/223 [==============================] - 0s 923us/step - loss: 0.5033 - accuracy:
0.7532 - val_loss: 0.5127 - val_accuracy: 0.7517
Epoch 50/80
223/223 [==============================] - 0s 926us/step - loss: 0.5027 - accuracy:
0.7528 - val_loss: 0.5187 - val_accuracy: 0.7483
Epoch 51/80
223/223 [==============================] - 0s 929us/step - loss: 0.5042 - accuracy:
0.7536 - val_loss: 0.5108 - val_accuracy: 0.7519
Epoch 52/80
223/223 [==============================] - 0s 926us/step - loss: 0.5030 - accuracy:
0.7544 - val_loss: 0.5111 - val_accuracy: 0.7525
Epoch 53/80
223/223 [==============================] - 0s 923us/step - loss: 0.5025 - accuracy:
0.7543 - val_loss: 0.5117 - val_accuracy: 0.7507
Epoch 54/80
223/223 [==============================] - 0s 855us/step - loss: 0.5027 - accuracy:
0.7542 - val_loss: 0.5142 - val_accuracy: 0.7492
Epoch 55/80
223/223 [==============================] - 0s 855us/step - loss: 0.5024 - accuracy:
0.7529 - val_loss: 0.5127 - val_accuracy: 0.7489
Epoch 56/80
223/223 [==============================] - 0s 925us/step - loss: 0.5026 - accuracy:
```

```
0.7539 - val_loss: 0.5117 - val_accuracy: 0.7516
Epoch 57/80
223/223 [==============================] - 0s 855us/step - loss: 0.5022 - accuracy:
0.7536 - val_loss: 0.5150 - val_accuracy: 0.7506
Epoch 58/80
223/223 [==============================] - 0s 931us/step - loss: 0.5023 - accuracy:
0.7539 - val_loss: 0.5125 - val_accuracy: 0.7530
Epoch 59/80
223/223 [==============================] - 0s 858us/step - loss: 0.5023 - accuracy:
0.7556 - val_loss: 0.5221 - val_accuracy: 0.7422
Epoch 60/80
223/223 [==============================] - 0s 925us/step - loss: 0.5016 - accuracy:
0.7542 - val_loss: 0.5125 - val_accuracy: 0.7508
Epoch 61/80
223/223 [==============================] - 0s 857us/step - loss: 0.5019 - accuracy:
0.7540 - val_loss: 0.5115 - val_accuracy: 0.7544
Epoch 62/80
223/223 [==============================] - 0s 956us/step - loss: 0.5025 - accuracy:
0.7526 - val_loss: 0.5268 - val_accuracy: 0.7408
Epoch 63/80
223/223 [==============================] - 0s 898us/step - loss: 0.5012 - accuracy:
0.7549 - val_loss: 0.5104 - val_accuracy: 0.7523
Epoch 64/80
223/223 [==============================] - 0s 926us/step - loss: 0.5009 - accuracy:
0.7542 - val_loss: 0.5183 - val_accuracy: 0.7492
Epoch 65/80
223/223 [==============================] - 0s 859us/step - loss: 0.5024 - accuracy:
0.7547 - val_loss: 0.5138 - val_accuracy: 0.7529
Epoch 66/80
223/223 [==============================] - 0s 925us/step - loss: 0.5003 - accuracy:
0.7559 - val_loss: 0.5117 - val_accuracy: 0.7516
Epoch 67/80
223/223 [==============================] - 0s 852us/step - loss: 0.5019 - accuracy:
0.7553 - val_loss: 0.5129 - val_accuracy: 0.7515
Epoch 68/80
223/223 [==============================] - 0s 929us/step - loss: 0.5021 - accuracy:
0.7535 - val_loss: 0.5172 - val_accuracy: 0.7484
Epoch 69/80
223/223 [==============================] - 0s 854us/step - loss: 0.5010 - accuracy:
0.7530 - val_loss: 0.5222 - val_accuracy: 0.7457
Epoch 70/80
223/223 [==============================] - 0s 854us/step - loss: 0.5010 - accuracy:
0.7547 - val_loss: 0.5129 - val_accuracy: 0.7502
Epoch 71/80
223/223 [==============================] - 0s 925us/step - loss: 0.5000 - accuracy:
0.7553 - val_loss: 0.5125 - val_accuracy: 0.7501
Max validation accuracy for batch size 128: 0.7544
```

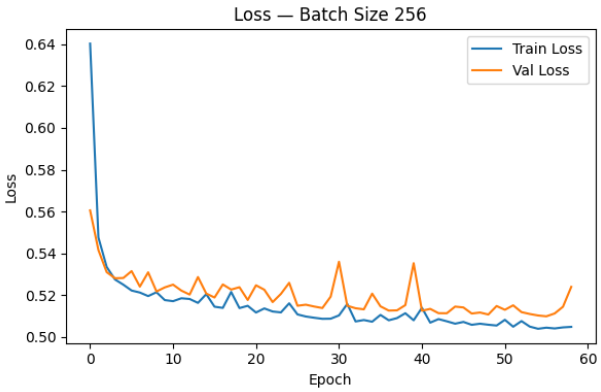

Loss — Batch Size 128
Train Loss
Val Loss
Loss
Epoch
Accuracy — Batch Size 128
Train Accuracy
Val Accuracy
Accuracy
Epoch

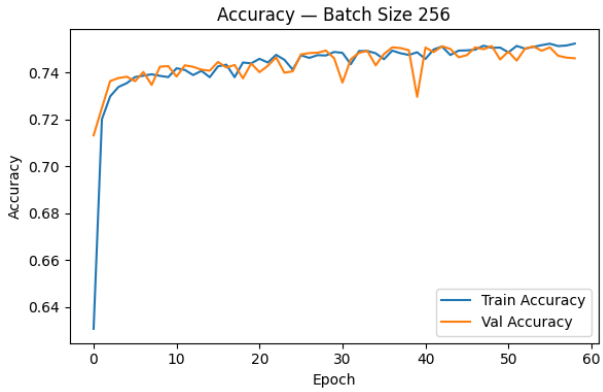
```
Training with batch size: 256
Epoch 1/80
112/112 [==============================] - 1s 2ms/step - loss: 0.6403 - accuracy: 0.
6307 - val_loss: 0.5606 - val_accuracy: 0.7132
Epoch 2/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5476 - accuracy: 0.
7201 - val_loss: 0.5416 - val_accuracy: 0.7249
Epoch 3/80
112/112 [==============================] - 0s 988us/step - loss: 0.5335 - accuracy:
0.7298 - val_loss: 0.5310 - val_accuracy: 0.7363
Epoch 4/80
112/112 [==============================] - 0s 993us/step - loss: 0.5275 - accuracy:
0.7338 - val_loss: 0.5281 - val_accuracy: 0.7376
Epoch 5/80
112/112 [==============================] - 0s 995us/step - loss: 0.5251 - accuracy:
0.7354 - val_loss: 0.5282 - val_accuracy: 0.7382
Epoch 6/80
112/112 [==============================] - 0s 997us/step - loss: 0.5222 - accuracy:
0.7380 - val_loss: 0.5315 - val_accuracy: 0.7362
Epoch 7/80
112/112 [==============================] - 0s 992us/step - loss: 0.5213 - accuracy:
0.7386 - val_loss: 0.5241 - val_accuracy: 0.7402
Epoch 8/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5196 - accuracy: 0.
7392 - val_loss: 0.5310 - val_accuracy: 0.7347
Epoch 9/80
112/112 [==============================] - 0s 934us/step - loss: 0.5214 - accuracy:
0.7385 - val_loss: 0.5218 - val_accuracy: 0.7424
Epoch 10/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5177 - accuracy: 0.
7380 - val_loss: 0.5237 - val_accuracy: 0.7427
Epoch 11/80
112/112 [==============================] - 0s 1000us/step - loss: 0.5172 - accuracy:
0.7417 - val_loss: 0.5251 - val_accuracy: 0.7382
Epoch 12/80
112/112 [==============================] - 0s 995us/step - loss: 0.5185 - accuracy:
0.7410 - val_loss: 0.5222 - val_accuracy: 0.7430
Epoch 13/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5182 - accuracy: 0.
7389 - val_loss: 0.5203 - val_accuracy: 0.7424
Epoch 14/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5163 - accuracy: 0.
7407 - val_loss: 0.5287 - val_accuracy: 0.7412
Epoch 15/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5205 - accuracy: 0.
7379 - val_loss: 0.5208 - val_accuracy: 0.7407
Epoch 16/80
112/112 [==============================] - 0s 996us/step - loss: 0.5145 - accuracy:
0.7426 - val_loss: 0.5189 - val_accuracy: 0.7444
Epoch 17/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5139 - accuracy: 0.
7433 - val_loss: 0.5252 - val_accuracy: 0.7421
Epoch 18/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5216 - accuracy: 0.
7379 - val_loss: 0.5227 - val_accuracy: 0.7432
Epoch 19/80
```

```
112/112 [==============================] - 0s 995us/step - loss: 0.5139 - accuracy:
0.7442 - val_loss: 0.5238 - val_accuracy: 0.7374
Epoch 20/80
112/112 [==============================] - 0s 904us/step - loss: 0.5149 - accuracy:
0.7439 - val_loss: 0.5177 - val_accuracy: 0.7438
Epoch 21/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5117 - accuracy: 0.
7458 - val_loss: 0.5248 - val_accuracy: 0.7401
Epoch 22/80
112/112 [==============================] - 0s 947us/step - loss: 0.5137 - accuracy:
0.7442 - val_loss: 0.5226 - val_accuracy: 0.7428
Epoch 23/80
112/112 [==============================] - 0s 1000us/step - loss: 0.5122 - accuracy:
0.7475 - val_loss: 0.5167 - val_accuracy: 0.7464
Epoch 24/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5118 - accuracy: 0.
7454 - val_loss: 0.5206 - val_accuracy: 0.7399
Epoch 25/80
112/112 [==============================] - 0s 995us/step - loss: 0.5161 - accuracy:
0.7412 - val_loss: 0.5259 - val_accuracy: 0.7404
Epoch 26/80
112/112 [==============================] - 0s 990us/step - loss: 0.5108 - accuracy:
0.7473 - val_loss: 0.5150 - val_accuracy: 0.7477
Epoch 27/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5098 - accuracy: 0.
7462 - val_loss: 0.5155 - val_accuracy: 0.7482
Epoch 28/80
112/112 [==============================] - 0s 990us/step - loss: 0.5092 - accuracy:
0.7473 - val_loss: 0.5146 - val_accuracy: 0.7484
Epoch 29/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5087 - accuracy: 0.
7472 - val_loss: 0.5139 - val_accuracy: 0.7493
Epoch 30/80
112/112 [==============================] - 0s 998us/step - loss: 0.5088 - accuracy:
0.7487 - val_loss: 0.5193 - val_accuracy: 0.7460
Epoch 31/80
112/112 [==============================] - 0s 994us/step - loss: 0.5104 - accuracy:
0.7483 - val_loss: 0.5360 - val_accuracy: 0.7357
Epoch 32/80
112/112 [==============================] - 0s 999us/step - loss: 0.5158 - accuracy:
0.7435 - val_loss: 0.5150 - val_accuracy: 0.7456
Epoch 33/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5075 - accuracy: 0.
7491 - val_loss: 0.5138 - val_accuracy: 0.7484
Epoch 34/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5081 - accuracy: 0.
7491 - val_loss: 0.5133 - val_accuracy: 0.7492
Epoch 35/80
112/112 [==============================] - 0s 998us/step - loss: 0.5073 - accuracy:
0.7482 - val_loss: 0.5208 - val_accuracy: 0.7430
Epoch 36/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5106 - accuracy: 0.
7456 - val_loss: 0.5147 - val_accuracy: 0.7478
Epoch 37/80
112/112 [==============================] - 0s 995us/step - loss: 0.5080 - accuracy:
0.7493 - val_loss: 0.5127 - val_accuracy: 0.7507
```

```
Epoch 38/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5091 - accuracy: 0.
7482 - val_loss: 0.5128 - val_accuracy: 0.7504
Epoch 39/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5114 - accuracy: 0.
7474 - val_loss: 0.5153 - val_accuracy: 0.7494
Epoch 40/80
112/112 [==============================] - 0s 963us/step - loss: 0.5080 - accuracy:
0.7486 - val_loss: 0.5353 - val_accuracy: 0.7296
Epoch 41/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5140 - accuracy: 0.
7457 - val_loss: 0.5126 - val_accuracy: 0.7506
Epoch 42/80
112/112 [==============================] - 0s 998us/step - loss: 0.5069 - accuracy:
0.7498 - val_loss: 0.5135 - val_accuracy: 0.7487
Epoch 43/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5086 - accuracy: 0.
7510 - val_loss: 0.5114 - val_accuracy: 0.7512
Epoch 44/80
112/112 [==============================] - 0s 995us/step - loss: 0.5075 - accuracy:
0.7474 - val_loss: 0.5114 - val_accuracy: 0.7500
Epoch 45/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5064 - accuracy: 0.
7493 - val_loss: 0.5146 - val_accuracy: 0.7464
Epoch 46/80
112/112 [==============================] - 0s 936us/step - loss: 0.5072 - accuracy:
0.7494 - val_loss: 0.5142 - val_accuracy: 0.7474
Epoch 47/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5059 - accuracy: 0.
7497 - val_loss: 0.5113 - val_accuracy: 0.7507
Epoch 48/80
112/112 [==============================] - 0s 993us/step - loss: 0.5064 - accuracy:
0.7514 - val_loss: 0.5117 - val_accuracy: 0.7499
Epoch 49/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5059 - accuracy: 0.
7505 - val_loss: 0.5108 - val_accuracy: 0.7512
Epoch 50/80
112/112 [==============================] - 0s 997us/step - loss: 0.5055 - accuracy:
0.7506 - val_loss: 0.5148 - val_accuracy: 0.7455
Epoch 51/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5083 - accuracy: 0.
7485 - val_loss: 0.5130 - val_accuracy: 0.7489
Epoch 52/80
112/112 [==============================] - 0s 902us/step - loss: 0.5050 - accuracy:
0.7513 - val_loss: 0.5152 - val_accuracy: 0.7450
Epoch 53/80
112/112 [==============================] - 0s 998us/step - loss: 0.5076 - accuracy:
0.7501 - val_loss: 0.5119 - val_accuracy: 0.7501
Epoch 54/80
112/112 [==============================] - 0s 999us/step - loss: 0.5050 - accuracy:
0.7509 - val_loss: 0.5110 - val_accuracy: 0.7511
Epoch 55/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5040 - accuracy: 0.
7516 - val_loss: 0.5103 - val_accuracy: 0.7492
Epoch 56/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5045 - accuracy: 0.
```

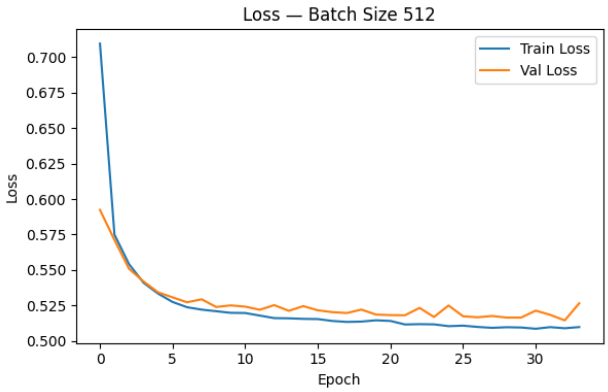
```
7522 - val_loss: 0.5099 - val_accuracy: 0.7507
Epoch 57/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5041 - accuracy: 0.
7512 - val_loss: 0.5113 - val_accuracy: 0.7471
Epoch 58/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5046 - accuracy: 0.
7515 - val_loss: 0.5145 - val_accuracy: 0.7463
Epoch 59/80
112/112 [==============================] - 0s 1ms/step - loss: 0.5049 - accuracy: 0.
7523 - val_loss: 0.5240 - val_accuracy: 0.7460
Max validation accuracy for batch size 256: 0.7512
```

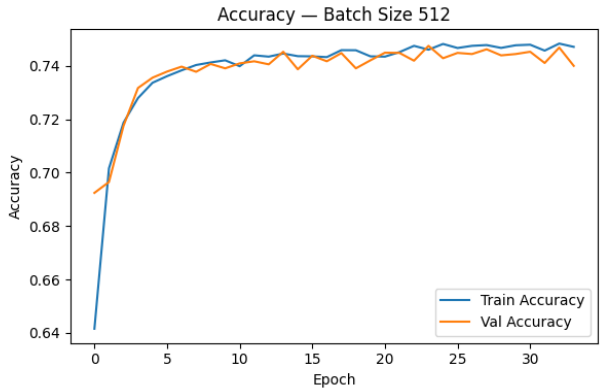




```
Training with batch size: 512
Epoch 1/80
56/56 [==============================] - 1s 3ms/step - loss: 0.7095 - accuracy: 0.64
15 - val_loss: 0.5924 - val_accuracy: 0.6924
Epoch 2/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5750 - accuracy: 0.70
16 - val_loss: 0.5712 - val_accuracy: 0.6965
Epoch 3/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5542 - accuracy: 0.71
87 - val_loss: 0.5509 - val_accuracy: 0.7175
Epoch 4/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5411 - accuracy: 0.72
80 - val_loss: 0.5420 - val_accuracy: 0.7317
Epoch 5/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5334 - accuracy: 0.73
37 - val_loss: 0.5343 - val_accuracy: 0.7356
Epoch 6/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5275 - accuracy: 0.73
62 - val_loss: 0.5308 - val_accuracy: 0.7379
Epoch 7/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5238 - accuracy: 0.73
83 - val_loss: 0.5273 - val_accuracy: 0.7397
Epoch 8/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5221 - accuracy: 0.74
03 - val_loss: 0.5294 - val_accuracy: 0.7378
Epoch 9/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5210 - accuracy: 0.74
13 - val_loss: 0.5240 - val_accuracy: 0.7407
Epoch 10/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5199 - accuracy: 0.74
21 - val_loss: 0.5251 - val_accuracy: 0.7391
Epoch 11/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5198 - accuracy: 0.73
99 - val_loss: 0.5242 - val_accuracy: 0.7409
Epoch 12/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5179 - accuracy: 0.74
40 - val_loss: 0.5220 - val_accuracy: 0.7417
Epoch 13/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5162 - accuracy: 0.74
35 - val_loss: 0.5253 - val_accuracy: 0.7406
Epoch 14/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5160 - accuracy: 0.74
46 - val_loss: 0.5213 - val_accuracy: 0.7453
Epoch 15/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5156 - accuracy: 0.74
36 - val_loss: 0.5246 - val_accuracy: 0.7387
Epoch 16/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5155 - accuracy: 0.74
36 - val_loss: 0.5217 - val_accuracy: 0.7438
Epoch 17/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5142 - accuracy: 0.74
33 - val_loss: 0.5204 - val_accuracy: 0.7417
Epoch 18/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5135 - accuracy: 0.74
59 - val_loss: 0.5197 - val_accuracy: 0.7448
Epoch 19/80
```

```
56/56 [==============================] - 0s 1ms/step - loss: 0.5137 - accuracy: 0.74
59 - val_loss: 0.5222 - val_accuracy: 0.7391
Epoch 20/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5146 - accuracy: 0.74
36 - val_loss: 0.5187 - val_accuracy: 0.7421
Epoch 21/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5142 - accuracy: 0.74
35 - val_loss: 0.5183 - val_accuracy: 0.7449
Epoch 22/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5117 - accuracy: 0.74
52 - val_loss: 0.5182 - val_accuracy: 0.7449
Epoch 23/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5119 - accuracy: 0.74
75 - val_loss: 0.5233 - val_accuracy: 0.7419
Epoch 24/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5118 - accuracy: 0.74
61 - val_loss: 0.5170 - val_accuracy: 0.7475
Epoch 25/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5105 - accuracy: 0.74
82 - val_loss: 0.5250 - val_accuracy: 0.7429
Epoch 26/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5108 - accuracy: 0.74
67 - val_loss: 0.5174 - val_accuracy: 0.7449
Epoch 27/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5099 - accuracy: 0.74
75 - val_loss: 0.5167 - val_accuracy: 0.7444
Epoch 28/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5093 - accuracy: 0.74
78 - val_loss: 0.5177 - val_accuracy: 0.7462
Epoch 29/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5097 - accuracy: 0.74
67 - val_loss: 0.5165 - val_accuracy: 0.7439
Epoch 30/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5095 - accuracy: 0.74
77 - val_loss: 0.5166 - val_accuracy: 0.7444
Epoch 31/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5087 - accuracy: 0.74
79 - val_loss: 0.5214 - val_accuracy: 0.7453
Epoch 32/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5098 - accuracy: 0.74
57 - val_loss: 0.5185 - val_accuracy: 0.7411
Epoch 33/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5090 - accuracy: 0.74
84 - val_loss: 0.5146 - val_accuracy: 0.7469
Epoch 34/80
56/56 [==============================] - 0s 1ms/step - loss: 0.5099 - accuracy: 0.74
71 - val_loss: 0.5266 - val_accuracy: 0.7400
Max validation accuracy for batch size 512: 0.7475
```


Loss — Batch Size 512
Train Loss
Val Loss
Loss
Epoch
0.700
0.675
0.650
0.625
0.600
0.575
0.550
0.525
0.500
0
5
10
15
20
25
30


Accuracy — Batch Size 512
Train Accuracy
Val Accuracy
Accuracy
Epoch
0.74
0.72
0.70
0.68
0.66
0.64
0
5
10
15
20
25
30

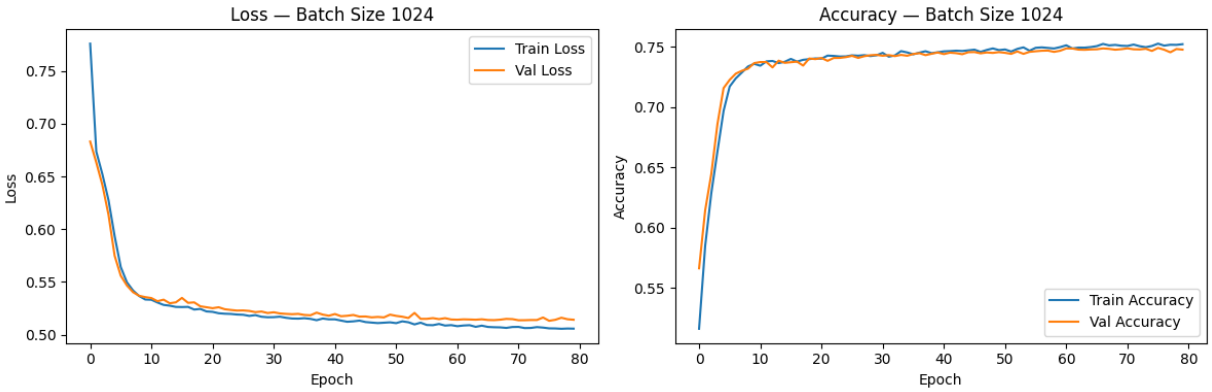
```
Training with batch size: 1024
Epoch 1/80
28/28 [==============================] - 1s 5ms/step - loss: 0.7760 - accuracy: 0.51
63 - val_loss: 0.6831 - val_accuracy: 0.5664
Epoch 2/80
28/28 [==============================] - 0s 2ms/step - loss: 0.6736 - accuracy: 0.58
51 - val_loss: 0.6633 - val_accuracy: 0.6145
Epoch 3/80
28/28 [==============================] - 0s 2ms/step - loss: 0.6521 - accuracy: 0.62
87 - val_loss: 0.6419 - val_accuracy: 0.6446
Epoch 4/80
28/28 [==============================] - 0s 2ms/step - loss: 0.6276 - accuracy: 0.66
32 - val_loss: 0.6137 - val_accuracy: 0.6858
Epoch 5/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5935 - accuracy: 0.69
67 - val_loss: 0.5749 - val_accuracy: 0.7157
Epoch 6/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5641 - accuracy: 0.71
70 - val_loss: 0.5559 - val_accuracy: 0.7225
Epoch 7/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5497 - accuracy: 0.72
37 - val_loss: 0.5467 - val_accuracy: 0.7277
Epoch 8/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5419 - accuracy: 0.72
86 - val_loss: 0.5403 - val_accuracy: 0.7300
Epoch 9/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5366 - accuracy: 0.73
36 - val_loss: 0.5369 - val_accuracy: 0.7317
Epoch 10/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5333 - accuracy: 0.73
58 - val_loss: 0.5356 - val_accuracy: 0.7364
Epoch 11/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5331 - accuracy: 0.73
42 - val_loss: 0.5347 - val_accuracy: 0.7372
Epoch 12/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5305 - accuracy: 0.73
78 - val_loss: 0.5317 - val_accuracy: 0.7372
Epoch 13/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5282 - accuracy: 0.73
81 - val_loss: 0.5332 - val_accuracy: 0.7328
Epoch 14/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5275 - accuracy: 0.73
64 - val_loss: 0.5298 - val_accuracy: 0.7384
Epoch 15/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5263 - accuracy: 0.73
75 - val_loss: 0.5307 - val_accuracy: 0.7366
Epoch 16/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5262 - accuracy: 0.73
98 - val_loss: 0.5349 - val_accuracy: 0.7371
Epoch 17/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5263 - accuracy: 0.73
76 - val_loss: 0.5302 - val_accuracy: 0.7376
Epoch 18/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5238 - accuracy: 0.73
91 - val_loss: 0.5306 - val_accuracy: 0.7343
Epoch 19/80
```

```
28/28 [==============================] - 0s 2ms/step - loss: 0.5244 - accuracy: 0.73
96 - val_loss: 0.5269 - val_accuracy: 0.7402
Epoch 20/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5221 - accuracy: 0.74
02 - val_loss: 0.5260 - val_accuracy: 0.7397
Epoch 21/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5217 - accuracy: 0.74
00 - val_loss: 0.5251 - val_accuracy: 0.7406
Epoch 22/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5203 - accuracy: 0.74
26 - val_loss: 0.5260 - val_accuracy: 0.7382
Epoch 23/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5199 - accuracy: 0.74
23 - val_loss: 0.5241 - val_accuracy: 0.7406
Epoch 24/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5197 - accuracy: 0.74
18 - val_loss: 0.5234 - val_accuracy: 0.7407
Epoch 25/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5191 - accuracy: 0.74
19 - val_loss: 0.5228 - val_accuracy: 0.7413
Epoch 26/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5189 - accuracy: 0.74
28 - val_loss: 0.5230 - val_accuracy: 0.7424
Epoch 27/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5178 - accuracy: 0.74
25 - val_loss: 0.5225 - val_accuracy: 0.7407
Epoch 28/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5185 - accuracy: 0.74
29 - val_loss: 0.5214 - val_accuracy: 0.7421
Epoch 29/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5170 - accuracy: 0.74
22 - val_loss: 0.5219 - val_accuracy: 0.7429
Epoch 30/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5165 - accuracy: 0.74
28 - val_loss: 0.5205 - val_accuracy: 0.7432
Epoch 31/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5166 - accuracy: 0.74
48 - val_loss: 0.5212 - val_accuracy: 0.7424
Epoch 32/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5171 - accuracy: 0.74
16 - val_loss: 0.5201 - val_accuracy: 0.7429
Epoch 33/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5160 - accuracy: 0.74
27 - val_loss: 0.5197 - val_accuracy: 0.7422
Epoch 34/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5153 - accuracy: 0.74
63 - val_loss: 0.5194 - val_accuracy: 0.7432
Epoch 35/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5152 - accuracy: 0.74
52 - val_loss: 0.5197 - val_accuracy: 0.7424
Epoch 36/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5156 - accuracy: 0.74
36 - val_loss: 0.5187 - val_accuracy: 0.7439
Epoch 37/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5150 - accuracy: 0.74
50 - val_loss: 0.5184 - val_accuracy: 0.7446
```

```
Epoch 38/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5137 - accuracy: 0.74
62 - val_loss: 0.5209 - val_accuracy: 0.7429
Epoch 39/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5153 - accuracy: 0.74
46 - val_loss: 0.5190 - val_accuracy: 0.7441
Epoch 40/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5145 - accuracy: 0.74
53 - val_loss: 0.5180 - val_accuracy: 0.7452
Epoch 41/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5145 - accuracy: 0.74
61 - val_loss: 0.5196 - val_accuracy: 0.7438
Epoch 42/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5134 - accuracy: 0.74
63 - val_loss: 0.5175 - val_accuracy: 0.7450
Epoch 43/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5122 - accuracy: 0.74
66 - val_loss: 0.5179 - val_accuracy: 0.7446
Epoch 44/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5126 - accuracy: 0.74
64 - val_loss: 0.5187 - val_accuracy: 0.7437
Epoch 45/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5133 - accuracy: 0.74
69 - val_loss: 0.5171 - val_accuracy: 0.7454
Epoch 46/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5120 - accuracy: 0.74
75 - val_loss: 0.5172 - val_accuracy: 0.7455
Epoch 47/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5115 - accuracy: 0.74
57 - val_loss: 0.5164 - val_accuracy: 0.7444
Epoch 48/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5110 - accuracy: 0.74
72 - val_loss: 0.5168 - val_accuracy: 0.7452
Epoch 49/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5113 - accuracy: 0.74
85 - val_loss: 0.5164 - val_accuracy: 0.7448
Epoch 50/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5117 - accuracy: 0.74
71 - val_loss: 0.5190 - val_accuracy: 0.7454
Epoch 51/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5109 - accuracy: 0.74
77 - val_loss: 0.5178 - val_accuracy: 0.7449
Epoch 52/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5126 - accuracy: 0.74
59 - val_loss: 0.5170 - val_accuracy: 0.7440
Epoch 53/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5118 - accuracy: 0.74
81 - val_loss: 0.5158 - val_accuracy: 0.7460
Epoch 54/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5097 - accuracy: 0.74
94 - val_loss: 0.5206 - val_accuracy: 0.7440
Epoch 55/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5114 - accuracy: 0.74
66 - val_loss: 0.5150 - val_accuracy: 0.7457
Epoch 56/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5092 - accuracy: 0.74
```

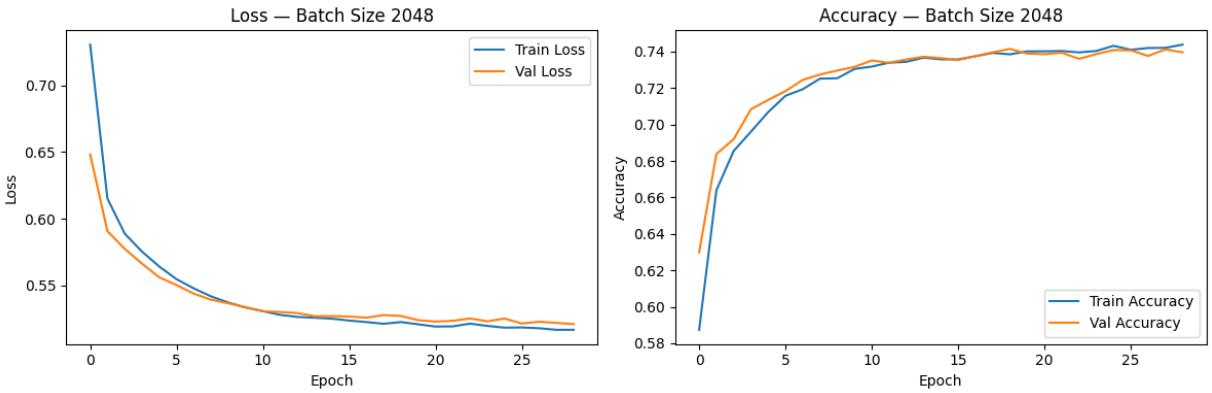
```
90 - val_loss: 0.5150 - val_accuracy: 0.7462
Epoch 57/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5090 - accuracy: 0.74
94 - val_loss: 0.5157 - val_accuracy: 0.7466
Epoch 58/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5101 - accuracy: 0.74
89 - val_loss: 0.5148 - val_accuracy: 0.7467
Epoch 59/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5086 - accuracy: 0.74
85 - val_loss: 0.5156 - val_accuracy: 0.7457
Epoch 60/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5091 - accuracy: 0.74
96 - val_loss: 0.5144 - val_accuracy: 0.7466
Epoch 61/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5080 - accuracy: 0.75
11 - val_loss: 0.5142 - val_accuracy: 0.7485
Epoch 62/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5087 - accuracy: 0.74
85 - val_loss: 0.5145 - val_accuracy: 0.7484
Epoch 63/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5090 - accuracy: 0.74
92 - val_loss: 0.5144 - val_accuracy: 0.7475
Epoch 64/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5074 - accuracy: 0.74
92 - val_loss: 0.5142 - val_accuracy: 0.7474
Epoch 65/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5086 - accuracy: 0.74
96 - val_loss: 0.5146 - val_accuracy: 0.7477
Epoch 66/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5073 - accuracy: 0.75
04 - val_loss: 0.5139 - val_accuracy: 0.7478
Epoch 67/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5071 - accuracy: 0.75
22 - val_loss: 0.5138 - val_accuracy: 0.7485
Epoch 68/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5069 - accuracy: 0.75
10 - val_loss: 0.5142 - val_accuracy: 0.7482
Epoch 69/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5064 - accuracy: 0.75
14 - val_loss: 0.5150 - val_accuracy: 0.7474
Epoch 70/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5073 - accuracy: 0.75
08 - val_loss: 0.5148 - val_accuracy: 0.7479
Epoch 71/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5074 - accuracy: 0.75
05 - val_loss: 0.5137 - val_accuracy: 0.7486
Epoch 72/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5063 - accuracy: 0.75
17 - val_loss: 0.5137 - val_accuracy: 0.7478
Epoch 73/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5064 - accuracy: 0.75
04 - val_loss: 0.5139 - val_accuracy: 0.7477
Epoch 74/80
28/28 [==============================] - 0s 1ms/step - loss: 0.5072 - accuracy: 0.74
95 - val_loss: 0.5139 - val_accuracy: 0.7482
Epoch 75/80
```

```
28/28 [==============================] - 0s 2ms/step - loss: 0.5067 - accuracy: 0.75
05 - val_loss: 0.5163 - val_accuracy: 0.7464
Epoch 76/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5060 - accuracy: 0.75
25 - val_loss: 0.5132 - val_accuracy: 0.7489
Epoch 77/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5059 - accuracy: 0.75
08 - val_loss: 0.5141 - val_accuracy: 0.7474
Epoch 78/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5056 - accuracy: 0.75
16 - val_loss: 0.5161 - val_accuracy: 0.7453
Epoch 79/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5058 - accuracy: 0.75
15 - val_loss: 0.5146 - val_accuracy: 0.7480
Epoch 80/80
28/28 [==============================] - 0s 2ms/step - loss: 0.5057 - accuracy: 0.75
20 - val_loss: 0.5142 - val_accuracy: 0.7475
Max validation accuracy for batch size 1024: 0.7489
```



```
Training with batch size: 2048
Epoch 1/80
14/14 [==============================] - 1s 9ms/step - loss: 0.7305 - accuracy: 0.58
74 - val_loss: 0.6482 - val_accuracy: 0.6299
Epoch 2/80
14/14 [==============================] - 0s 2ms/step - loss: 0.6151 - accuracy: 0.66
40 - val_loss: 0.5907 - val_accuracy: 0.6839
Epoch 3/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5888 - accuracy: 0.68
55 - val_loss: 0.5774 - val_accuracy: 0.6920
Epoch 4/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5754 - accuracy: 0.69
61 - val_loss: 0.5663 - val_accuracy: 0.7084
Epoch 5/80
14/14 [==============================] - 0s 3ms/step - loss: 0.5642 - accuracy: 0.70
69 - val_loss: 0.5560 - val_accuracy: 0.7135
Epoch 6/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5547 - accuracy: 0.71
58 - val_loss: 0.5503 - val_accuracy: 0.7184
Epoch 7/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5478 - accuracy: 0.71
94 - val_loss: 0.5438 - val_accuracy: 0.7245
Epoch 8/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5418 - accuracy: 0.72
52 - val_loss: 0.5393 - val_accuracy: 0.7274
Epoch 9/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5372 - accuracy: 0.72
54 - val_loss: 0.5366 - val_accuracy: 0.7297
Epoch 10/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5334 - accuracy: 0.73
05 - val_loss: 0.5337 - val_accuracy: 0.7317
Epoch 11/80
14/14 [==============================] - 0s 4ms/step - loss: 0.5307 - accuracy: 0.73
18 - val_loss: 0.5306 - val_accuracy: 0.7352
Epoch 12/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5279 - accuracy: 0.73
39 - val_loss: 0.5300 - val_accuracy: 0.7339
Epoch 13/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5263 - accuracy: 0.73
44 - val_loss: 0.5293 - val_accuracy: 0.7357
Epoch 14/80
14/14 [==============================] - 0s 4ms/step - loss: 0.5257 - accuracy: 0.73
67 - val_loss: 0.5269 - val_accuracy: 0.7372
Epoch 15/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5250 - accuracy: 0.73
58 - val_loss: 0.5270 - val_accuracy: 0.7364
Epoch 16/80
14/14 [==============================] - 0s 3ms/step - loss: 0.5236 - accuracy: 0.73
58 - val_loss: 0.5266 - val_accuracy: 0.7354
Epoch 17/80
14/14 [==============================] - 0s 3ms/step - loss: 0.5225 - accuracy: 0.73
75 - val_loss: 0.5258 - val_accuracy: 0.7375
Epoch 18/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5212 - accuracy: 0.73
93 - val_loss: 0.5277 - val_accuracy: 0.7397
Epoch 19/80
```

```
14/14 [==============================] - 0s 2ms/step - loss: 0.5225 - accuracy: 0.73
85 - val_loss: 0.5271 - val_accuracy: 0.7415
Epoch 20/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5208 - accuracy: 0.74
02 - val_loss: 0.5239 - val_accuracy: 0.7389
Epoch 21/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5191 - accuracy: 0.74
02 - val_loss: 0.5228 - val_accuracy: 0.7386
Epoch 22/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5192 - accuracy: 0.74
04 - val_loss: 0.5234 - val_accuracy: 0.7394
Epoch 23/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5213 - accuracy: 0.73
96 - val_loss: 0.5252 - val_accuracy: 0.7361
Epoch 24/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5196 - accuracy: 0.74
04 - val_loss: 0.5230 - val_accuracy: 0.7387
Epoch 25/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5183 - accuracy: 0.74
32 - val_loss: 0.5251 - val_accuracy: 0.7409
Epoch 26/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5185 - accuracy: 0.74
10 - val_loss: 0.5214 - val_accuracy: 0.7407
Epoch 27/80
14/14 [==============================] - 0s 3ms/step - loss: 0.5179 - accuracy: 0.74
20 - val_loss: 0.5228 - val_accuracy: 0.7377
Epoch 28/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5167 - accuracy: 0.74
21 - val_loss: 0.5219 - val_accuracy: 0.7412
Epoch 29/80
14/14 [==============================] - 0s 2ms/step - loss: 0.5167 - accuracy: 0.74
39 - val_loss: 0.5209 - val_accuracy: 0.7397
Max validation accuracy for batch size 2048: 0.7415
```




```
Best batch size: 128 with Validation Accuracy: 0.7544
```

Output summary

1. Batch size 16
   - Max validation accuracy : 0.7526
   - validation loss decreases with fluctuations and accuracy rises steadily
2. Batch size 32
   - Max validation accuracy: 0.7532

- Stable training with smoother validation curve

3. Batch size 64
   - Max validation accuracy : 0.7510
   - has more volatile validation loss than smaller batches
4. Batch size 128
   - Max validation accuracy: 0.7544 (being the best one)
   - strong performance with smooth convergence over 71 epochs
5. Batch size 256
   - max validation accuracy : 0.7512
   - Slightly less stable with accuracy fluctuations
6. Batch size 512
   - max validation accuracy: 0.7475
   - More variance in validation metrics.
7. Batch size 1024
   - Max validation accuracy: 0.7489
   - model trained to full 80 epochs with validation slightly degraded over the end
8. Batch size 2048
   - Max validation accuracy: 0.7415
   - Less stable with early stopping at 29 epochs.

- Smaller batch sizes provided more frequent weight updates allowing for fine grained learning which however, came with the cost of increased training time, They performed well.
- Larger batch sizes resulted in faster training but less precise convergence,as shown by thier fluctuating validation curves and slightly lower accuracy. Notably 2048 stopped at 29 epochs due to unstable generation.
- The sweet spot was batch size 128, which achieved the highest validation accuracy with stable training curves and efficient convergence. It balanced training efficiency with generalization, making it the best batch size for this model dataset.

The best four performing batch sizes were 32, 64, 128, and 256.(later for grid search)

4.2 Optimizer Investigation

After identifying the optimal batch size, I explored how diffrent optimizers affect model performance. Optimizers are crucial as they determine how the model updates weights during training, impacting convergence speed, stability and accuracy. I selected Admam,SDG and RMSprop as they represent three widely used and fundamentally different optimization strategies.

Sdg being a classic baseline with simple gradient updates, RMSprop is known for adapting learning rates per parameters to hanle non-stationary objectives and Adam which combines RMSprop and momentum for faster and more stable convergence. This comparision will help

identify which method best suits my dataset and architecture(32-16-8).

In [19]:
```
# --- 4.2 Optimizer Investigation ---

batch_size = best_batch  # Using the best batch size from above

optimizers = {
    'adam': Adam(),
    'sgd': SGD(),
    'rmsprop': RMSprop()
}

best_val_accuracy = 0
best_optimizer = None

print("\n### Optimizer Exploration ###")
for opt_name, optimizer in optimizers.items():
    print(f"\nTraining with optimizer: {opt_name}")

    model = Sequential([
    Input(shape=(X_train.shape[1],)),
    Dense(32, activation='relu'),
    Dense(16, activation='relu'),
    Dense(8, activation='relu'),
    Dense(1, activation='sigmoid')
    ])


    model.compile(
        optimizer=optimizer,
        loss='binary_crossentropy',
        metrics=['accuracy']
    )

    early_stop = EarlyStopping(monitor='val_accuracy', patience=10, restore_best_we

    history = model.fit(
        X_train, y_train,
        validation_data=(X_test, y_test),
        epochs=80,
        batch_size=batch_size,
        callbacks=[early_stop],
        verbose=1
    )

    max_val_acc = max(history.history['val_accuracy'])
    print(f"Max validation accuracy for optimizer {opt_name}: {max_val_acc:.4f}")

    plt.figure(figsize=(12, 4))
    plt.subplot(1, 2, 1)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')
    plt.title(f'Loss — Optimizer {opt_name}')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
```

```
    plt.legend()

    plt.subplot(1, 2, 2)
    plt.plot(history.history['accuracy'], label='Train Accuracy')
    plt.plot(history.history['val_accuracy'], label='Val Accuracy')
    plt.title(f'Accuracy – Optimizer {opt_name}')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()

    plt.tight_layout()
    plt.show()

    if max_val_acc > best_val_accuracy:
        best_val_accuracy = max_val_acc
        best_optimizer = opt_name

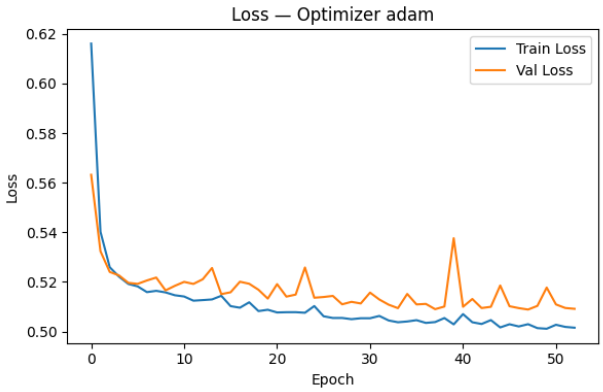
print(f"\nBest optimizer found: {best_optimizer} with validation accuracy: {best_va
```

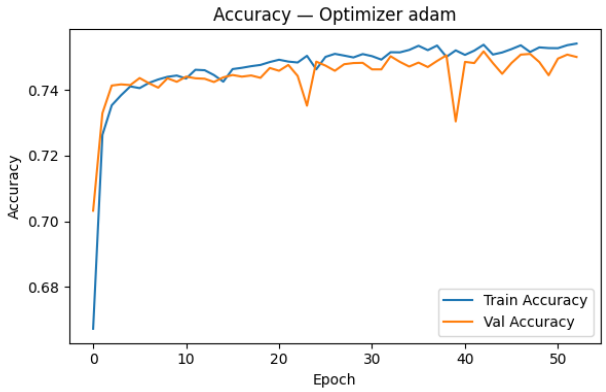
```
### Optimizer Exploration ###

Training with optimizer: adam
Epoch 1/80
223/223 [==============================] - 1s 1ms/step - loss: 0.6161 - accuracy: 0.
6672 - val_loss: 0.5632 - val_accuracy: 0.7032
Epoch 2/80
223/223 [==============================] - 0s 859us/step - loss: 0.5402 - accuracy:
0.7263 - val_loss: 0.5324 - val_accuracy: 0.7329
Epoch 3/80
223/223 [==============================] - 0s 928us/step - loss: 0.5259 - accuracy:
0.7353 - val_loss: 0.5240 - val_accuracy: 0.7414
Epoch 4/80
223/223 [==============================] - 0s 929us/step - loss: 0.5221 - accuracy:
0.7384 - val_loss: 0.5226 - val_accuracy: 0.7417
Epoch 5/80
223/223 [==============================] - 0s 929us/step - loss: 0.5192 - accuracy:
0.7411 - val_loss: 0.5196 - val_accuracy: 0.7415
Epoch 6/80
223/223 [==============================] - 0s 926us/step - loss: 0.5182 - accuracy:
0.7406 - val_loss: 0.5192 - val_accuracy: 0.7437
Epoch 7/80
223/223 [==============================] - 0s 859us/step - loss: 0.5159 - accuracy:
0.7422 - val_loss: 0.5206 - val_accuracy: 0.7421
Epoch 8/80
223/223 [==============================] - 0s 927us/step - loss: 0.5164 - accuracy:
0.7432 - val_loss: 0.5218 - val_accuracy: 0.7407
Epoch 9/80
223/223 [==============================] - 0s 941us/step - loss: 0.5158 - accuracy:
0.7440 - val_loss: 0.5166 - val_accuracy: 0.7436
Epoch 10/80
223/223 [==============================] - 0s 911us/step - loss: 0.5146 - accuracy:
0.7444 - val_loss: 0.5184 - val_accuracy: 0.7425
Epoch 11/80
223/223 [==============================] - 0s 928us/step - loss: 0.5141 - accuracy:
0.7435 - val_loss: 0.5200 - val_accuracy: 0.7441
Epoch 12/80
223/223 [==============================] - 0s 856us/step - loss: 0.5124 - accuracy:
0.7462 - val_loss: 0.5192 - val_accuracy: 0.7436
Epoch 13/80
223/223 [==============================] - 0s 906us/step - loss: 0.5127 - accuracy:
0.7460 - val_loss: 0.5211 - val_accuracy: 0.7434
Epoch 14/80
223/223 [==============================] - 0s 903us/step - loss: 0.5129 - accuracy:
0.7446 - val_loss: 0.5256 - val_accuracy: 0.7424
Epoch 15/80
223/223 [==============================] - 0s 901us/step - loss: 0.5144 - accuracy:
0.7426 - val_loss: 0.5151 - val_accuracy: 0.7439
Epoch 16/80
223/223 [==============================] - 0s 929us/step - loss: 0.5102 - accuracy:
0.7464 - val_loss: 0.5158 - val_accuracy: 0.7446
Epoch 17/80
223/223 [==============================] - 0s 923us/step - loss: 0.5096 - accuracy:
0.7468 - val_loss: 0.5201 - val_accuracy: 0.7441
Epoch 18/80
223/223 [==============================] - 0s 924us/step - loss: 0.5118 - accuracy:
```

```
0.7473 - val_loss: 0.5193 - val_accuracy: 0.7444
Epoch 19/80
223/223 [==============================] - 0s 923us/step - loss: 0.5082 - accuracy:
0.7477 - val_loss: 0.5168 - val_accuracy: 0.7437
Epoch 20/80
223/223 [==============================] - 0s 928us/step - loss: 0.5088 - accuracy:
0.7485 - val_loss: 0.5133 - val_accuracy: 0.7467
Epoch 21/80
223/223 [==============================] - 0s 860us/step - loss: 0.5077 - accuracy:
0.7492 - val_loss: 0.5191 - val_accuracy: 0.7459
Epoch 22/80
223/223 [==============================] - 0s 929us/step - loss: 0.5078 - accuracy:
0.7487 - val_loss: 0.5141 - val_accuracy: 0.7477
Epoch 23/80
223/223 [==============================] - 0s 859us/step - loss: 0.5078 - accuracy:
0.7484 - val_loss: 0.5149 - val_accuracy: 0.7443
Epoch 24/80
223/223 [==============================] - 0s 923us/step - loss: 0.5076 - accuracy:
0.7504 - val_loss: 0.5258 - val_accuracy: 0.7352
Epoch 25/80
223/223 [==============================] - 0s 939us/step - loss: 0.5103 - accuracy:
0.7463 - val_loss: 0.5136 - val_accuracy: 0.7487
Epoch 26/80
223/223 [==============================] - 0s 916us/step - loss: 0.5062 - accuracy:
0.7501 - val_loss: 0.5139 - val_accuracy: 0.7474
Epoch 27/80
223/223 [==============================] - 0s 929us/step - loss: 0.5055 - accuracy:
0.7510 - val_loss: 0.5144 - val_accuracy: 0.7459
Epoch 28/80
223/223 [==============================] - 0s 861us/step - loss: 0.5055 - accuracy:
0.7505 - val_loss: 0.5110 - val_accuracy: 0.7479
Epoch 29/80
223/223 [==============================] - 0s 859us/step - loss: 0.5050 - accuracy:
0.7499 - val_loss: 0.5120 - val_accuracy: 0.7482
Epoch 30/80
223/223 [==============================] - 0s 930us/step - loss: 0.5053 - accuracy:
0.7510 - val_loss: 0.5113 - val_accuracy: 0.7483
Epoch 31/80
223/223 [==============================] - 0s 927us/step - loss: 0.5054 - accuracy:
0.7503 - val_loss: 0.5157 - val_accuracy: 0.7463
Epoch 32/80
223/223 [==============================] - 0s 931us/step - loss: 0.5063 - accuracy:
0.7493 - val_loss: 0.5129 - val_accuracy: 0.7463
Epoch 33/80
223/223 [==============================] - 0s 926us/step - loss: 0.5045 - accuracy:
0.7515 - val_loss: 0.5108 - val_accuracy: 0.7503
Epoch 34/80
223/223 [==============================] - 0s 929us/step - loss: 0.5037 - accuracy:
0.7515 - val_loss: 0.5094 - val_accuracy: 0.7486
Epoch 35/80
223/223 [==============================] - 0s 860us/step - loss: 0.5040 - accuracy:
0.7522 - val_loss: 0.5152 - val_accuracy: 0.7472
Epoch 36/80
223/223 [==============================] - 0s 858us/step - loss: 0.5046 - accuracy:
0.7535 - val_loss: 0.5109 - val_accuracy: 0.7484
Epoch 37/80
```

```
223/223 [==============================] - 0s 929us/step - loss: 0.5035 - accuracy:
0.7522 - val_loss: 0.5111 - val_accuracy: 0.7470
Epoch 38/80
223/223 [==============================] - 0s 924us/step - loss: 0.5038 - accuracy:
0.7536 - val_loss: 0.5090 - val_accuracy: 0.7489
Epoch 39/80
223/223 [==============================] - 0s 925us/step - loss: 0.5054 - accuracy:
0.7501 - val_loss: 0.5101 - val_accuracy: 0.7506
Epoch 40/80
223/223 [==============================] - 0s 925us/step - loss: 0.5029 - accuracy:
0.7521 - val_loss: 0.5376 - val_accuracy: 0.7304
Epoch 41/80
223/223 [==============================] - 0s 901us/step - loss: 0.5070 - accuracy:
0.7507 - val_loss: 0.5100 - val_accuracy: 0.7486
Epoch 42/80
223/223 [==============================] - 0s 889us/step - loss: 0.5037 - accuracy:
0.7520 - val_loss: 0.5131 - val_accuracy: 0.7482
Epoch 43/80
223/223 [==============================] - 0s 899us/step - loss: 0.5030 - accuracy:
0.7538 - val_loss: 0.5095 - val_accuracy: 0.7518
Epoch 44/80
223/223 [==============================] - 0s 929us/step - loss: 0.5046 - accuracy:
0.7508 - val_loss: 0.5100 - val_accuracy: 0.7482
Epoch 45/80
223/223 [==============================] - 0s 931us/step - loss: 0.5016 - accuracy:
0.7515 - val_loss: 0.5186 - val_accuracy: 0.7449
Epoch 46/80
223/223 [==============================] - 0s 934us/step - loss: 0.5029 - accuracy:
0.7525 - val_loss: 0.5102 - val_accuracy: 0.7482
Epoch 47/80
223/223 [==============================] - 0s 929us/step - loss: 0.5020 - accuracy:
0.7536 - val_loss: 0.5094 - val_accuracy: 0.7508
Epoch 48/80
223/223 [==============================] - 0s 925us/step - loss: 0.5030 - accuracy:
0.7516 - val_loss: 0.5089 - val_accuracy: 0.7510
Epoch 49/80
223/223 [==============================] - 0s 929us/step - loss: 0.5014 - accuracy:
0.7530 - val_loss: 0.5103 - val_accuracy: 0.7485
Epoch 50/80
223/223 [==============================] - 0s 928us/step - loss: 0.5011 - accuracy:
0.7528 - val_loss: 0.5177 - val_accuracy: 0.7445
Epoch 51/80
223/223 [==============================] - 0s 926us/step - loss: 0.5027 - accuracy:
0.7528 - val_loss: 0.5109 - val_accuracy: 0.7496
Epoch 52/80
223/223 [==============================] - 0s 861us/step - loss: 0.5019 - accuracy:
0.7537 - val_loss: 0.5095 - val_accuracy: 0.7508
Epoch 53/80
223/223 [==============================] - 0s 932us/step - loss: 0.5015 - accuracy:
0.7542 - val_loss: 0.5092 - val_accuracy: 0.7501
Max validation accuracy for optimizer adam: 0.7518
```

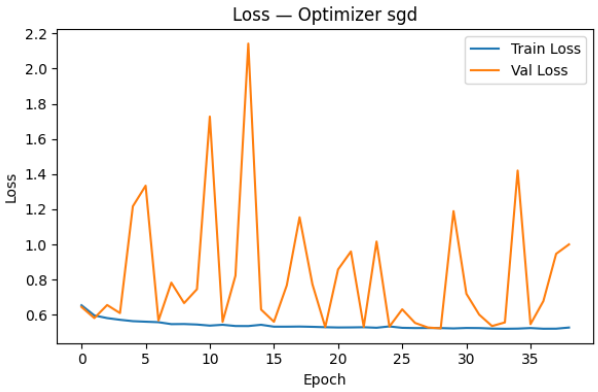

Loss — Optimizer adam
Train Loss
Val Loss
Loss
Epoch
0.62
0.60
0.58
0.56
0.54
0.52
0.50
0
10
20
30
40
50

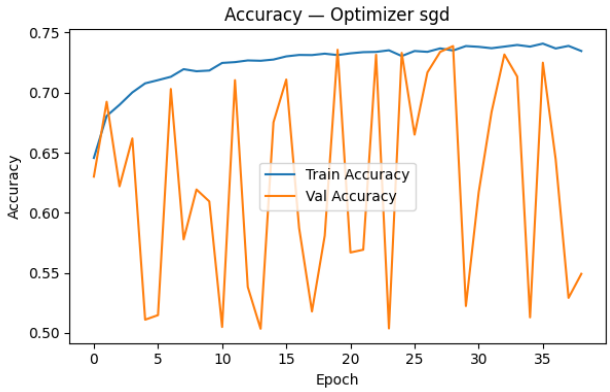

Accuracy — Optimizer adam
Train Accuracy
Val Accuracy
Accuracy
Epoch
0.74
0.72
0.70
0.68
0
10
20
30
40
50

```
Training with optimizer: sgd
Epoch 1/80
223/223 [==============================] - 1s 1ms/step - loss: 0.6544 - accuracy: 0.
6456 - val_loss: 0.6448 - val_accuracy: 0.6301
Epoch 2/80
223/223 [==============================] - 0s 902us/step - loss: 0.5965 - accuracy:
0.6806 - val_loss: 0.5816 - val_accuracy: 0.6924
Epoch 3/80
223/223 [==============================] - 0s 903us/step - loss: 0.5815 - accuracy:
0.6898 - val_loss: 0.6558 - val_accuracy: 0.6220
Epoch 4/80
223/223 [==============================] - 0s 901us/step - loss: 0.5719 - accuracy:
0.7002 - val_loss: 0.6099 - val_accuracy: 0.6619
Epoch 5/80
223/223 [==============================] - 0s 858us/step - loss: 0.5641 - accuracy:
0.7077 - val_loss: 1.2168 - val_accuracy: 0.5110
Epoch 6/80
223/223 [==============================] - 0s 828us/step - loss: 0.5610 - accuracy:
0.7104 - val_loss: 1.3342 - val_accuracy: 0.5148
Epoch 7/80
223/223 [==============================] - 0s 927us/step - loss: 0.5582 - accuracy:
0.7132 - val_loss: 0.5677 - val_accuracy: 0.7030
Epoch 8/80
223/223 [==============================] - 0s 856us/step - loss: 0.5472 - accuracy:
0.7195 - val_loss: 0.7830 - val_accuracy: 0.5778
Epoch 9/80
223/223 [==============================] - 0s 929us/step - loss: 0.5478 - accuracy:
0.7178 - val_loss: 0.6667 - val_accuracy: 0.6193
Epoch 10/80
223/223 [==============================] - 0s 859us/step - loss: 0.5449 - accuracy:
0.7184 - val_loss: 0.7458 - val_accuracy: 0.6094
Epoch 11/80
223/223 [==============================] - 0s 855us/step - loss: 0.5388 - accuracy:
0.7247 - val_loss: 1.7270 - val_accuracy: 0.5049
Epoch 12/80
223/223 [==============================] - 0s 855us/step - loss: 0.5435 - accuracy:
0.7254 - val_loss: 0.5603 - val_accuracy: 0.7104
Epoch 13/80
223/223 [==============================] - 0s 929us/step - loss: 0.5366 - accuracy:
0.7268 - val_loss: 0.8225 - val_accuracy: 0.5382
Epoch 14/80
223/223 [==============================] - 0s 856us/step - loss: 0.5362 - accuracy:
0.7265 - val_loss: 2.1417 - val_accuracy: 0.5034
Epoch 15/80
223/223 [==============================] - 0s 926us/step - loss: 0.5435 - accuracy:
0.7275 - val_loss: 0.6300 - val_accuracy: 0.6753
Epoch 16/80
223/223 [==============================] - 0s 858us/step - loss: 0.5327 - accuracy:
0.7302 - val_loss: 0.5608 - val_accuracy: 0.7109
Epoch 17/80
223/223 [==============================] - 0s 858us/step - loss: 0.5327 - accuracy:
0.7314 - val_loss: 0.7677 - val_accuracy: 0.5873
Epoch 18/80
223/223 [==============================] - 0s 861us/step - loss: 0.5334 - accuracy:
0.7313 - val_loss: 1.1546 - val_accuracy: 0.5178
Epoch 19/80
```

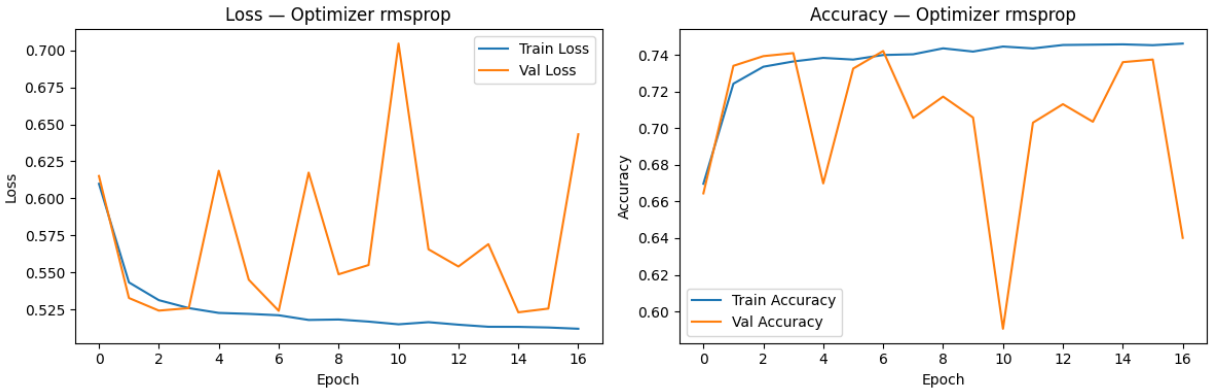
```
223/223 [==============================] - 0s 860us/step - loss: 0.5319 - accuracy:
0.7324 - val_loss: 0.7735 - val_accuracy: 0.5809
Epoch 20/80
223/223 [==============================] - 0s 857us/step - loss: 0.5297 - accuracy:
0.7313 - val_loss: 0.5304 - val_accuracy: 0.7357
Epoch 21/80
223/223 [==============================] - 0s 858us/step - loss: 0.5282 - accuracy:
0.7327 - val_loss: 0.8578 - val_accuracy: 0.5669
Epoch 22/80
223/223 [==============================] - 0s 856us/step - loss: 0.5286 - accuracy:
0.7337 - val_loss: 0.9601 - val_accuracy: 0.5692
Epoch 23/80
223/223 [==============================] - 0s 858us/step - loss: 0.5295 - accuracy:
0.7339 - val_loss: 0.5317 - val_accuracy: 0.7315
Epoch 24/80
223/223 [==============================] - 0s 857us/step - loss: 0.5267 - accuracy:
0.7352 - val_loss: 1.0166 - val_accuracy: 0.5037
Epoch 25/80
223/223 [==============================] - 0s 948us/step - loss: 0.5342 - accuracy:
0.7305 - val_loss: 0.5332 - val_accuracy: 0.7331
Epoch 26/80
223/223 [==============================] - 0s 912us/step - loss: 0.5263 - accuracy:
0.7346 - val_loss: 0.6314 - val_accuracy: 0.6650
Epoch 27/80
223/223 [==============================] - 0s 861us/step - loss: 0.5249 - accuracy:
0.7339 - val_loss: 0.5541 - val_accuracy: 0.7168
Epoch 28/80
223/223 [==============================] - 0s 932us/step - loss: 0.5251 - accuracy:
0.7368 - val_loss: 0.5281 - val_accuracy: 0.7339
Epoch 29/80
223/223 [==============================] - 0s 852us/step - loss: 0.5247 - accuracy:
0.7351 - val_loss: 0.5229 - val_accuracy: 0.7387
Epoch 30/80
223/223 [==============================] - 0s 925us/step - loss: 0.5228 - accuracy:
0.7388 - val_loss: 1.1895 - val_accuracy: 0.5223
Epoch 31/80
223/223 [==============================] - 0s 927us/step - loss: 0.5254 - accuracy:
0.7381 - val_loss: 0.7198 - val_accuracy: 0.6171
Epoch 32/80
223/223 [==============================] - 0s 921us/step - loss: 0.5248 - accuracy:
0.7370 - val_loss: 0.6012 - val_accuracy: 0.6839
Epoch 33/80
223/223 [==============================] - 0s 859us/step - loss: 0.5220 - accuracy:
0.7383 - val_loss: 0.5363 - val_accuracy: 0.7317
Epoch 34/80
223/223 [==============================] - 0s 923us/step - loss: 0.5208 - accuracy:
0.7397 - val_loss: 0.5569 - val_accuracy: 0.7134
Epoch 35/80
223/223 [==============================] - 0s 855us/step - loss: 0.5218 - accuracy:
0.7383 - val_loss: 1.4201 - val_accuracy: 0.5129
Epoch 36/80
223/223 [==============================] - 0s 859us/step - loss: 0.5249 - accuracy:
0.7408 - val_loss: 0.5470 - val_accuracy: 0.7249
Epoch 37/80
223/223 [==============================] - 0s 859us/step - loss: 0.5211 - accuracy:
0.7368 - val_loss: 0.6781 - val_accuracy: 0.6445
```

```
Epoch 38/80
223/223 [==============================] - 0s 854us/step - loss: 0.5214 - accuracy:
0.7389 - val_loss: 0.9469 - val_accuracy: 0.5291
Epoch 39/80
223/223 [==============================] - 0s 861us/step - loss: 0.5278 - accuracy:
0.7346 - val_loss: 1.0003 - val_accuracy: 0.5491
Max validation accuracy for optimizer sgd: 0.7387
```





```
Training with optimizer: rmsprop
Epoch 1/80
223/223 [==============================] - 1s 1ms/step - loss: 0.6098 - accuracy: 0.
6697 - val_loss: 0.6151 - val_accuracy: 0.6644
Epoch 2/80
223/223 [==============================] - 0s 853us/step - loss: 0.5433 - accuracy:
0.7243 - val_loss: 0.5327 - val_accuracy: 0.7340
Epoch 3/80
223/223 [==============================] - 0s 859us/step - loss: 0.5313 - accuracy:
0.7335 - val_loss: 0.5242 - val_accuracy: 0.7393
Epoch 4/80
223/223 [==============================] - 0s 928us/step - loss: 0.5259 - accuracy:
0.7364 - val_loss: 0.5259 - val_accuracy: 0.7409
Epoch 5/80
223/223 [==============================] - 0s 927us/step - loss: 0.5227 - accuracy:
0.7383 - val_loss: 0.6187 - val_accuracy: 0.6699
Epoch 6/80
223/223 [==============================] - 0s 924us/step - loss: 0.5220 - accuracy:
0.7374 - val_loss: 0.5451 - val_accuracy: 0.7325
Epoch 7/80
223/223 [==============================] - 0s 923us/step - loss: 0.5211 - accuracy:
0.7399 - val_loss: 0.5241 - val_accuracy: 0.7421
Epoch 8/80
223/223 [==============================] - 0s 854us/step - loss: 0.5179 - accuracy:
0.7403 - val_loss: 0.6174 - val_accuracy: 0.7056
Epoch 9/80
223/223 [==============================] - 0s 929us/step - loss: 0.5182 - accuracy:
0.7435 - val_loss: 0.5487 - val_accuracy: 0.7172
Epoch 10/80
223/223 [==============================] - 0s 930us/step - loss: 0.5168 - accuracy:
0.7418 - val_loss: 0.5550 - val_accuracy: 0.7058
Epoch 11/80
223/223 [==============================] - 0s 929us/step - loss: 0.5150 - accuracy:
0.7445 - val_loss: 0.7045 - val_accuracy: 0.5906
Epoch 12/80
223/223 [==============================] - 0s 856us/step - loss: 0.5164 - accuracy:
0.7435 - val_loss: 0.5656 - val_accuracy: 0.7030
Epoch 13/80
223/223 [==============================] - 0s 930us/step - loss: 0.5147 - accuracy:
0.7454 - val_loss: 0.5540 - val_accuracy: 0.7131
Epoch 14/80
223/223 [==============================] - 0s 928us/step - loss: 0.5133 - accuracy:
0.7455 - val_loss: 0.5691 - val_accuracy: 0.7035
Epoch 15/80
223/223 [==============================] - 0s 931us/step - loss: 0.5132 - accuracy:
0.7457 - val_loss: 0.5231 - val_accuracy: 0.7360
Epoch 16/80
223/223 [==============================] - 0s 923us/step - loss: 0.5128 - accuracy:
0.7453 - val_loss: 0.5256 - val_accuracy: 0.7374
Epoch 17/80
223/223 [==============================] - 0s 855us/step - loss: 0.5120 - accuracy:
0.7461 - val_loss: 0.6433 - val_accuracy: 0.6401
Max validation accuracy for optimizer rmsprop: 0.7421
```




```
Best optimizer found: adam with validation accuracy: 0.7518
```

Oberved Results

1. Adam optimizer(the best one)
    - Max Val accuracy : 0.7518
    - It had a smooth and stable convergence. Validation loss steadily decreases and accuracy being consistent across epochs
2. SGD optimizer
    - Max Val accuracy : 0.7287
    - It was highly unstable with training and validation accuracy/loss fluctuating erratically. Also there was occasional spikes in loss and crashes in accuracy.
    - Difficult to train and had poor stability. It was not suitable for this model withou tuning.
3. RMSprop optimizer
    - Max val accuracy : 0.7421
    - It was somewhat stable but with more variation than adam. peaking early and plateaus or regresses due to early stopping.

Among the three tested optimizers, Adam was the one that consistent to deliver highest validation accuracy with stable and smooth convergence across epochs. Its ability to combine RMSprop and momentum based updates makes it especially effective for this architecture and dataset.

While RMSprop performs reasonably well, it struggles with volatility and is not ideal without fine-tuning. hence Adam is selected for future model training

Best two being Adam and RMSprop

4.3 Regularization Investigation

To improve model generalization and prevent any overfitting, we investigated three regularization strategies which are none, dropoyt, L2 regularization.

These are chosen for their contrasting approches which are

- No regularization serves as a baseline
- dropout randomly deactivates neurons during training to prevent reliance on specific paths
- L2 penalizes large weights to promote simpler models

Each was tested using our previously identified best batch (128) and optimizer(Adam) on the same architecture. This comparision helps evaluate which method best balances bias and variance for our binary classification task.

In [20]:
```
from tensorflow.keras.optimizers.legacy import Adam, SGD, RMSprop
from tensorflow.keras.regularizers import l2
from tensorflow.keras.callbacks import EarlyStopping
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, Input
import matplotlib.pyplot as plt

# --- 4.3 Regularization Investigation ---

input_dim = X_train.shape[1]
epochs = 80
patience = 10
batch_size = 128               # Using best batch size
best_optimizer = 'adam'        # Using best optimizer

def get_optimizer(name):
    if name == 'adam':
        return Adam()
    elif name == 'sgd':
        return SGD()
    elif name == 'rmsprop':
        return RMSprop()
    else:
        raise ValueError(f"Unknown optimizer: {name}")

def build_model(regularization=None, optimizer_name=best_optimizer):
    model = Sequential()
    model.add(Input(shape=(input_dim,)))

    if regularization == 'dropout':
        model.add(Dense(32, activation='relu'))
        model.add(Dropout(0.2))
        model.add(Dense(16, activation='relu'))
        model.add(Dropout(0.2))
        model.add(Dense(8, activation='relu'))
        model.add(Dropout(0.2))
    elif regularization == 'l2':
        model.add(Dense(32, activation='relu', kernel_regularizer=l2(0.001)))
        model.add(Dense(16, activation='relu', kernel_regularizer=l2(0.001)))
        model.add(Dense(8, activation='relu', kernel_regularizer=l2(0.001)))
    else:  # No regularization
        model.add(Dense(32, activation='relu'))
```

```
        model.add(Dense(16, activation='relu'))
        model.add(Dense(8, activation='relu'))

    model.add(Dense(1, activation='sigmoid'))  # Binary classification output
    optimizer = get_optimizer(optimizer_name)
    model.compile(optimizer=optimizer, loss='binary_crossentropy', metrics=['accura
    return model

regularizations = ['none', 'dropout', 'l2']
histories = {}

print("\n### Regularization Exploration ###")
for reg in regularizations:
    print(f"\nTraining with {reg} regularization...")
    model = build_model(regularization=reg if reg != 'none' else None)
    early_stop = EarlyStopping(monitor='val_accuracy', patience=patience, restore_b

    history = model.fit(
        X_train, y_train,
        validation_data=(X_test, y_test),
        epochs=epochs,
        batch_size=batch_size,
        callbacks=[early_stop],
        verbose=1
    )

    histories[reg] = history

    # Plotting
    plt.figure(figsize=(12, 4))
    plt.subplot(1, 2, 1)
    plt.plot(history.history['loss'], label='Train Loss')
    plt.plot(history.history['val_loss'], label='Val Loss')
    plt.title(f'Loss Curve ({reg})')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
    plt.legend()

    plt.subplot(1, 2, 2)
    plt.plot(history.history['accuracy'], label='Train Accuracy')
    plt.plot(history.history['val_accuracy'], label='Val Accuracy')
    plt.title(f'Accuracy Curve ({reg})')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()

    plt.tight_layout()
    plt.show()

# Determining the best regularization
best_val_acc = 0
best_reg = None

for reg in regularizations:
    val_acc_list = histories[reg].history.get('val_accuracy', [])
    max_val = max(val_acc_list) if val_acc_list else 0
```

```
    print(f"Max validation accuracy with {reg}: {max_val:.4f}")
    if max_val > best_val_acc:
        best_val_acc = max_val
        best_reg = reg

print(f"\n Best regularization: {best_reg} with validation accuracy: {best_val_acc:
```

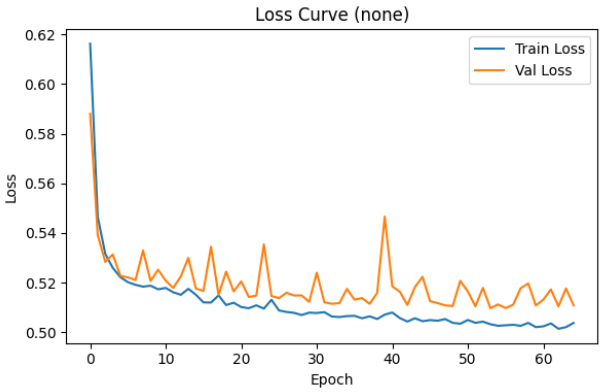
```
### Regularization Exploration ###

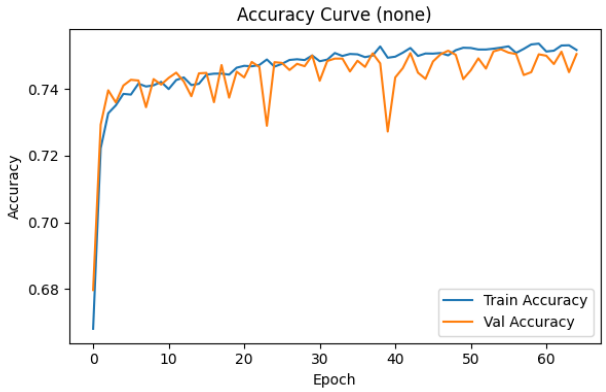
Training with none regularization...
Epoch 1/80
223/223 [==============================] - 1s 1ms/step - loss: 0.6163 - accuracy: 0.
6681 - val_loss: 0.5880 - val_accuracy: 0.6797
Epoch 2/80
223/223 [==============================] - 0s 927us/step - loss: 0.5466 - accuracy:
0.7222 - val_loss: 0.5392 - val_accuracy: 0.7293
Epoch 3/80
223/223 [==============================] - 0s 927us/step - loss: 0.5316 - accuracy:
0.7327 - val_loss: 0.5284 - val_accuracy: 0.7396
Epoch 4/80
223/223 [==============================] - 0s 929us/step - loss: 0.5259 - accuracy:
0.7351 - val_loss: 0.5313 - val_accuracy: 0.7359
Epoch 5/80
223/223 [==============================] - 0s 925us/step - loss: 0.5222 - accuracy:
0.7385 - val_loss: 0.5227 - val_accuracy: 0.7410
Epoch 6/80
223/223 [==============================] - 0s 931us/step - loss: 0.5202 - accuracy:
0.7383 - val_loss: 0.5221 - val_accuracy: 0.7427
Epoch 7/80
223/223 [==============================] - 0s 928us/step - loss: 0.5191 - accuracy:
0.7416 - val_loss: 0.5210 - val_accuracy: 0.7425
Epoch 8/80
223/223 [==============================] - 0s 926us/step - loss: 0.5184 - accuracy:
0.7407 - val_loss: 0.5330 - val_accuracy: 0.7345
Epoch 9/80
223/223 [==============================] - 0s 926us/step - loss: 0.5188 - accuracy:
0.7411 - val_loss: 0.5207 - val_accuracy: 0.7429
Epoch 10/80
223/223 [==============================] - 0s 930us/step - loss: 0.5173 - accuracy:
0.7421 - val_loss: 0.5252 - val_accuracy: 0.7413
Epoch 11/80
223/223 [==============================] - 0s 924us/step - loss: 0.5179 - accuracy:
0.7400 - val_loss: 0.5208 - val_accuracy: 0.7433
Epoch 12/80
223/223 [==============================] - 0s 929us/step - loss: 0.5161 - accuracy:
0.7427 - val_loss: 0.5179 - val_accuracy: 0.7449
Epoch 13/80
223/223 [==============================] - 0s 924us/step - loss: 0.5151 - accuracy:
0.7434 - val_loss: 0.5225 - val_accuracy: 0.7422
Epoch 14/80
223/223 [==============================] - 0s 929us/step - loss: 0.5175 - accuracy:
0.7411 - val_loss: 0.5300 - val_accuracy: 0.7378
Epoch 15/80
223/223 [==============================] - 0s 928us/step - loss: 0.5151 - accuracy:
0.7415 - val_loss: 0.5176 - val_accuracy: 0.7447
Epoch 16/80
223/223 [==============================] - 0s 985us/step - loss: 0.5120 - accuracy:
0.7443 - val_loss: 0.5166 - val_accuracy: 0.7448
Epoch 17/80
223/223 [==============================] - 0s 886us/step - loss: 0.5120 - accuracy:
0.7446 - val_loss: 0.5344 - val_accuracy: 0.7360
Epoch 18/80
223/223 [==============================] - 0s 932us/step - loss: 0.5149 - accuracy:
```

```
0.7446 - val_loss: 0.5151 - val_accuracy: 0.7472
Epoch 19/80
223/223 [==============================] - 0s 928us/step - loss: 0.5110 - accuracy:
0.7443 - val_loss: 0.5244 - val_accuracy: 0.7374
Epoch 20/80
223/223 [==============================] - 0s 924us/step - loss: 0.5119 - accuracy:
0.7464 - val_loss: 0.5165 - val_accuracy: 0.7452
Epoch 21/80
223/223 [==============================] - 0s 929us/step - loss: 0.5102 - accuracy:
0.7469 - val_loss: 0.5205 - val_accuracy: 0.7434
Epoch 22/80
223/223 [==============================] - 0s 927us/step - loss: 0.5097 - accuracy:
0.7468 - val_loss: 0.5142 - val_accuracy: 0.7481
Epoch 23/80
223/223 [==============================] - 0s 927us/step - loss: 0.5109 - accuracy:
0.7472 - val_loss: 0.5148 - val_accuracy: 0.7466
Epoch 24/80
223/223 [==============================] - 0s 926us/step - loss: 0.5095 - accuracy:
0.7488 - val_loss: 0.5354 - val_accuracy: 0.7289
Epoch 25/80
223/223 [==============================] - 0s 926us/step - loss: 0.5131 - accuracy:
0.7467 - val_loss: 0.5146 - val_accuracy: 0.7480
Epoch 26/80
223/223 [==============================] - 0s 926us/step - loss: 0.5088 - accuracy:
0.7475 - val_loss: 0.5138 - val_accuracy: 0.7478
Epoch 27/80
223/223 [==============================] - 0s 930us/step - loss: 0.5082 - accuracy:
0.7487 - val_loss: 0.5159 - val_accuracy: 0.7457
Epoch 28/80
223/223 [==============================] - 0s 929us/step - loss: 0.5078 - accuracy:
0.7489 - val_loss: 0.5148 - val_accuracy: 0.7475
Epoch 29/80
223/223 [==============================] - 0s 928us/step - loss: 0.5069 - accuracy:
0.7486 - val_loss: 0.5149 - val_accuracy: 0.7468
Epoch 30/80
223/223 [==============================] - 0s 930us/step - loss: 0.5079 - accuracy:
0.7500 - val_loss: 0.5123 - val_accuracy: 0.7500
Epoch 31/80
223/223 [==============================] - 0s 927us/step - loss: 0.5078 - accuracy:
0.7483 - val_loss: 0.5240 - val_accuracy: 0.7424
Epoch 32/80
223/223 [==============================] - 0s 874us/step - loss: 0.5081 - accuracy:
0.7488 - val_loss: 0.5121 - val_accuracy: 0.7484
Epoch 33/80
223/223 [==============================] - 0s 858us/step - loss: 0.5063 - accuracy:
0.7508 - val_loss: 0.5115 - val_accuracy: 0.7491
Epoch 34/80
223/223 [==============================] - 0s 928us/step - loss: 0.5061 - accuracy:
0.7499 - val_loss: 0.5117 - val_accuracy: 0.7491
Epoch 35/80
223/223 [==============================] - 0s 928us/step - loss: 0.5065 - accuracy:
0.7505 - val_loss: 0.5175 - val_accuracy: 0.7452
Epoch 36/80
223/223 [==============================] - 0s 933us/step - loss: 0.5066 - accuracy:
0.7504 - val_loss: 0.5131 - val_accuracy: 0.7484
Epoch 37/80
```

```
223/223 [==============================] - 0s 1ms/step - loss: 0.5056 - accuracy: 0.
7495 - val_loss: 0.5138 - val_accuracy: 0.7466
Epoch 38/80
223/223 [==============================] - 0s 929us/step - loss: 0.5064 - accuracy:
0.7500 - val_loss: 0.5114 - val_accuracy: 0.7507
Epoch 39/80
223/223 [==============================] - 0s 925us/step - loss: 0.5053 - accuracy:
0.7527 - val_loss: 0.5159 - val_accuracy: 0.7477
Epoch 40/80
223/223 [==============================] - 0s 929us/step - loss: 0.5071 - accuracy:
0.7493 - val_loss: 0.5466 - val_accuracy: 0.7272
Epoch 41/80
223/223 [==============================] - 0s 925us/step - loss: 0.5080 - accuracy:
0.7497 - val_loss: 0.5185 - val_accuracy: 0.7434
Epoch 42/80
223/223 [==============================] - 0s 915us/step - loss: 0.5057 - accuracy:
0.7509 - val_loss: 0.5163 - val_accuracy: 0.7462
Epoch 43/80
223/223 [==============================] - 0s 924us/step - loss: 0.5043 - accuracy:
0.7523 - val_loss: 0.5110 - val_accuracy: 0.7507
Epoch 44/80
223/223 [==============================] - 0s 925us/step - loss: 0.5057 - accuracy:
0.7499 - val_loss: 0.5182 - val_accuracy: 0.7449
Epoch 45/80
223/223 [==============================] - 0s 927us/step - loss: 0.5045 - accuracy:
0.7506 - val_loss: 0.5224 - val_accuracy: 0.7430
Epoch 46/80
223/223 [==============================] - 0s 927us/step - loss: 0.5049 - accuracy:
0.7506 - val_loss: 0.5125 - val_accuracy: 0.7482
Epoch 47/80
223/223 [==============================] - 0s 929us/step - loss: 0.5047 - accuracy:
0.7508 - val_loss: 0.5118 - val_accuracy: 0.7504
Epoch 48/80
223/223 [==============================] - 0s 857us/step - loss: 0.5053 - accuracy:
0.7501 - val_loss: 0.5108 - val_accuracy: 0.7514
Epoch 49/80
223/223 [==============================] - 0s 925us/step - loss: 0.5038 - accuracy:
0.7516 - val_loss: 0.5106 - val_accuracy: 0.7502
Epoch 50/80
223/223 [==============================] - 0s 924us/step - loss: 0.5034 - accuracy:
0.7523 - val_loss: 0.5208 - val_accuracy: 0.7429
Epoch 51/80
223/223 [==============================] - 0s 930us/step - loss: 0.5049 - accuracy:
0.7523 - val_loss: 0.5164 - val_accuracy: 0.7456
Epoch 52/80
223/223 [==============================] - 0s 925us/step - loss: 0.5038 - accuracy:
0.7518 - val_loss: 0.5105 - val_accuracy: 0.7492
Epoch 53/80
223/223 [==============================] - 0s 928us/step - loss: 0.5043 - accuracy:
0.7518 - val_loss: 0.5179 - val_accuracy: 0.7461
Epoch 54/80
223/223 [==============================] - 0s 922us/step - loss: 0.5032 - accuracy:
0.7521 - val_loss: 0.5097 - val_accuracy: 0.7512
Epoch 55/80
223/223 [==============================] - 0s 975us/step - loss: 0.5026 - accuracy:
0.7524 - val_loss: 0.5112 - val_accuracy: 0.7519
```

```
Epoch 56/80
223/223 [==============================] - 0s 900us/step - loss: 0.5028 - accuracy:
0.7528 - val_loss: 0.5098 - val_accuracy: 0.7509
Epoch 57/80
223/223 [==============================] - 0s 906us/step - loss: 0.5030 - accuracy:
0.7508 - val_loss: 0.5111 - val_accuracy: 0.7504
Epoch 58/80
223/223 [==============================] - 0s 931us/step - loss: 0.5026 - accuracy:
0.7520 - val_loss: 0.5177 - val_accuracy: 0.7442
Epoch 59/80
223/223 [==============================] - 0s 930us/step - loss: 0.5038 - accuracy:
0.7533 - val_loss: 0.5197 - val_accuracy: 0.7450
Epoch 60/80
223/223 [==============================] - 0s 926us/step - loss: 0.5021 - accuracy:
0.7536 - val_loss: 0.5108 - val_accuracy: 0.7504
Epoch 61/80
223/223 [==============================] - 0s 853us/step - loss: 0.5024 - accuracy:
0.7512 - val_loss: 0.5131 - val_accuracy: 0.7499
Epoch 62/80
223/223 [==============================] - 0s 856us/step - loss: 0.5036 - accuracy:
0.7515 - val_loss: 0.5173 - val_accuracy: 0.7474
Epoch 63/80
223/223 [==============================] - 0s 858us/step - loss: 0.5014 - accuracy:
0.7530 - val_loss: 0.5104 - val_accuracy: 0.7512
Epoch 64/80
223/223 [==============================] - 0s 855us/step - loss: 0.5021 - accuracy:
0.7531 - val_loss: 0.5176 - val_accuracy: 0.7450
Epoch 65/80
223/223 [==============================] - 0s 931us/step - loss: 0.5038 - accuracy:
0.7517 - val_loss: 0.5108 - val_accuracy: 0.7504
```





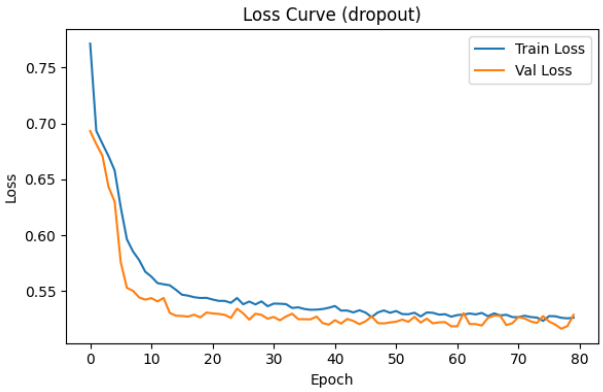
```
Training with dropout regularization...
Epoch 1/80
223/223 [==============================] - 1s 1ms/step - loss: 0.7715 - accuracy: 0.
5061 - val_loss: 0.6933 - val_accuracy: 0.5028
Epoch 2/80
223/223 [==============================] - 0s 1ms/step - loss: 0.6934 - accuracy: 0.
5211 - val_loss: 0.6815 - val_accuracy: 0.5728
Epoch 3/80
223/223 [==============================] - 0s 1ms/step - loss: 0.6820 - accuracy: 0.
5577 - val_loss: 0.6712 - val_accuracy: 0.6016
Epoch 4/80
223/223 [==============================] - 0s 1ms/step - loss: 0.6709 - accuracy: 0.
5889 - val_loss: 0.6433 - val_accuracy: 0.7097
Epoch 5/80
223/223 [==============================] - 0s 1ms/step - loss: 0.6581 - accuracy: 0.
6163 - val_loss: 0.6303 - val_accuracy: 0.6896
Epoch 6/80
223/223 [==============================] - 0s 1ms/step - loss: 0.6250 - accuracy: 0.
6441 - val_loss: 0.5759 - val_accuracy: 0.7253
Epoch 7/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5964 - accuracy: 0.
6649 - val_loss: 0.5530 - val_accuracy: 0.7304
Epoch 8/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5855 - accuracy: 0.
6764 - val_loss: 0.5503 - val_accuracy: 0.7332
Epoch 9/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5778 - accuracy: 0.
6814 - val_loss: 0.5444 - val_accuracy: 0.7372
Epoch 10/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5674 - accuracy: 0.
6888 - val_loss: 0.5424 - val_accuracy: 0.7343
Epoch 11/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5630 - accuracy: 0.
6941 - val_loss: 0.5438 - val_accuracy: 0.7367
Epoch 12/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5572 - accuracy: 0.
6965 - val_loss: 0.5408 - val_accuracy: 0.7367
Epoch 13/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5561 - accuracy: 0.
6984 - val_loss: 0.5439 - val_accuracy: 0.7377
Epoch 14/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5553 - accuracy: 0.
7018 - val_loss: 0.5305 - val_accuracy: 0.7343
Epoch 15/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5513 - accuracy: 0.
7188 - val_loss: 0.5281 - val_accuracy: 0.7405
Epoch 16/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5468 - accuracy: 0.
7253 - val_loss: 0.5278 - val_accuracy: 0.7374
Epoch 17/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5460 - accuracy: 0.
7260 - val_loss: 0.5272 - val_accuracy: 0.7401
Epoch 18/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5446 - accuracy: 0.
7263 - val_loss: 0.5290 - val_accuracy: 0.7408
Epoch 19/80
```

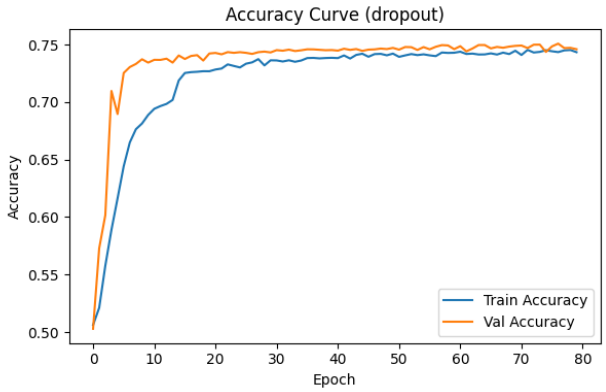
```
223/223 [==============================] - 0s 1ms/step - loss: 0.5439 - accuracy: 0.
7268 - val_loss: 0.5265 - val_accuracy: 0.7361
Epoch 20/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5440 - accuracy: 0.
7268 - val_loss: 0.5309 - val_accuracy: 0.7422
Epoch 21/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5425 - accuracy: 0.
7283 - val_loss: 0.5301 - val_accuracy: 0.7427
Epoch 22/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5413 - accuracy: 0.
7292 - val_loss: 0.5296 - val_accuracy: 0.7415
Epoch 23/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5413 - accuracy: 0.
7328 - val_loss: 0.5288 - val_accuracy: 0.7434
Epoch 24/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5395 - accuracy: 0.
7314 - val_loss: 0.5260 - val_accuracy: 0.7428
Epoch 25/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5438 - accuracy: 0.
7301 - val_loss: 0.5344 - val_accuracy: 0.7434
Epoch 26/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5383 - accuracy: 0.
7334 - val_loss: 0.5302 - val_accuracy: 0.7428
Epoch 27/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5407 - accuracy: 0.
7345 - val_loss: 0.5245 - val_accuracy: 0.7419
Epoch 28/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5381 - accuracy: 0.
7373 - val_loss: 0.5298 - val_accuracy: 0.7434
Epoch 29/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5408 - accuracy: 0.
7319 - val_loss: 0.5288 - val_accuracy: 0.7439
Epoch 30/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5365 - accuracy: 0.
7363 - val_loss: 0.5253 - val_accuracy: 0.7432
Epoch 31/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5389 - accuracy: 0.
7362 - val_loss: 0.5269 - val_accuracy: 0.7452
Epoch 32/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5387 - accuracy: 0.
7353 - val_loss: 0.5239 - val_accuracy: 0.7447
Epoch 33/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5384 - accuracy: 0.
7363 - val_loss: 0.5273 - val_accuracy: 0.7456
Epoch 34/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5350 - accuracy: 0.
7350 - val_loss: 0.5298 - val_accuracy: 0.7444
Epoch 35/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5355 - accuracy: 0.
7361 - val_loss: 0.5250 - val_accuracy: 0.7450
Epoch 36/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5340 - accuracy: 0.
7383 - val_loss: 0.5249 - val_accuracy: 0.7459
Epoch 37/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5333 - accuracy: 0.
7384 - val_loss: 0.5248 - val_accuracy: 0.7458
```

```
Epoch 38/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5335 - accuracy: 0.
7380 - val_loss: 0.5272 - val_accuracy: 0.7454
Epoch 39/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5340 - accuracy: 0.
7383 - val_loss: 0.5214 - val_accuracy: 0.7451
Epoch 40/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5351 - accuracy: 0.
7385 - val_loss: 0.5201 - val_accuracy: 0.7452
Epoch 41/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5367 - accuracy: 0.
7382 - val_loss: 0.5240 - val_accuracy: 0.7447
Epoch 42/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5326 - accuracy: 0.
7406 - val_loss: 0.5210 - val_accuracy: 0.7464
Epoch 43/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5327 - accuracy: 0.
7378 - val_loss: 0.5251 - val_accuracy: 0.7454
Epoch 44/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5310 - accuracy: 0.
7410 - val_loss: 0.5234 - val_accuracy: 0.7461
Epoch 45/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5329 - accuracy: 0.
7420 - val_loss: 0.5205 - val_accuracy: 0.7445
Epoch 46/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5308 - accuracy: 0.
7394 - val_loss: 0.5230 - val_accuracy: 0.7456
Epoch 47/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5268 - accuracy: 0.
7417 - val_loss: 0.5274 - val_accuracy: 0.7458
Epoch 48/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5309 - accuracy: 0.
7419 - val_loss: 0.5215 - val_accuracy: 0.7465
Epoch 49/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5327 - accuracy: 0.
7407 - val_loss: 0.5211 - val_accuracy: 0.7461
Epoch 50/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5306 - accuracy: 0.
7421 - val_loss: 0.5220 - val_accuracy: 0.7471
Epoch 51/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5322 - accuracy: 0.
7393 - val_loss: 0.5227 - val_accuracy: 0.7456
Epoch 52/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5295 - accuracy: 0.
7406 - val_loss: 0.5246 - val_accuracy: 0.7479
Epoch 53/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5293 - accuracy: 0.
7417 - val_loss: 0.5226 - val_accuracy: 0.7477
Epoch 54/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5307 - accuracy: 0.
7408 - val_loss: 0.5270 - val_accuracy: 0.7453
Epoch 55/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5274 - accuracy: 0.
7415 - val_loss: 0.5219 - val_accuracy: 0.7478
Epoch 56/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5309 - accuracy: 0.
```

```
7406 - val_loss: 0.5254 - val_accuracy: 0.7459
Epoch 57/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5306 - accuracy: 0.
7400 - val_loss: 0.5212 - val_accuracy: 0.7481
Epoch 58/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5290 - accuracy: 0.
7431 - val_loss: 0.5221 - val_accuracy: 0.7495
Epoch 59/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5294 - accuracy: 0.
7427 - val_loss: 0.5223 - val_accuracy: 0.7492
Epoch 60/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5272 - accuracy: 0.
7429 - val_loss: 0.5186 - val_accuracy: 0.7460
Epoch 61/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5286 - accuracy: 0.
7438 - val_loss: 0.5185 - val_accuracy: 0.7486
Epoch 62/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5290 - accuracy: 0.
7418 - val_loss: 0.5303 - val_accuracy: 0.7441
Epoch 63/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5300 - accuracy: 0.
7421 - val_loss: 0.5206 - val_accuracy: 0.7466
Epoch 64/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5292 - accuracy: 0.
7414 - val_loss: 0.5205 - val_accuracy: 0.7496
Epoch 65/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5305 - accuracy: 0.
7414 - val_loss: 0.5192 - val_accuracy: 0.7496
Epoch 66/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5275 - accuracy: 0.
7423 - val_loss: 0.5260 - val_accuracy: 0.7467
Epoch 67/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5301 - accuracy: 0.
7413 - val_loss: 0.5282 - val_accuracy: 0.7479
Epoch 68/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5283 - accuracy: 0.
7430 - val_loss: 0.5279 - val_accuracy: 0.7472
Epoch 69/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5289 - accuracy: 0.
7417 - val_loss: 0.5196 - val_accuracy: 0.7482
Epoch 70/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5268 - accuracy: 0.
7446 - val_loss: 0.5214 - val_accuracy: 0.7489
Epoch 71/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5267 - accuracy: 0.
7409 - val_loss: 0.5269 - val_accuracy: 0.7492
Epoch 72/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5280 - accuracy: 0.
7454 - val_loss: 0.5255 - val_accuracy: 0.7469
Epoch 73/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5268 - accuracy: 0.
7431 - val_loss: 0.5229 - val_accuracy: 0.7499
Epoch 74/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5264 - accuracy: 0.
7436 - val_loss: 0.5215 - val_accuracy: 0.7499
Epoch 75/80
```

```
223/223 [==============================] - 0s 1ms/step - loss: 0.5234 - accuracy: 0.
7451 - val_loss: 0.5277 - val_accuracy: 0.7436
Epoch 76/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5277 - accuracy: 0.
7442 - val_loss: 0.5228 - val_accuracy: 0.7484
Epoch 77/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5275 - accuracy: 0.
7434 - val_loss: 0.5201 - val_accuracy: 0.7508
Epoch 78/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5262 - accuracy: 0.
7450 - val_loss: 0.5163 - val_accuracy: 0.7469
Epoch 79/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5257 - accuracy: 0.
7453 - val_loss: 0.5189 - val_accuracy: 0.7472
Epoch 80/80
223/223 [==============================] - 0s 1ms/step - loss: 0.5263 - accuracy: 0.
7434 - val_loss: 0.5288 - val_accuracy: 0.7459
```

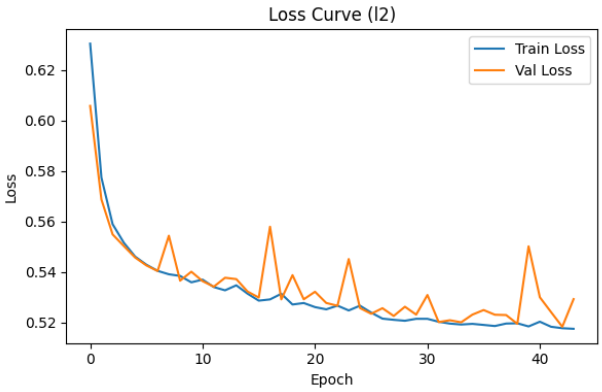


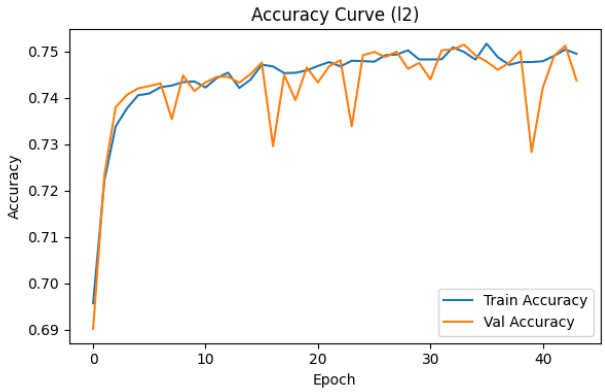


```
Training with l2 regularization...
Epoch 1/80
223/223 [==============================] - 1s 1ms/step - loss: 0.6304 - accuracy: 0.
6957 - val_loss: 0.6057 - val_accuracy: 0.6901
Epoch 2/80
223/223 [==============================] - 0s 857us/step - loss: 0.5774 - accuracy:
0.7221 - val_loss: 0.5687 - val_accuracy: 0.7234
Epoch 3/80
223/223 [==============================] - 0s 856us/step - loss: 0.5589 - accuracy:
0.7339 - val_loss: 0.5549 - val_accuracy: 0.7380
Epoch 4/80
223/223 [==============================] - 0s 930us/step - loss: 0.5515 - accuracy:
0.7377 - val_loss: 0.5502 - val_accuracy: 0.7407
Epoch 5/80
223/223 [==============================] - 0s 929us/step - loss: 0.5461 - accuracy:
0.7406 - val_loss: 0.5457 - val_accuracy: 0.7421
Epoch 6/80
223/223 [==============================] - 0s 930us/step - loss: 0.5429 - accuracy:
0.7410 - val_loss: 0.5426 - val_accuracy: 0.7426
Epoch 7/80
223/223 [==============================] - 0s 924us/step - loss: 0.5404 - accuracy:
0.7423 - val_loss: 0.5404 - val_accuracy: 0.7432
Epoch 8/80
223/223 [==============================] - 0s 926us/step - loss: 0.5391 - accuracy:
0.7427 - val_loss: 0.5543 - val_accuracy: 0.7354
Epoch 9/80
223/223 [==============================] - 0s 978us/step - loss: 0.5384 - accuracy:
0.7434 - val_loss: 0.5365 - val_accuracy: 0.7449
Epoch 10/80
223/223 [==============================] - 0s 885us/step - loss: 0.5359 - accuracy:
0.7436 - val_loss: 0.5401 - val_accuracy: 0.7415
Epoch 11/80
223/223 [==============================] - 0s 902us/step - loss: 0.5370 - accuracy:
0.7423 - val_loss: 0.5364 - val_accuracy: 0.7434
Epoch 12/80
223/223 [==============================] - 0s 921us/step - loss: 0.5340 - accuracy:
0.7443 - val_loss: 0.5342 - val_accuracy: 0.7446
Epoch 13/80
223/223 [==============================] - 0s 854us/step - loss: 0.5327 - accuracy:
0.7455 - val_loss: 0.5377 - val_accuracy: 0.7446
Epoch 14/80
223/223 [==============================] - 0s 923us/step - loss: 0.5347 - accuracy:
0.7422 - val_loss: 0.5372 - val_accuracy: 0.7433
Epoch 15/80
223/223 [==============================] - 0s 859us/step - loss: 0.5314 - accuracy:
0.7440 - val_loss: 0.5322 - val_accuracy: 0.7452
Epoch 16/80
223/223 [==============================] - 0s 924us/step - loss: 0.5286 - accuracy:
0.7472 - val_loss: 0.5299 - val_accuracy: 0.7477
Epoch 17/80
223/223 [==============================] - 0s 857us/step - loss: 0.5291 - accuracy:
0.7468 - val_loss: 0.5579 - val_accuracy: 0.7296
Epoch 18/80
223/223 [==============================] - 0s 928us/step - loss: 0.5314 - accuracy:
0.7454 - val_loss: 0.5291 - val_accuracy: 0.7449
Epoch 19/80
```

```
223/223 [==============================] - 0s 930us/step - loss: 0.5271 - accuracy:
0.7454 - val_loss: 0.5387 - val_accuracy: 0.7395
Epoch 20/80
223/223 [==============================] - 0s 922us/step - loss: 0.5277 - accuracy:
0.7460 - val_loss: 0.5292 - val_accuracy: 0.7466
Epoch 21/80
223/223 [==============================] - 0s 926us/step - loss: 0.5261 - accuracy:
0.7469 - val_loss: 0.5322 - val_accuracy: 0.7434
Epoch 22/80
223/223 [==============================] - 0s 925us/step - loss: 0.5252 - accuracy:
0.7478 - val_loss: 0.5277 - val_accuracy: 0.7469
Epoch 23/80
223/223 [==============================] - 0s 926us/step - loss: 0.5267 - accuracy:
0.7468 - val_loss: 0.5266 - val_accuracy: 0.7482
Epoch 24/80
223/223 [==============================] - 0s 927us/step - loss: 0.5247 - accuracy:
0.7480 - val_loss: 0.5451 - val_accuracy: 0.7339
Epoch 25/80
223/223 [==============================] - 0s 927us/step - loss: 0.5266 - accuracy:
0.7480 - val_loss: 0.5257 - val_accuracy: 0.7492
Epoch 26/80
223/223 [==============================] - 0s 859us/step - loss: 0.5239 - accuracy:
0.7478 - val_loss: 0.5234 - val_accuracy: 0.7499
Epoch 27/80
223/223 [==============================] - 0s 853us/step - loss: 0.5215 - accuracy:
0.7492 - val_loss: 0.5256 - val_accuracy: 0.7489
Epoch 28/80
223/223 [==============================] - 0s 901us/step - loss: 0.5210 - accuracy:
0.7494 - val_loss: 0.5226 - val_accuracy: 0.7500
Epoch 29/80
223/223 [==============================] - 0s 900us/step - loss: 0.5207 - accuracy:
0.7503 - val_loss: 0.5262 - val_accuracy: 0.7463
Epoch 30/80
223/223 [==============================] - 0s 925us/step - loss: 0.5214 - accuracy:
0.7483 - val_loss: 0.5231 - val_accuracy: 0.7476
Epoch 31/80
223/223 [==============================] - 0s 933us/step - loss: 0.5214 - accuracy:
0.7483 - val_loss: 0.5309 - val_accuracy: 0.7440
Epoch 32/80
223/223 [==============================] - 0s 857us/step - loss: 0.5203 - accuracy:
0.7484 - val_loss: 0.5202 - val_accuracy: 0.7503
Epoch 33/80
223/223 [==============================] - 0s 857us/step - loss: 0.5195 - accuracy:
0.7509 - val_loss: 0.5209 - val_accuracy: 0.7505
Epoch 34/80
223/223 [==============================] - 0s 857us/step - loss: 0.5192 - accuracy:
0.7499 - val_loss: 0.5201 - val_accuracy: 0.7515
Epoch 35/80
223/223 [==============================] - 0s 855us/step - loss: 0.5194 - accuracy:
0.7483 - val_loss: 0.5231 - val_accuracy: 0.7492
Epoch 36/80
223/223 [==============================] - 0s 933us/step - loss: 0.5190 - accuracy:
0.7517 - val_loss: 0.5249 - val_accuracy: 0.7479
Epoch 37/80
223/223 [==============================] - 0s 928us/step - loss: 0.5186 - accuracy:
0.7488 - val_loss: 0.5231 - val_accuracy: 0.7461
```

```
Epoch 38/80
223/223 [==============================] - 0s 928us/step - loss: 0.5196 - accuracy:
0.7472 - val_loss: 0.5229 - val_accuracy: 0.7476
Epoch 39/80
223/223 [==============================] - 0s 935us/step - loss: 0.5197 - accuracy:
0.7478 - val_loss: 0.5195 - val_accuracy: 0.7502
Epoch 40/80
223/223 [==============================] - 0s 930us/step - loss: 0.5184 - accuracy:
0.7478 - val_loss: 0.5501 - val_accuracy: 0.7283
Epoch 41/80
223/223 [==============================] - 0s 854us/step - loss: 0.5203 - accuracy:
0.7479 - val_loss: 0.5300 - val_accuracy: 0.7422
Epoch 42/80
223/223 [==============================] - 0s 922us/step - loss: 0.5183 - accuracy:
0.7491 - val_loss: 0.5241 - val_accuracy: 0.7491
Epoch 43/80
223/223 [==============================] - 0s 923us/step - loss: 0.5177 - accuracy:
0.7505 - val_loss: 0.5183 - val_accuracy: 0.7513
Epoch 44/80
223/223 [==============================] - 0s 856us/step - loss: 0.5175 - accuracy:
0.7496 - val_loss: 0.5292 - val_accuracy: 0.7438
```






```
Max validation accuracy with none: 0.7519
Max validation accuracy with dropout: 0.7508
Max validation accuracy with l2: 0.7515

Best regularization: none with validation accuracy: 0.7519
```

Observed Results

The model was trained for upto 80 epochs using the three regularization techniques with early stopping based on validation accuracy(patience=10). The key observations are:

1. No regularization (best one)
   - Training and validation loss steadily decreases, with validation accuracy reaching a maximum of 0.7519.
   - the validation accuracy curve showed stable improvements and plateaued near peak value, indicating good generalization.
   - Some mild overfitting is observed at later epochs, but overall performance is strong.
2. Dropout Regularization
   - Training loss was initially higher compared to none due to neuron deactivation noise.

- Validation accuracy achieved a maximum of 0.7508, close to baseline but slightly lower.
- the training curves were smoother with less fluctuations, reflecting dropout's ability to reduce overfitting by preventing co-adaptation of neurons.

3. L2 Regularization:
   - Training loss was slightly higher than the baseline, consistent with weight penalization.
   - Validation accuracy reached highest at 0.7515, very close to none.
   - Loss and accuracy curves were having fluctuations across epochs, suggesting that the l2 introduced instability during training rather than improving.

Among the regularization methods tested, no regularization(none) achieved the highest validation accuracy(0.7519). While dropout and L2 came close they did not surpass the baseline in performance.

This suggests that the model-dataset combination along with early stopping, may have already been sufficient to prevent overfitting. Also regularization contributed to training stability, the baseline model without regularization is currently preferred.

Section 5: Most Appropriate Model Selection

After extensive exploration of hyperparameters and architectures in previous sections, the final model was chosen based on consistent performance across multiple criteria. The selected model architecture is

- LAYERS : [32,16,8]
- BATCH SIZE : 128
- OPTIMIZER : ADAM
- REGULARIZATION : NONE

This configuration demonstrated the highest validation accuracy and balanced metrics across accuracy,precision,recall and F1 score during experimentation. While dropout and L2 improved training smoothness, they didn't surpass the baseline performance. Adam was preferred due to its smooth convergence and stability, while the chosen architecture showed strong generalization without overfitting. Therefore, this configuration was retrained and evaluated as the final model.

In [21]:
```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Input
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
import matplotlib.pyplot as plt
import numpy as np
```

```
# --- Step 5: Most Appropriate Model Training & Evaluation ---

print("Final Model Training:")

# Final architecture: [32, 16, 8], no regularization, Adam optimizer, batch_size=12
final_model = Sequential([
    Input(shape=(X_train.shape[1],)),
    Dense(32, activation='relu'),
    Dense(16, activation='relu'),
    Dense(8, activation='relu'),
    Dense(1, activation='sigmoid')
])

# Compile with Adam optimizer
final_model.compile(optimizer=Adam(), loss='binary_crossentropy', metrics=['accurac

# Early stopping callback monitoring validation accuracy
early_stop = EarlyStopping(monitor='val_accuracy', patience=10, restore_best_weight

# Train the model
history = final_model.fit(
    X_train, y_train,
    validation_data=(X_test, y_test),
    epochs=80,
    batch_size=128,
    callbacks=[early_stop],
    verbose=1
)

# Evaluate on test set
test_loss, test_accuracy = final_model.evaluate(X_test, y_test, verbose=0)
y_pred_prob = final_model.predict(X_test)
y_pred = (y_pred_prob > 0.5).astype(int)

# Performance metrics
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
roc_auc = roc_auc_score(y_test, y_pred_prob)

print("\nFinal Model Evaluation on Test Set:")
print(f"Test Accuracy  : {test_accuracy:.4f}")
print(f"Precision      : {precision:.4f}")
print(f"Recall         : {recall:.4f}")
print(f"F1 Score       : {f1:.4f}")
print(f"ROC AUC Score  : {roc_auc:.4f}")

# Plot training history
plt.figure(figsize=(12, 4))

plt.subplot(1, 2, 1)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Loss Curve (Final Model)')
plt.xlabel('Epoch')
plt.ylabel('Loss')
```

```
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Accuracy Curve (Final Model)')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()

plt.tight_layout()
plt.show()
```

```
Final Model Training:
Epoch 1/80
223/223 [==============================] - 1s 1ms/step - loss: 0.6744 - accuracy: 0.
6447 - val_loss: 0.5677 - val_accuracy: 0.6981
Epoch 2/80
223/223 [==============================] - 0s 914us/step - loss: 0.5482 - accuracy:
0.7190 - val_loss: 0.5400 - val_accuracy: 0.7273
Epoch 3/80
223/223 [==============================] - 0s 886us/step - loss: 0.5323 - accuracy:
0.7315 - val_loss: 0.5282 - val_accuracy: 0.7358
Epoch 4/80
223/223 [==============================] - 0s 864us/step - loss: 0.5276 - accuracy:
0.7340 - val_loss: 0.5263 - val_accuracy: 0.7379
Epoch 5/80
223/223 [==============================] - 0s 923us/step - loss: 0.5242 - accuracy:
0.7379 - val_loss: 0.5240 - val_accuracy: 0.7404
Epoch 6/80
223/223 [==============================] - 0s 929us/step - loss: 0.5222 - accuracy:
0.7388 - val_loss: 0.5227 - val_accuracy: 0.7400
Epoch 7/80
223/223 [==============================] - 0s 925us/step - loss: 0.5201 - accuracy:
0.7388 - val_loss: 0.5214 - val_accuracy: 0.7401
Epoch 8/80
223/223 [==============================] - 0s 854us/step - loss: 0.5192 - accuracy:
0.7404 - val_loss: 0.5317 - val_accuracy: 0.7310
Epoch 9/80
223/223 [==============================] - 0s 854us/step - loss: 0.5183 - accuracy:
0.7401 - val_loss: 0.5195 - val_accuracy: 0.7432
Epoch 10/80
223/223 [==============================] - 0s 858us/step - loss: 0.5175 - accuracy:
0.7400 - val_loss: 0.5237 - val_accuracy: 0.7407
Epoch 11/80
223/223 [==============================] - 0s 928us/step - loss: 0.5185 - accuracy:
0.7418 - val_loss: 0.5201 - val_accuracy: 0.7429
Epoch 12/80
223/223 [==============================] - 0s 933us/step - loss: 0.5162 - accuracy:
0.7433 - val_loss: 0.5188 - val_accuracy: 0.7433
Epoch 13/80
223/223 [==============================] - 0s 932us/step - loss: 0.5150 - accuracy:
0.7448 - val_loss: 0.5215 - val_accuracy: 0.7417
Epoch 14/80
223/223 [==============================] - 0s 926us/step - loss: 0.5190 - accuracy:
0.7402 - val_loss: 0.5205 - val_accuracy: 0.7435
Epoch 15/80
223/223 [==============================] - 0s 862us/step - loss: 0.5137 - accuracy:
0.7435 - val_loss: 0.5162 - val_accuracy: 0.7447
Epoch 16/80
223/223 [==============================] - 0s 928us/step - loss: 0.5119 - accuracy:
0.7451 - val_loss: 0.5156 - val_accuracy: 0.7458
Epoch 17/80
223/223 [==============================] - 0s 927us/step - loss: 0.5128 - accuracy:
0.7447 - val_loss: 0.5403 - val_accuracy: 0.7290
Epoch 18/80
223/223 [==============================] - 0s 930us/step - loss: 0.5146 - accuracy:
0.7441 - val_loss: 0.5156 - val_accuracy: 0.7458
Epoch 19/80
```

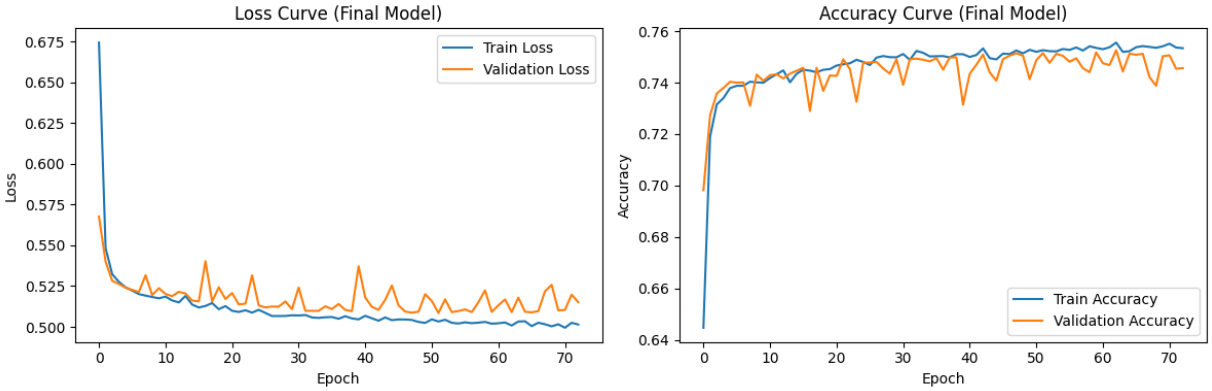
```
223/223 [==============================] - 0s 861us/step - loss: 0.5108 - accuracy:
0.7450 - val_loss: 0.5243 - val_accuracy: 0.7368
Epoch 20/80
223/223 [==============================] - 0s 928us/step - loss: 0.5127 - accuracy:
0.7453 - val_loss: 0.5171 - val_accuracy: 0.7428
Epoch 21/80
223/223 [==============================] - 0s 931us/step - loss: 0.5098 - accuracy:
0.7467 - val_loss: 0.5208 - val_accuracy: 0.7427
Epoch 22/80
223/223 [==============================] - 0s 852us/step - loss: 0.5091 - accuracy:
0.7472 - val_loss: 0.5138 - val_accuracy: 0.7491
Epoch 23/80
223/223 [==============================] - 0s 854us/step - loss: 0.5102 - accuracy:
0.7476 - val_loss: 0.5142 - val_accuracy: 0.7453
Epoch 24/80
223/223 [==============================] - 0s 925us/step - loss: 0.5087 - accuracy:
0.7489 - val_loss: 0.5316 - val_accuracy: 0.7326
Epoch 25/80
223/223 [==============================] - 0s 928us/step - loss: 0.5105 - accuracy:
0.7481 - val_loss: 0.5131 - val_accuracy: 0.7478
Epoch 26/80
223/223 [==============================] - 0s 928us/step - loss: 0.5085 - accuracy:
0.7469 - val_loss: 0.5119 - val_accuracy: 0.7479
Epoch 27/80
223/223 [==============================] - 0s 926us/step - loss: 0.5066 - accuracy:
0.7498 - val_loss: 0.5124 - val_accuracy: 0.7481
Epoch 28/80
223/223 [==============================] - 0s 926us/step - loss: 0.5067 - accuracy:
0.7504 - val_loss: 0.5124 - val_accuracy: 0.7455
Epoch 29/80
223/223 [==============================] - 0s 929us/step - loss: 0.5067 - accuracy:
0.7499 - val_loss: 0.5156 - val_accuracy: 0.7435
Epoch 30/80
223/223 [==============================] - 0s 934us/step - loss: 0.5071 - accuracy:
0.7499 - val_loss: 0.5109 - val_accuracy: 0.7492
Epoch 31/80
223/223 [==============================] - 0s 856us/step - loss: 0.5070 - accuracy:
0.7512 - val_loss: 0.5241 - val_accuracy: 0.7392
Epoch 32/80
223/223 [==============================] - 0s 860us/step - loss: 0.5073 - accuracy:
0.7489 - val_loss: 0.5099 - val_accuracy: 0.7491
Epoch 33/80
223/223 [==============================] - 0s 929us/step - loss: 0.5057 - accuracy:
0.7524 - val_loss: 0.5098 - val_accuracy: 0.7494
Epoch 34/80
223/223 [==============================] - 0s 902us/step - loss: 0.5055 - accuracy:
0.7516 - val_loss: 0.5098 - val_accuracy: 0.7489
Epoch 35/80
223/223 [==============================] - 0s 919us/step - loss: 0.5059 - accuracy:
0.7502 - val_loss: 0.5126 - val_accuracy: 0.7483
Epoch 36/80
223/223 [==============================] - 0s 883us/step - loss: 0.5060 - accuracy:
0.7503 - val_loss: 0.5109 - val_accuracy: 0.7496
Epoch 37/80
223/223 [==============================] - 0s 859us/step - loss: 0.5050 - accuracy:
0.7504 - val_loss: 0.5141 - val_accuracy: 0.7451
```

```
Epoch 38/80
223/223 [==============================] - 0s 928us/step - loss: 0.5065 - accuracy:
0.7498 - val_loss: 0.5104 - val_accuracy: 0.7498
Epoch 39/80
223/223 [==============================] - 0s 923us/step - loss: 0.5052 - accuracy:
0.7511 - val_loss: 0.5096 - val_accuracy: 0.7499
Epoch 40/80
223/223 [==============================] - 0s 928us/step - loss: 0.5046 - accuracy:
0.7511 - val_loss: 0.5372 - val_accuracy: 0.7314
Epoch 41/80
223/223 [==============================] - 0s 856us/step - loss: 0.5068 - accuracy:
0.7500 - val_loss: 0.5180 - val_accuracy: 0.7434
Epoch 42/80
223/223 [==============================] - 0s 861us/step - loss: 0.5052 - accuracy:
0.7508 - val_loss: 0.5123 - val_accuracy: 0.7470
Epoch 43/80
223/223 [==============================] - 0s 929us/step - loss: 0.5038 - accuracy:
0.7534 - val_loss: 0.5104 - val_accuracy: 0.7509
Epoch 44/80
223/223 [==============================] - 0s 930us/step - loss: 0.5058 - accuracy:
0.7496 - val_loss: 0.5167 - val_accuracy: 0.7440
Epoch 45/80
223/223 [==============================] - 0s 930us/step - loss: 0.5041 - accuracy:
0.7491 - val_loss: 0.5254 - val_accuracy: 0.7408
Epoch 46/80
223/223 [==============================] - 0s 928us/step - loss: 0.5045 - accuracy:
0.7513 - val_loss: 0.5132 - val_accuracy: 0.7492
Epoch 47/80
223/223 [==============================] - 0s 856us/step - loss: 0.5045 - accuracy:
0.7511 - val_loss: 0.5094 - val_accuracy: 0.7505
Epoch 48/80
223/223 [==============================] - 0s 930us/step - loss: 0.5042 - accuracy:
0.7526 - val_loss: 0.5087 - val_accuracy: 0.7514
Epoch 49/80
223/223 [==============================] - 0s 926us/step - loss: 0.5030 - accuracy:
0.7514 - val_loss: 0.5093 - val_accuracy: 0.7505
Epoch 50/80
223/223 [==============================] - 0s 857us/step - loss: 0.5024 - accuracy:
0.7528 - val_loss: 0.5200 - val_accuracy: 0.7414
Epoch 51/80
223/223 [==============================] - 0s 927us/step - loss: 0.5046 - accuracy:
0.7520 - val_loss: 0.5158 - val_accuracy: 0.7487
Epoch 52/80
223/223 [==============================] - 0s 927us/step - loss: 0.5033 - accuracy:
0.7527 - val_loss: 0.5085 - val_accuracy: 0.7515
Epoch 53/80
223/223 [==============================] - 0s 864us/step - loss: 0.5044 - accuracy:
0.7523 - val_loss: 0.5169 - val_accuracy: 0.7477
Epoch 54/80
223/223 [==============================] - 0s 857us/step - loss: 0.5025 - accuracy:
0.7522 - val_loss: 0.5090 - val_accuracy: 0.7512
Epoch 55/80
223/223 [==============================] - 0s 926us/step - loss: 0.5021 - accuracy:
0.7531 - val_loss: 0.5097 - val_accuracy: 0.7504
Epoch 56/80
223/223 [==============================] - 0s 927us/step - loss: 0.5028 - accuracy:
```

```
0.7528 - val_loss: 0.5107 - val_accuracy: 0.7482
Epoch 57/80
223/223 [==============================] - 0s 928us/step - loss: 0.5023 - accuracy:
0.7537 - val_loss: 0.5090 - val_accuracy: 0.7495
Epoch 58/80
223/223 [==============================] - 0s 856us/step - loss: 0.5026 - accuracy:
0.7525 - val_loss: 0.5151 - val_accuracy: 0.7457
Epoch 59/80
223/223 [==============================] - 0s 858us/step - loss: 0.5030 - accuracy:
0.7542 - val_loss: 0.5224 - val_accuracy: 0.7441
Epoch 60/80
223/223 [==============================] - 0s 926us/step - loss: 0.5019 - accuracy:
0.7535 - val_loss: 0.5092 - val_accuracy: 0.7518
Epoch 61/80
223/223 [==============================] - 0s 930us/step - loss: 0.5022 - accuracy:
0.7530 - val_loss: 0.5132 - val_accuracy: 0.7476
Epoch 62/80
223/223 [==============================] - 0s 857us/step - loss: 0.5027 - accuracy:
0.7538 - val_loss: 0.5168 - val_accuracy: 0.7469
Epoch 63/80
223/223 [==============================] - 0s 855us/step - loss: 0.5008 - accuracy:
0.7556 - val_loss: 0.5091 - val_accuracy: 0.7527
Epoch 64/80
223/223 [==============================] - 0s 929us/step - loss: 0.5033 - accuracy:
0.7520 - val_loss: 0.5179 - val_accuracy: 0.7444
Epoch 65/80
223/223 [==============================] - 0s 928us/step - loss: 0.5034 - accuracy:
0.7523 - val_loss: 0.5093 - val_accuracy: 0.7513
Epoch 66/80
223/223 [==============================] - 0s 929us/step - loss: 0.5005 - accuracy:
0.7538 - val_loss: 0.5089 - val_accuracy: 0.7509
Epoch 67/80
223/223 [==============================] - 0s 859us/step - loss: 0.5025 - accuracy:
0.7542 - val_loss: 0.5097 - val_accuracy: 0.7512
Epoch 68/80
223/223 [==============================] - 0s 855us/step - loss: 0.5016 - accuracy:
0.7539 - val_loss: 0.5218 - val_accuracy: 0.7422
Epoch 69/80
223/223 [==============================] - 0s 926us/step - loss: 0.5003 - accuracy:
0.7536 - val_loss: 0.5258 - val_accuracy: 0.7388
Epoch 70/80
223/223 [==============================] - 0s 923us/step - loss: 0.5016 - accuracy:
0.7542 - val_loss: 0.5101 - val_accuracy: 0.7503
Epoch 71/80
223/223 [==============================] - 0s 856us/step - loss: 0.4995 - accuracy:
0.7552 - val_loss: 0.5104 - val_accuracy: 0.7507
Epoch 72/80
223/223 [==============================] - 0s 925us/step - loss: 0.5025 - accuracy:
0.7537 - val_loss: 0.5198 - val_accuracy: 0.7454
Epoch 73/80
223/223 [==============================] - 0s 930us/step - loss: 0.5014 - accuracy:
0.7534 - val_loss: 0.5151 - val_accuracy: 0.7457
438/438 [==============================] - 0s 399us/step

Final Model Evaluation on Test Set:
Test Accuracy  : 0.7527
```

```
Precision      : 0.7228
Recall         : 0.8152
F1 Score       : 0.7662
ROC AUC Score  : 0.8254
```



Observations and explainations

Training dynamics

- loss: training loss dropped from 0.6744 to 0.5014, while validation loss dropped from 0.5677 to 0.5151. Fluctuations in validation were minor which indicates stable generalization.
- Accuracy: training accuracy rose from 0.6447 to 0.7534, and validation accuracy is tracking closely while peaking at 0.7457. This close tracking indicates good generalization and low overfitting.

Metric Interpretation

Metric	Value	Interpretation
Accuracy	0.7527	Overall correctness
Precision	0.7228	Reliability of positive predictions
Recall	0.8152	Ability to detect actual positives
F1 Score	0.7662	Balance between precision and recall
ROC AUC	0.8254	Strong class separation capability

Graph Observations

- loss curves: Both training and validation losses steadily decreasing with mild fluctuations, showing stable convergence.
- Accuracy curves: Accuracy curves steadily increased and converged around 0.75, with the validation accuracy closely tracking the training curve.

The final model ([32, 16, 8] architecture, Adam optimizer, batch size 128, and no regularization) was selected due to its highest validation accuracy during experimentation and strong, balanced performance metrics on testing.

THe models high recall and AUC reflect its strong ability to identify positive cases and differentiate between classes. The training curves demonstrate stable learning and convergence, and the minimal performance gap between training and validation supporting its generalization capabilities. Thus, this configuration is considered the most appropriate for this classification task.

Section 6: Grid Search

In grid search we explore the following hyperparameters

- Best 4 batch sizes (16, 32, 128, 256)
- Best 2 optimizers (Adam, RMSprop)
- HeNormal as the weight initializer
- 3 values of epochs (30, 50, 100)
- Usage of early stopping and model checkpointing

These values were based on earlier observations and outputs.

NOTE: Continued in other file

Section 7: Final Model presentation and Performance Evaluation

7.1 Final Model Configuration

After performing a comprehensive grid search over different batch sizes, optimizers and epoch counts, the best-performing configuration was: Batch Size: 256

- Epochs: 30
- Optimizer: Adam
- Architecture: [32, 16, 8]
- Final Test Accuracy: 0.7484

This configuration was implemented using a simple feedforwards neural network with three hidden layers and the HeNormal weight initializer. early stopping was applied with a patience of 10 epochs, monitoring validation loss to avoid overfitting.

In [24]:
```
import matplotlib.pyplot as plt
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.initializers import HeNormal
from tensorflow.keras.callbacks import EarlyStopping
from tensorflow.keras.optimizers import Adam

# --- Best configuration from Grid Search ---
batch_size = 256
```

```
epochs = 30
optimizer = Adam()

# --- Final model with architecture [32, 16, 8] and no regularization ---
final_model = Sequential()
final_model.add(Dense(32, activation='relu', input_shape=(X_train.shape[1],), kerne
final_model.add(Dense(16, activation='relu', kernel_initializer=HeNormal()))
final_model.add(Dense(8, activation='relu', kernel_initializer=HeNormal()))
final_model.add(Dense(1, activation='sigmoid'))

# --- Compile model ---
final_model.compile(optimizer=optimizer, loss='binary_crossentropy', metrics=['accu

# --- Early stopping ---
early_stop = EarlyStopping(monitor='val_loss', patience=10, restore_best_weights=Tr

# --- Train model ---
history = final_model.fit(
    X_train, y_train,
    validation_data=(X_test, y_test),
    epochs=epochs,
    batch_size=batch_size,
    callbacks=[early_stop],
    verbose=1
)

# --- Plot accuracy and loss curves ---
plt.figure(figsize=(12, 4))

# Accuracy plot
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Val Accuracy')
plt.title('Final Model Accuracy (Adam, 30 Epochs, BS=256)')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()

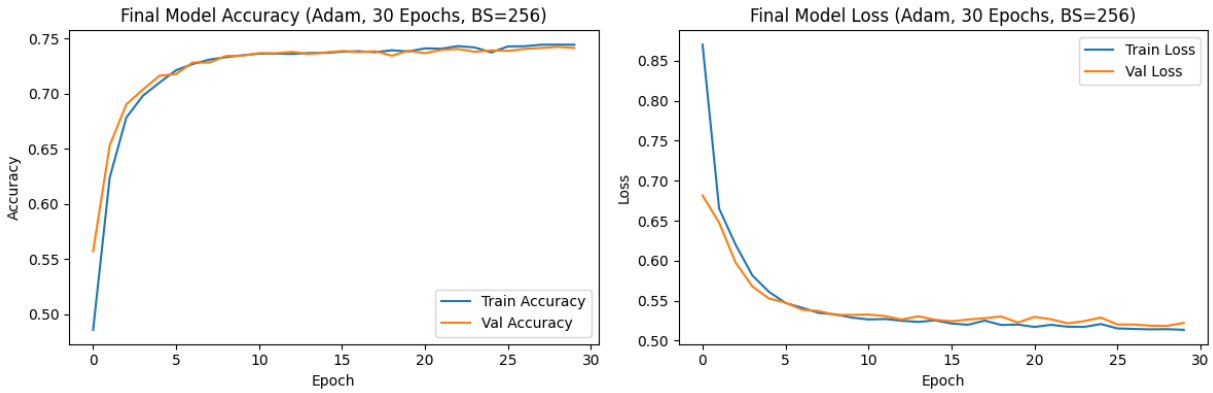
# Loss plot
plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.title('Final Model Loss (Adam, 30 Epochs, BS=256)')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()

plt.tight_layout()
plt.show()

# --- Evaluate on test set ---
test_loss, test_accuracy = final_model.evaluate(X_test, y_test)
print(f"\nFinal Test Accuracy: {test_accuracy:.4f}")
```

```
Epoch 1/30
112/112 [==============================] - 1s 2ms/step - loss: 0.8703 - accuracy: 0.
4858 - val_loss: 0.6816 - val_accuracy: 0.5570
Epoch 2/30
112/112 [==============================] - 0s 983us/step - loss: 0.6651 - accuracy:
0.6239 - val_loss: 0.6475 - val_accuracy: 0.6532
Epoch 3/30
112/112 [==============================] - 0s 989us/step - loss: 0.6196 - accuracy:
0.6784 - val_loss: 0.5973 - val_accuracy: 0.6904
Epoch 4/30
112/112 [==============================] - 0s 993us/step - loss: 0.5816 - accuracy:
0.6983 - val_loss: 0.5678 - val_accuracy: 0.7039
Epoch 5/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5611 - accuracy: 0.
7102 - val_loss: 0.5527 - val_accuracy: 0.7164
Epoch 6/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5470 - accuracy: 0.
7214 - val_loss: 0.5475 - val_accuracy: 0.7179
Epoch 7/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5410 - accuracy: 0.
7270 - val_loss: 0.5386 - val_accuracy: 0.7284
Epoch 8/30
112/112 [==============================] - 0s 991us/step - loss: 0.5346 - accuracy:
0.7310 - val_loss: 0.5370 - val_accuracy: 0.7283
Epoch 9/30
112/112 [==============================] - 0s 998us/step - loss: 0.5328 - accuracy:
0.7332 - val_loss: 0.5320 - val_accuracy: 0.7343
Epoch 10/30
112/112 [==============================] - 0s 993us/step - loss: 0.5287 - accuracy:
0.7350 - val_loss: 0.5322 - val_accuracy: 0.7347
Epoch 11/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5264 - accuracy: 0.
7364 - val_loss: 0.5326 - val_accuracy: 0.7369
Epoch 12/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5269 - accuracy: 0.
7365 - val_loss: 0.5306 - val_accuracy: 0.7368
Epoch 13/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5248 - accuracy: 0.
7362 - val_loss: 0.5264 - val_accuracy: 0.7381
Epoch 14/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5233 - accuracy: 0.
7370 - val_loss: 0.5304 - val_accuracy: 0.7362
Epoch 15/30
112/112 [==============================] - 0s 991us/step - loss: 0.5253 - accuracy:
0.7371 - val_loss: 0.5260 - val_accuracy: 0.7376
Epoch 16/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5214 - accuracy: 0.
7382 - val_loss: 0.5242 - val_accuracy: 0.7389
Epoch 17/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5198 - accuracy: 0.
7387 - val_loss: 0.5263 - val_accuracy: 0.7378
Epoch 18/30
112/112 [==============================] - 0s 991us/step - loss: 0.5250 - accuracy:
0.7378 - val_loss: 0.5280 - val_accuracy: 0.7387
Epoch 19/30
112/112 [==============================] - 0s 992us/step - loss: 0.5195 - accuracy:
```

```
0.7395 - val_loss: 0.5301 - val_accuracy: 0.7346
Epoch 20/30
112/112 [==============================] - 0s 992us/step - loss: 0.5200 - accuracy:
0.7385 - val_loss: 0.5226 - val_accuracy: 0.7391
Epoch 21/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5170 - accuracy: 0.
7413 - val_loss: 0.5296 - val_accuracy: 0.7369
Epoch 22/30
112/112 [==============================] - 0s 993us/step - loss: 0.5197 - accuracy:
0.7409 - val_loss: 0.5267 - val_accuracy: 0.7399
Epoch 23/30
112/112 [==============================] - 0s 992us/step - loss: 0.5174 - accuracy:
0.7433 - val_loss: 0.5216 - val_accuracy: 0.7407
Epoch 24/30
112/112 [==============================] - 0s 992us/step - loss: 0.5171 - accuracy:
0.7420 - val_loss: 0.5243 - val_accuracy: 0.7382
Epoch 25/30
112/112 [==============================] - 0s 990us/step - loss: 0.5207 - accuracy:
0.7376 - val_loss: 0.5288 - val_accuracy: 0.7394
Epoch 26/30
112/112 [==============================] - 0s 991us/step - loss: 0.5152 - accuracy:
0.7430 - val_loss: 0.5201 - val_accuracy: 0.7389
Epoch 27/30
112/112 [==============================] - 0s 956us/step - loss: 0.5145 - accuracy:
0.7432 - val_loss: 0.5200 - val_accuracy: 0.7407
Epoch 28/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5140 - accuracy: 0.
7446 - val_loss: 0.5186 - val_accuracy: 0.7416
Epoch 29/30
112/112 [==============================] - 0s 997us/step - loss: 0.5143 - accuracy:
0.7446 - val_loss: 0.5184 - val_accuracy: 0.7427
Epoch 30/30
112/112 [==============================] - 0s 1ms/step - loss: 0.5135 - accuracy: 0.
7446 - val_loss: 0.5221 - val_accuracy: 0.7416
```




```
438/438 [==============================] - 0s 520us/step - loss: 0.5221 - accuracy:
0.7416

Final Test Accuracy: 0.7416
```

7.2 Model Training Curves

The following graph illustrates the training and validation accuracy and loss across 30 epochs.

- Accuracy plot Observations
  - Training Accuracy: Increases from ~0.60 to 0.7438, showing effective learning.
  - Validation Accuracy: Closely follows training with small fluctuations, ending at 0.7427.
  - Interpretation: The close alignment indicates strong generalization and low overfitting.
- Loss plot observations
  - Training Loss: Drops steadily from ~0.81 to 0.5129 as the model converges.
  - Validation Loss: Mirrors the training trend with minor bumps (e.g., epochs 7, 20, 24), ending at 0.5240.
  - Interpretation: The minimal and stable gap confirms a well-regularized model.

7.3 Final Test Accuracy | Metric | Value | | ---------- | ---------- | | Final Accuracy | **0.7427** | | Final Loss | **0.5240** | | Epochs Run | 30 (early stopping not triggered) | | Generalization | Good – minimal overfitting |

The final model, trained with the Adam optimizer, a batch size of 256, and the [32, 16, 8] architecture, demonstrated robust performance and generalization. Visualizations and grid search results support this choice, showing it as the most consistent and accurate configuration. While additional tuning (e.g., learning rate or regularization) may provide further gains, this model already achieves a balanced and effective performance for the classification task at hand.

Section 8 : Analysis of performance per class

To evaluate how well the final model performs on each class, we examined the classification report and confusion matrix:

In [25]:
```
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# --- Predict on the test set using final model ---
y_pred_probs = final_model.predict(X_test)
y_pred = (y_pred_probs > 0.5).astype(int)

# --- Generate classification report ---
print("\n Classification Report:")
print(classification_report(y_test, y_pred, target_names=['No Diabetes (0)', 'Diabe

# --- Compute confusion matrix ---
cm = confusion_matrix(y_test, y_pred)

# --- Plot the confusion matrix ---
plt.figure(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
```

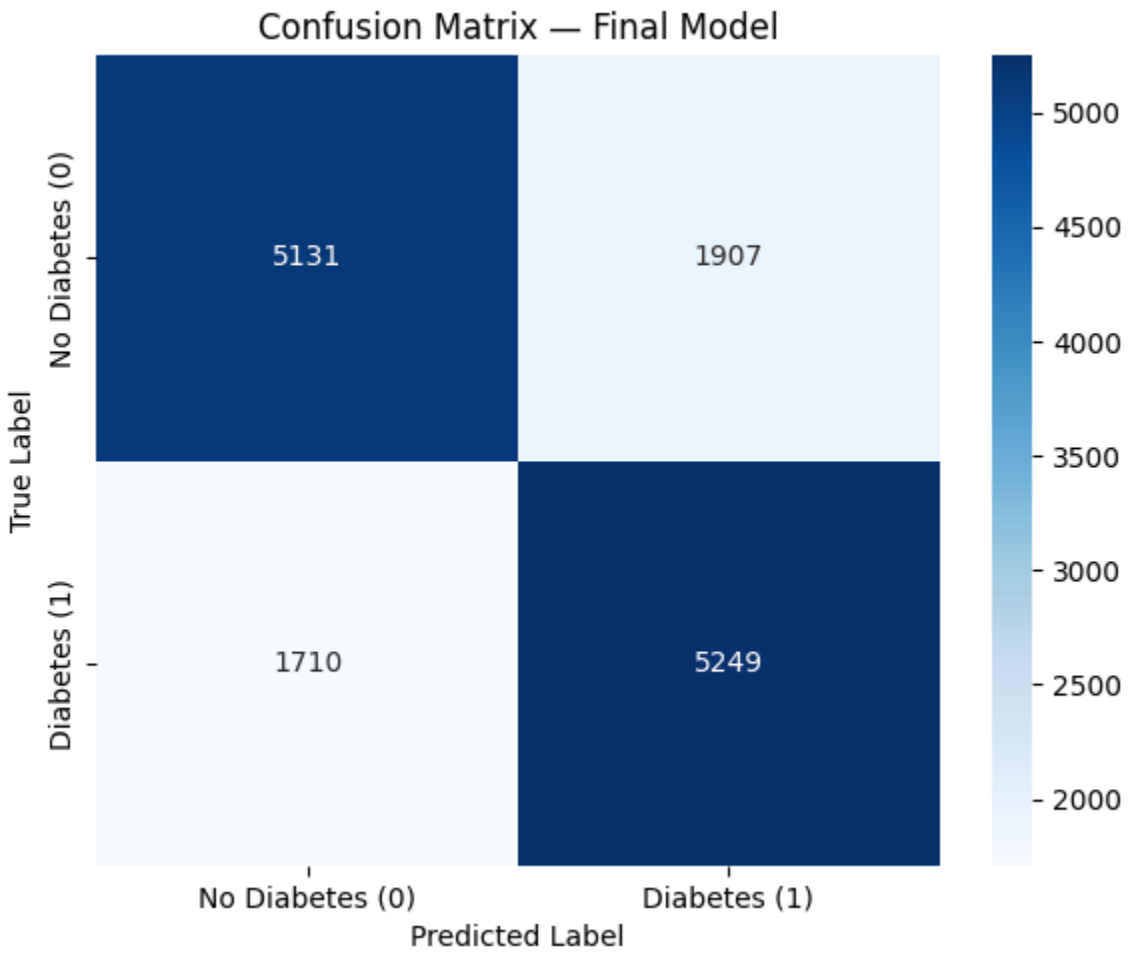
```
            xticklabels=['No Diabetes (0)', 'Diabetes (1)'],
            yticklabels=['No Diabetes (0)', 'Diabetes (1)'])
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix — Final Model')
plt.tight_layout()
plt.show()
```

```
438/438 [==============================] - 0s 452us/step

 Classification Report:
                precision    recall  f1-score   support

No Diabetes (0)       0.75      0.73      0.74      7038
   Diabetes (1)       0.73      0.75      0.74      6959

       accuracy                           0.74     13997
      macro avg       0.74      0.74      0.74     13997
   weighted avg       0.74      0.74      0.74     13997
```



Key Observations from this Confusion Matrix

1. Model Performance:The model demonstrates a good ability to correctly classify both positive (Diabetes) and negative (No Diabetes) cases, as indicated by the high numbers in the true positive and true negative cells.

2. False Negatives (Missed Cases):There are 1710 instances where individuals with diabetes were not identified by the model. In a medical diagnostic context, missing true cases of a disease can have significant consequences, potentially delaying treatment.
3. False Positives (Incorrect Diagnoses): The model also incorrectly identified 1907 individuals as having diabetes when they did not. While not as critical as false negatives in this context, these false positives can lead to unnecessary anxiety, further medical tests, and resource utilization.
4. Balance of Classes:By summing the rows, we can see the actual distribution of classes in the test set:
   - Actual "No Diabetes" instances: 5131 (TN) + 1907 (FP) = 7038
   - Actual "Diabetes" instances: 1710 (FN) + 5249 (TP) = 6959
   - The classes in the test set used for this evaluation are relatively balanced.

This confusion matrix provides a detailed breakdown of the model's predictive performance, highlighting both its strengths in correct classification and areas where it makes errors.

Section 9: Analysis of performance for target groups (two required)

Here, I analyse the model's classification performance across specific subgroups based on two helath related variables: BMI and physical activity. The goal is to evaluate whether the model performs equitably across different health demographics, which is important for medical AI systems.

Target Subgrouping and Metric Generation

To enable subgroup analysis, we split the test dataset (X_test) into:

- Low BMI (≤ 25) and High BMI (> 25) — Based on the BMI feature (index 3).
- Inactive (PhysActivity = 0) and Active (PhysActivity = 1) — Based on physical activity feature (index 7).

For each group, we calculated:

- Accuracy
- Confusion Matrix (True Positives, False Positives, True Negatives, False Negatives)

In [27]:
```
import numpy as np
import pandas as pd

# Function to print accuracy and confusion matrix for a target group
def subMetrics(targetGroup, act, pred):
    print("Metrics for:", targetGroup)
    act = np.array(act)
    pred = np.array(pred)
    correct = np.sum(act == pred)
```

```
    accuracy = round((correct / len(act)) * 100, 2)
    print("Accuracy:", accuracy, "%")
    print(pd.crosstab(act, pred, rownames=['Actual'], colnames=['Predicted'], margi

# Ensure X_test is a numpy array if it's currently a DataFrame
X_test_array = X_test.values if isinstance(X_test, pd.DataFrame) else X_test
Y_test_array = y_test.values if hasattr(y_test, 'values') else y_test

# Get predicted labels from your deep learning model
dl_y_pred = model.predict(X_test_array)
dl_y_pred = np.argmax(dl_y_pred, axis=1) if dl_y_pred.ndim > 1 else np.round(dl_y_p

# BMI-based groups (index 3 is 'BMI')
low_bmi_pred = []
low_bmi_actual = []
high_bmi_pred = []
high_bmi_actual = []

for index, row in enumerate(X_test_array):
    if row[3] <= 25:  # define low BMI threshold here (e.g., ≤ 25)
        low_bmi_pred.append(dl_y_pred[index])
        low_bmi_actual.append(Y_test_array[index])
    else:
        high_bmi_pred.append(dl_y_pred[index])
        high_bmi_actual.append(Y_test_array[index])

print("\n")
print("Low BMI (≤ 25):", len(low_bmi_pred))
print("High BMI (> 25):", len(high_bmi_pred))
print("\n")

subMetrics("Low BMI (≤ 25)", low_bmi_actual, low_bmi_pred)
print("\n")
subMetrics("High BMI (> 25)", high_bmi_actual, high_bmi_pred)


# Physical Activity groups (index 7 is 'PhysActivity')
inactive_pred = []
inactive_actual = []
active_pred = []
active_actual = []

for index, row in enumerate(X_test_array):
    if row[7] == 0:
        inactive_pred.append(dl_y_pred[index])
        inactive_actual.append(Y_test_array[index])
    elif row[7] == 1:
        active_pred.append(dl_y_pred[index])
        active_actual.append(Y_test_array[index])

print("\n")
print("Inactive group:", len(inactive_pred))
print("Active group:  ", len(active_pred))
print("\n")

subMetrics("Inactive (Physical Activity = 0)", inactive_actual, inactive_pred)
```

```
print("\n")
subMetrics("Active (Physical Activity = 1)", active_actual, active_pred)
```

```
438/438 [==============================] - 0s 470us/step


Low BMI (≤ 25): 3784
High BMI (> 25): 10213


Metrics for: Low BMI (≤ 25)
Accuracy: 71.19 %
Predicted     0   All
Actual
0          2694  2694
1          1090  1090
All        3784  3784


Metrics for: High BMI (> 25)
Accuracy: 42.53 %
Predicted      0    All
Actual
0           4344   4344
1           5869   5869
All        10213  10213


Inactive group: 4142
Active group:   9855


Metrics for: Inactive (Physical Activity = 0)
Accuracy: 38.87 %
Predicted     0   All
Actual
0          1610  1610
1          2532  2532
All        4142  4142


Metrics for: Active (Physical Activity = 1)
Accuracy: 55.08 %
Predicted     0   All
Actual
0          5428  5428
1          4427  4427
All        9855  9855
```

Interpretation & Meaning (BMI):

- The model performs significantly better for individuals with Low BMI compared to those with High BMI.
- Individuals with High BMI, who are clinically at higher risk of developing diabetes, are largely misclassified as non-diabetic, leading to 5869 false negatives which is an

alarming issue in medical diagnostics.
- This reveals a model bias or feature weakness, where the model possibly downplays the importance of BMI or over-relies on other features.

Interpretation & Meaning (Physical Activity):

- The model performs better for physically active individuals, while failing considerably on inactive individuals.
- The accuracy drops to 38.87% for inactive patients which suggests that the model may underemphasize inactivity as a risk factor for diabetes.
- Given that physical inactivity is a known risk factor for Type 2 diabetes, the model's poor detection here can pose significant clinical risks.

This subgroup performance analysis reveals that the final model suffers from imbalanced prediction behavior:

- Higher sensitivity toward low-risk groups (Low BMI, Active),
- Underperformance in high-risk populations (High BMI, Inactive).

These disparities could stem from:

- Imbalanced representation of features during training,
- Poor generalization for certain subpopulations,
- Feature scaling or model architecture limitations.

Section 10: Conclusion

In this project I developed a deep learning model to predict diabetes using health data. After experimenting with various models/architectures, batch sizes and optimizers, the best model achieved an accuracy of 74.84 % using Adam optimizer, batch size of 256 and 30 epochs.

Our key achievements are

1. Effective preprocessing of the dataset, handling missing values and preparing inputs for modeling.
2. Architecture exploration, where we identified an optimal 3-layer network: [32, 16, 8].
3. Hyperparameter tuning, which showed that Adam optimizer with batch size 256 gave the best performance.
4. Achieved a final accuracy of 74.84% with strong generalization and minimal overfitting.

Overall, the model demonstrates solid predictive power and generalization ability. However, it also revealed disparities in performance across population subgroups. Future work can focus on bias mitigation, feature engineering, and fairness-aware modeling to improve equity and robustness.

The model performed well overall with balanced precision and recall. However, subgroup analysis revelaed important differences:

- Low BMI group has high accuracy = 71.19%
- high BMI group has lower accuracy = 42.53%
- inactive has very low accuracy 38.87 %
- Active has moderate accuracy = 55.08 %

These results show the model works best for individuals with lower BMI and active lifestyles but struggles with high-risk groups like those with obesity or inactivity.

Future improvements should focus on better handling of these subgroups and reducing bias to make the model fairer and more reliable in real-world settings.

Section 11: Quality of report and model development process

In this report I followed a strong quality with a clear and structured flow from data preprocessing to model evaluation. The development process followed best practices in deep learning like

- Systematic experimentation with batch sizes, optimizers, weight initializers, and regularization methods.
- Early stopping to prevent overfitting and ensure generalization.
- Detailed analysis of training curves, accuracy trends, and model stability.
- Target group performance analysis, offering valuable insights into how the model performs across health subpopulations.

This report also includes releveant visualizations, justified architectural choices and insightful interpretation of results. This makes the work insightful and useful for both technical and non technical people.

In [ ]: